Signals and Systems

Lecture 7: Bode Diagrams

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Partly adapted from the materials provided on
the MIT OpenCourseWare

Review: Convolution

Last time, we saw how a linear, time-invariant (LTI) system can be characterized by its unit-sample/impulse response.

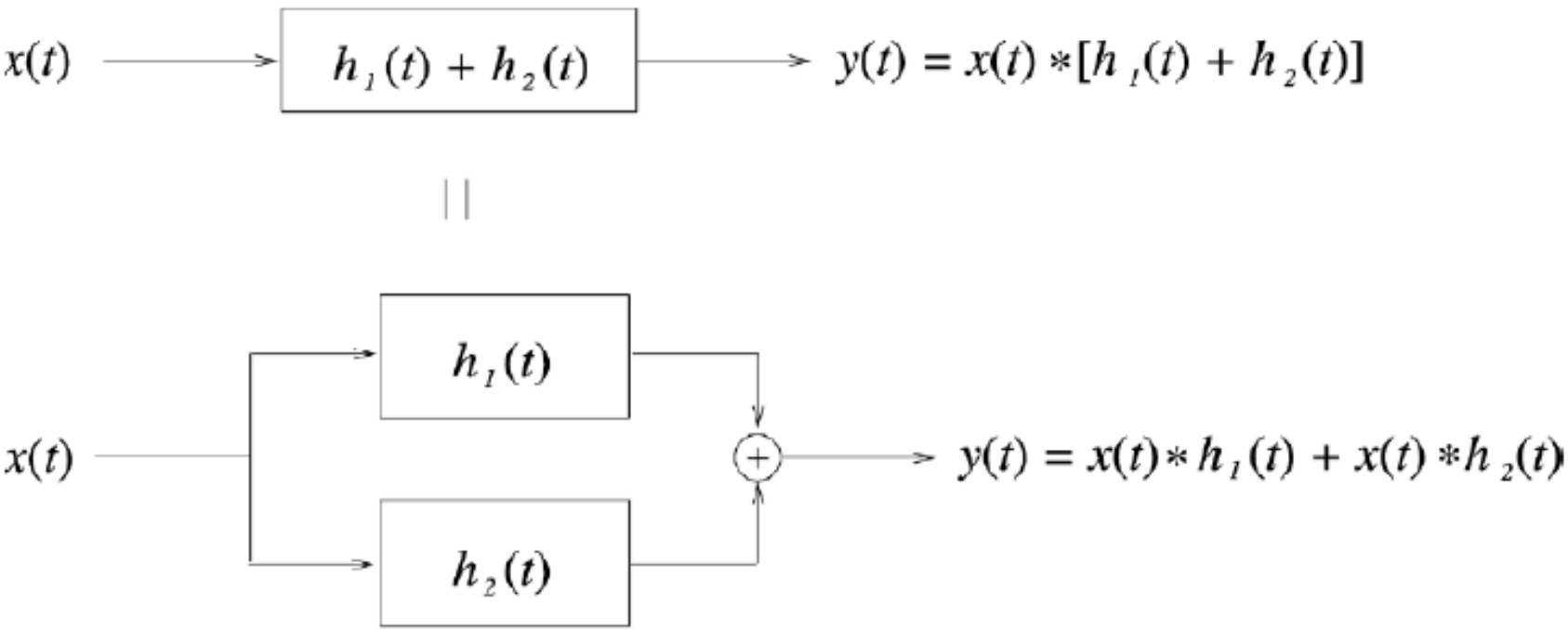
DT: $$y[n] = (x * h)[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

CT: $$y(t) = (x * h)(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$$

Characterizing a system by its unit-sample/impulse response is especially insightful for some systems.

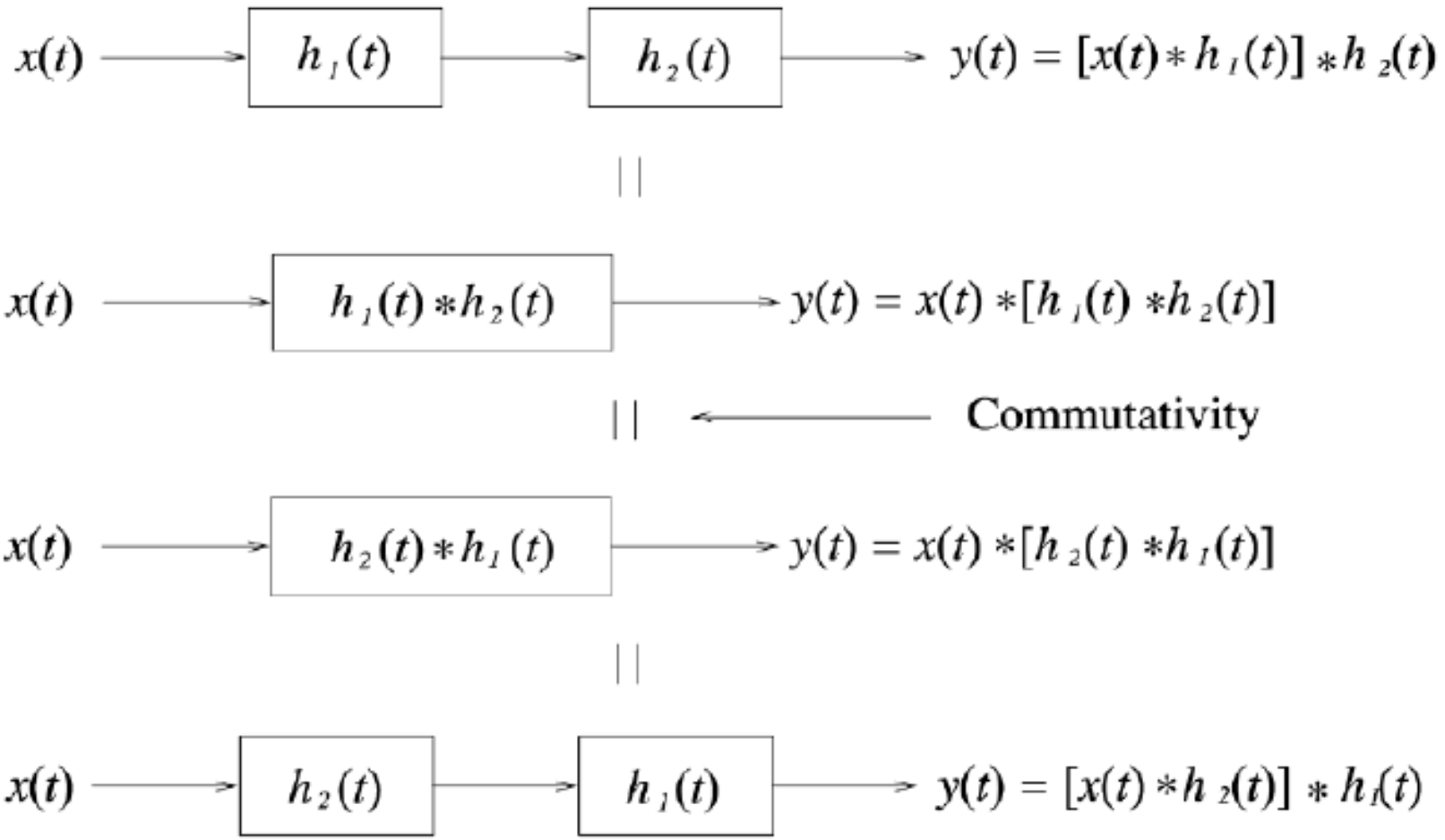
Review: Properties of LTI Systems

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Review: Properties of LTI Systems

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Review: Properties of LTI Systems

- LTI systems with and without memory

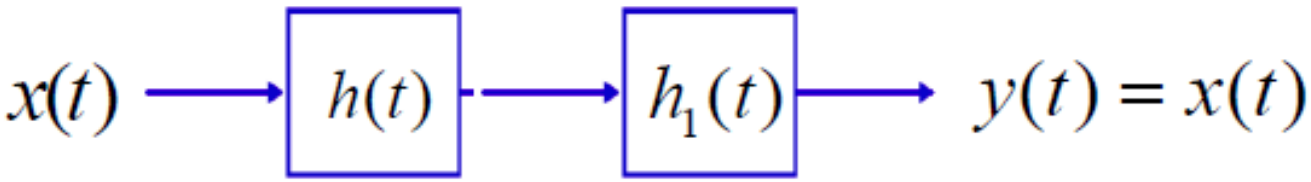


$$y(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau$$

Memoryless $\Leftrightarrow$ $h(t) = 0, t \neq 0$

Review: Properties of LTI Systems

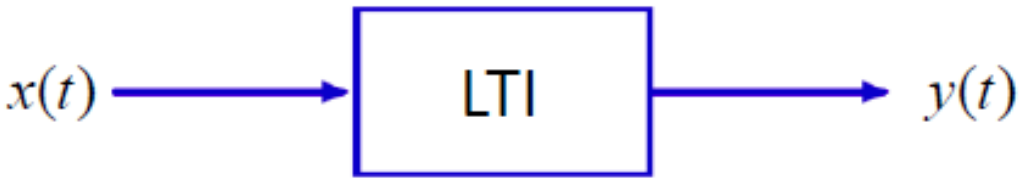
- Invertibility of LTI systems



$$h(t) * h_1(t) = \delta(t)$$

Review: Properties of LTI Systems

- Causality for LTI systems



$$y(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau$$

$$\text{CT LTI system is causal} \Leftrightarrow h(t) = 0, t < 0$$

Review: Properties of LTI Systems

- Stability for LTI systems

$$\text{CT LTI system is stable} \Leftrightarrow \int_{-\infty}^{+\infty} |h(\tau)| d\tau < \infty$$

Sufficient condition:

$$\text{For } |x(t)| \le x_{\max} < \infty.$$

$$|y(t)| = \left| \int_{-\infty}^{+\infty} x(\tau) h(t-\tau) d\tau \right| \le x_{\max} \left| \int_{-\infty}^{+\infty} h(t-\tau) d\tau \right| < \infty.$$

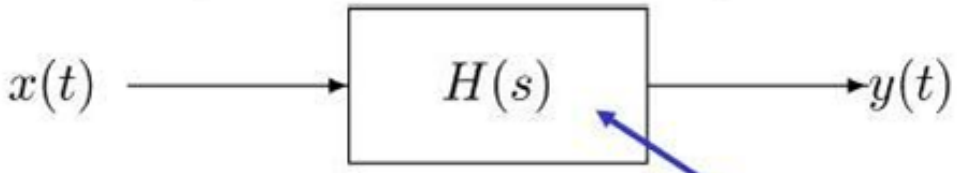
Necessary condition:

$$\text{Suppose} \quad \int_{-\infty}^{+\infty} |h(\tau)| d\tau = \infty$$

$$\text{Let} \quad x(t) = h^*(-t)/|h^*(-t)|, \quad \text{then} \quad |x(t)| \equiv 1 \quad \text{bounded}$$

$$\text{But} \quad y(0) = \int_{-\infty}^{+\infty} x(\tau) h(-\tau) d\tau = \int_{-\infty}^{+\infty} \frac{h^*(-\tau) h(-\tau)}{|h(-\tau)|} d\tau = \int_{-\infty}^{+\infty} |h(-\tau)| d\tau = \infty$$

Review: Properties of LTI Systems



H(s) = "system function"

$$Y(s) = H(s)X(s)$$

1) System is stable $\Leftrightarrow \int_{-\infty}^{\infty} |h(t)|dt < \infty$ $\Leftrightarrow$ ROC of $H(s)$ includes $j\omega$ axis

2) Causality $\Rightarrow$ $h(t)$ right-sided signal $\Rightarrow$ ROC of $H(s)$ is a right-half plane

Question:

If the ROC of $H(s)$ is a right-half plane, is the system causal?

Ex. $H(s) = \dfrac{e^{sT}}{s+1}, \quad \Re e\{s\} > -1 \Rightarrow h(t)$ right-sided

$$h(t) = \mathcal{L}^{-1}\left\{\frac{e^{sT}}{s+1}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\}_{t\to t+T} = e^{-t}u(t)|_{t\to t+T}$$

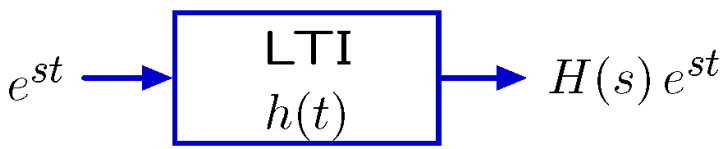
$$= e^{-(t+T)}u(t+T) \neq 0 \quad \text{at} \quad t < 0$$ Non-causal

Review: Frequency Response

Complex exponentials are eigenfunctions of LTI systems.

If $x(t) = e^{st}$ and $h(t)$ is the impulse response then

$$y(t) = (h * x)(t) = \int_{-\infty}^{\infty} h(\tau) e^{s(t-\tau)} d\tau = e^{st} \int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau = H(s)\, e^{st}$$



Eternal sinusoids are sums of complex exponentials.

$$\cos \omega_0 t = \frac{1}{2} \left(e^{j\omega_0 t} + e^{-j\omega_0 t} \right)$$

Review: Frequency Response

Response to eternal sinusoids.

Let $x(t) = \cos \omega_0 t$ (for all time), which can be written as

$$x(t) = \frac{1}{2}\left(e^{j\omega_0 t} + e^{-j\omega_0 t}\right)$$

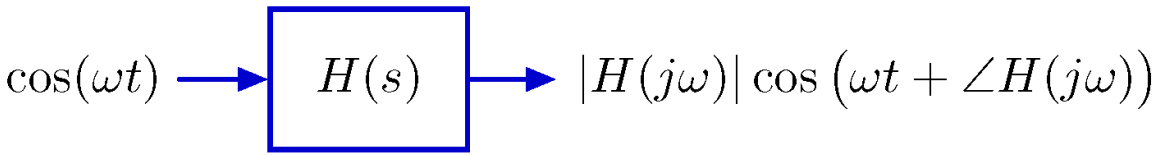
The response to a sum is the sum of the responses,

$$\begin{aligned} y(t) &= \frac{1}{2}\left(H(j\omega_0)e^{j\omega_0 t} + H(-j\omega_0)e^{-j\omega_0 t}\right) \\ &= \mathrm{Re}\left\{H(j\omega_0)e^{j\omega_0 t}\right\} \\ &= \mathrm{Re}\left\{|H(j\omega_0)|e^{j\angle H(j\omega_0)}e^{j\omega_0 t}\right\} \\ &= |H(j\omega_0)|\mathrm{Re}\left\{e^{j\omega_0 t + j\angle H(j\omega_0)}\right\} \end{aligned}$$

$$y(t) = |H(j\omega_0)|\cos\left(\omega_0 t + \angle\left(H(j\omega_0)\right)\right).$$

Review: Frequency Response

The magnitude and phase of the response of a system to an eternal cosine signal is the magnitude and phase of the system function evaluated at $s = j\omega$.



Review: Vector Diagrams

The value of $H(s)$ at a point $s = s_0$ can be determined by combining the contributions of the vectors associated with each of the poles and zeros.

$$H(s_0) = K\frac{(s_0 - z_0)(s_0 - z_1)(s_0 - z_2)\cdots}{(s_0 - p_0)(s_0 - p_1)(s_0 - p_2)\cdots}$$

The magnitude is determined by the product of the magnitudes.

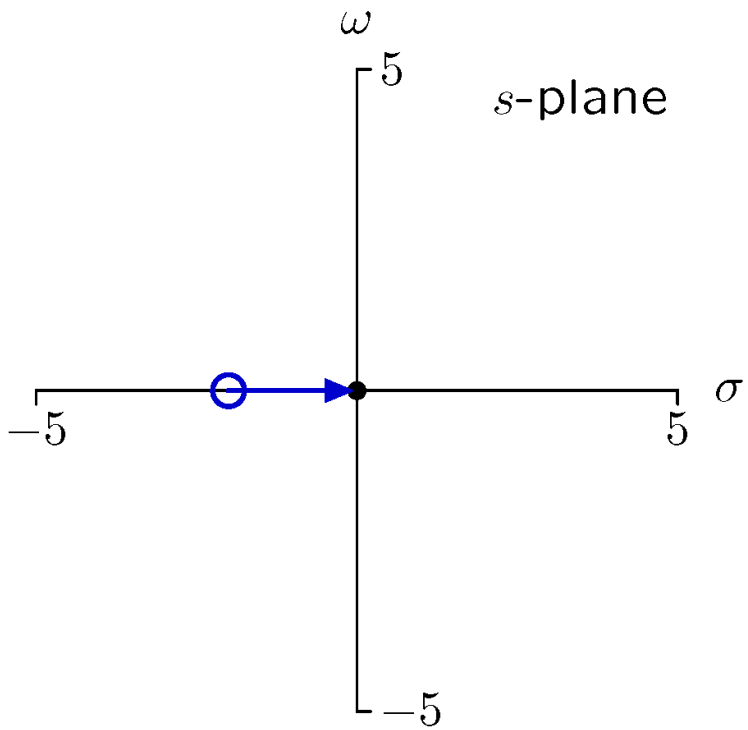
$$|H(s_0)| = |K|\frac{|(s_0 - z_0)||(s_0 - z_1)||(s_0 - z_2)|\cdots}{|(s_0 - p_0)||(s_0 - p_1)||(s_0 - p_2)|\cdots}$$

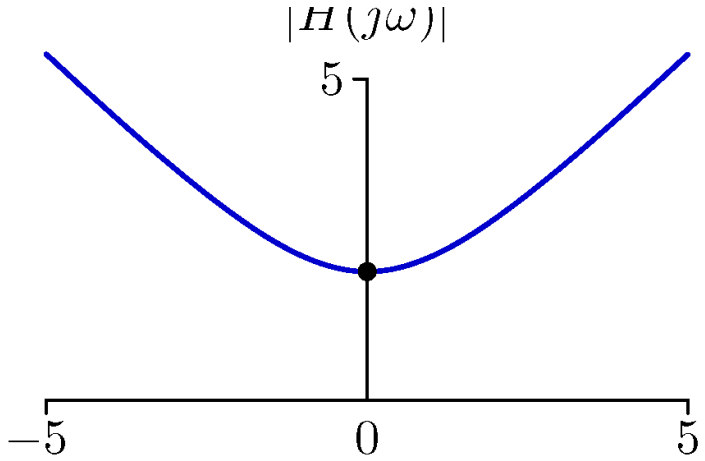
The angle is determined by the sum of the angles.

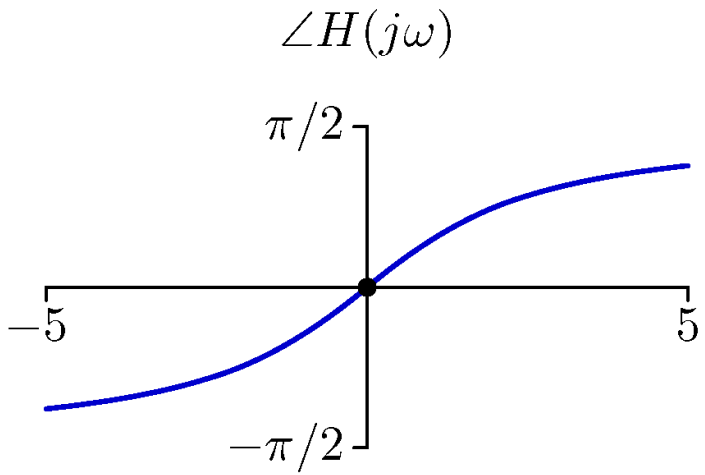
$$\angle H(s_0) = \angle K + \angle(s_0 - z_0) + \angle(s_0 - z_1) + \cdots - \angle(s_0 - p_0) - \angle(s_0 - p_1) - \cdots$$

Review: Vector Diagrams

$$H(s) = s - z_1$$

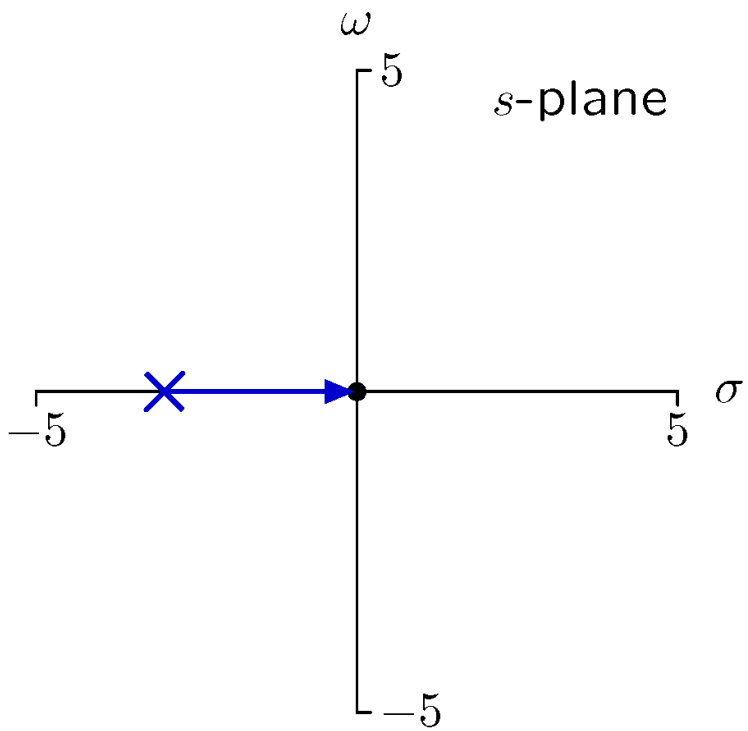


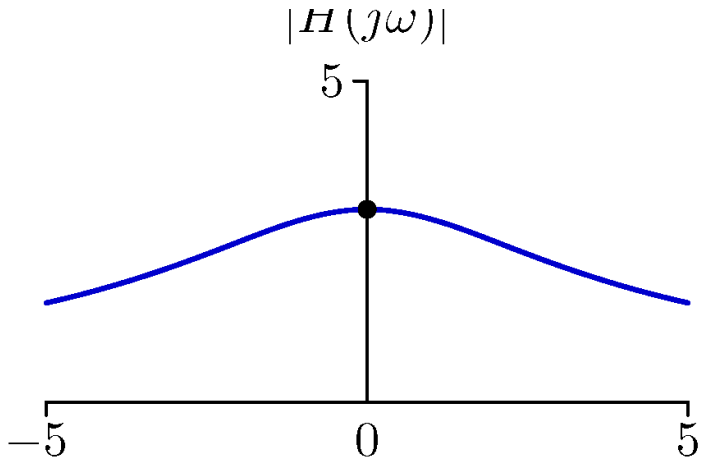


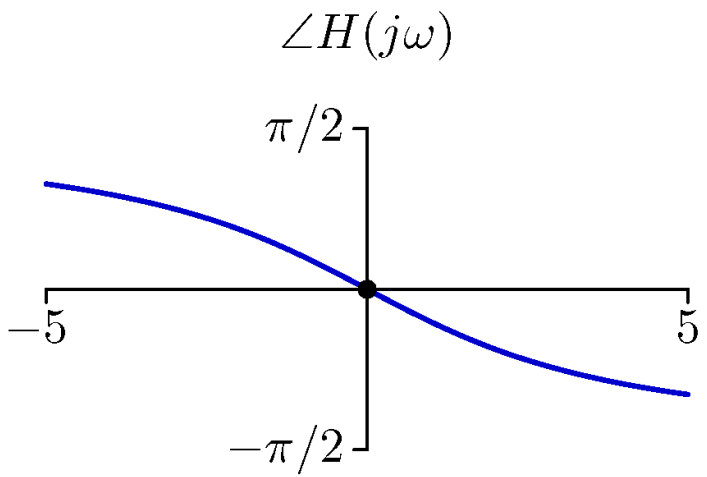


Review: Vector Diagrams

$$H(s) = \frac{9}{s - p_1}$$

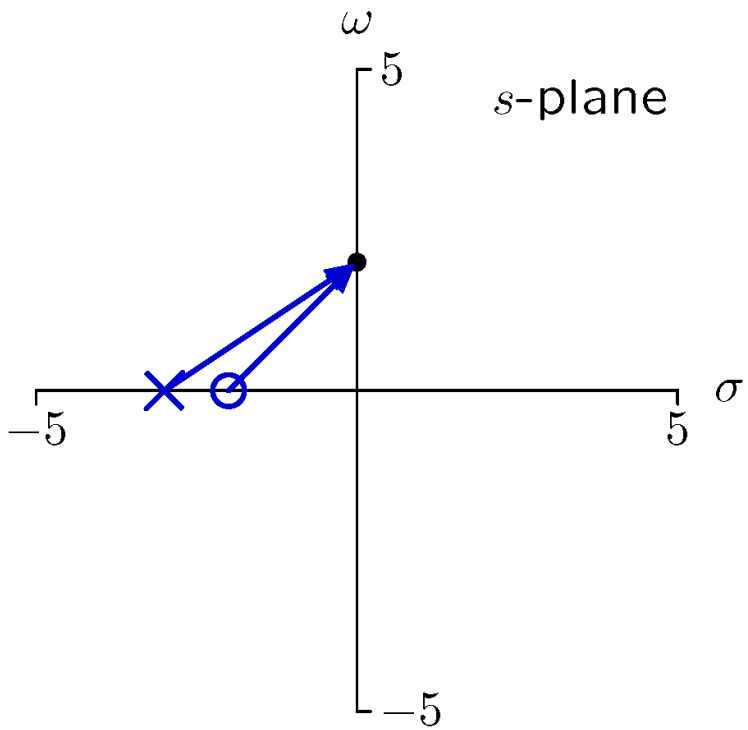


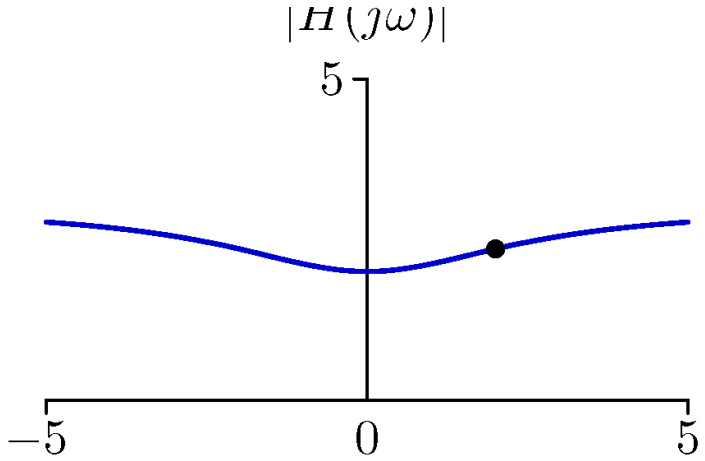


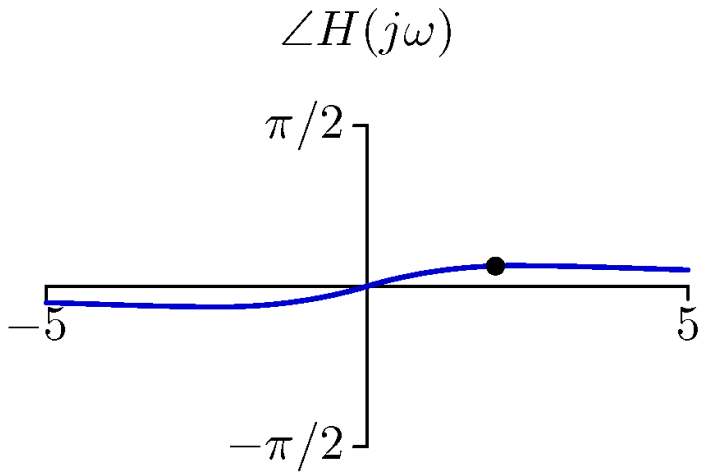


Review: Vector Diagrams

$$H(s) = 3\frac{s - z_1}{s - p1}$$

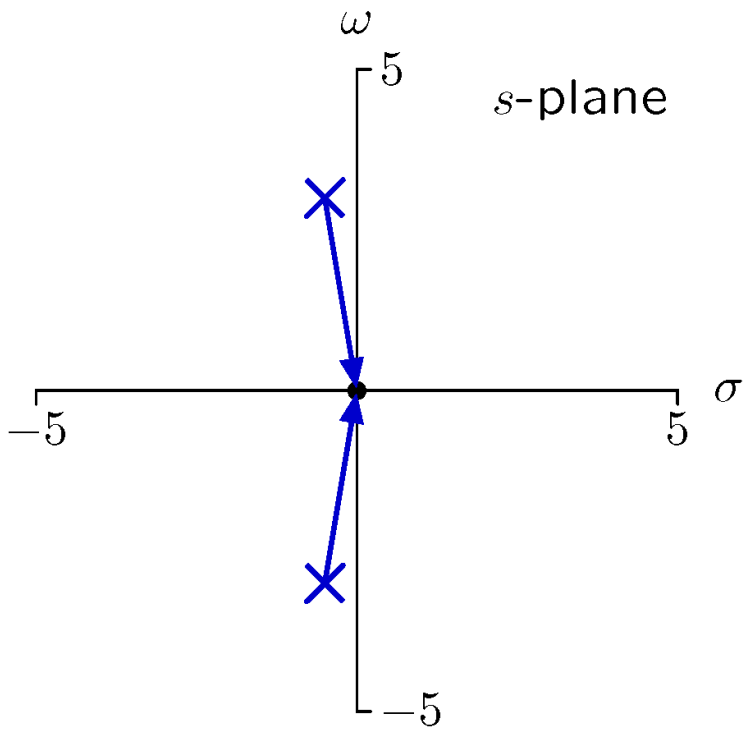


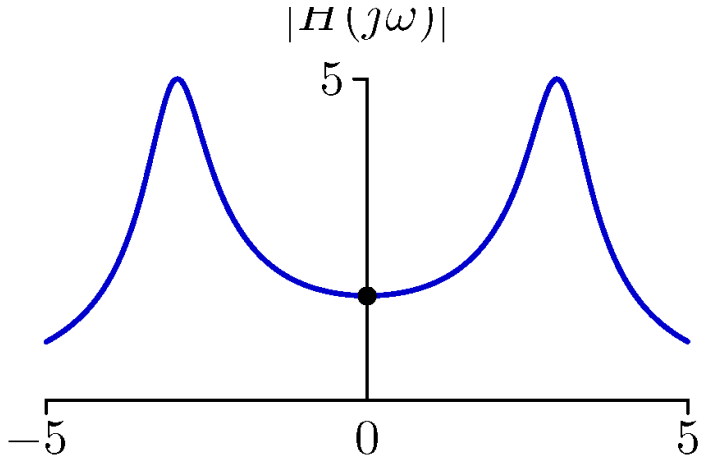


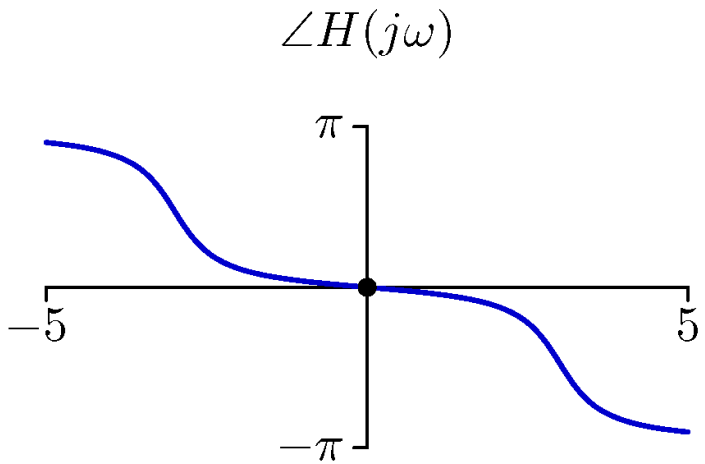


Review: Vector Diagrams

$$H(s) = \frac{15}{(s-p_1)(s-p_2)}$$

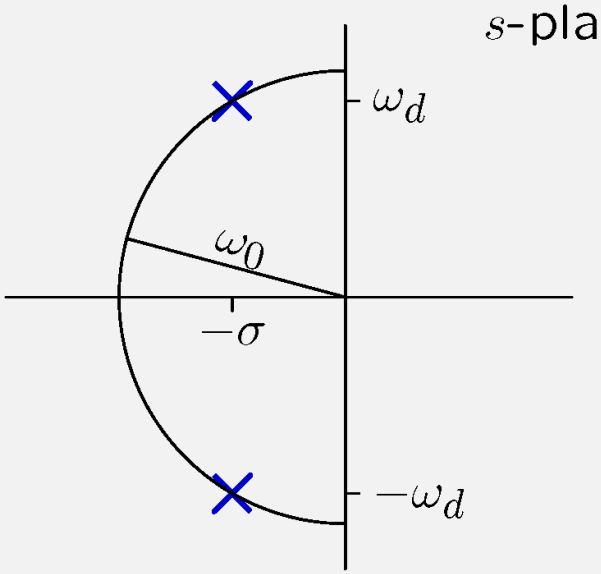






Check Yourself

Consider the system represented by the following poles.

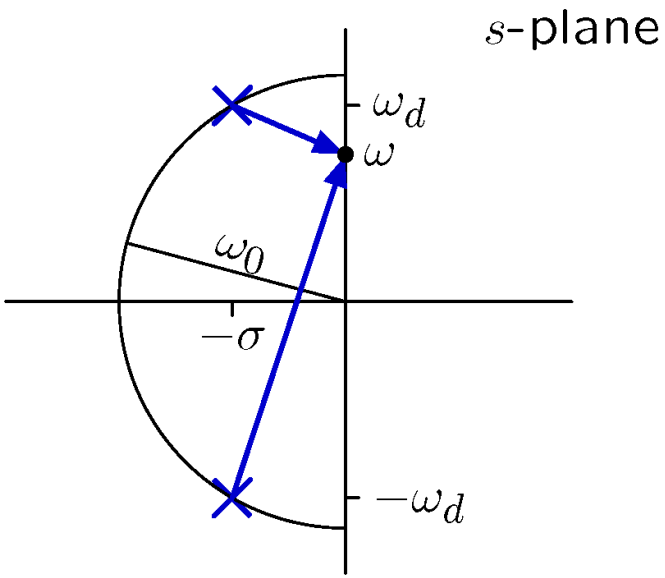


Find the frequency ω at which the magnitude of the response $y(t)$ is greatest if $x(t) = \cos \omega t$.

1. $\omega = \omega_d$
2. $\omega_d < \omega < \omega_0$
3. $0 < \omega < \omega_d$
4. none of the above

Check Yourself: Frequency Response

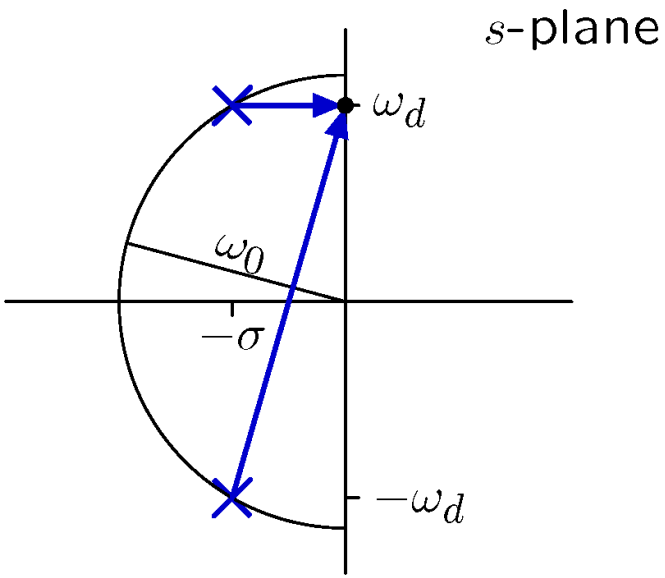
Analyze with vectors.



The product of the lengths is $\left(\sqrt{(\omega+\omega_d)^2+\sigma^2}\right)\left(\sqrt{(\omega-\omega_d)^2+\sigma^2}\right)$.

Check Yourself: Frequency Response

Analyze with vectors.

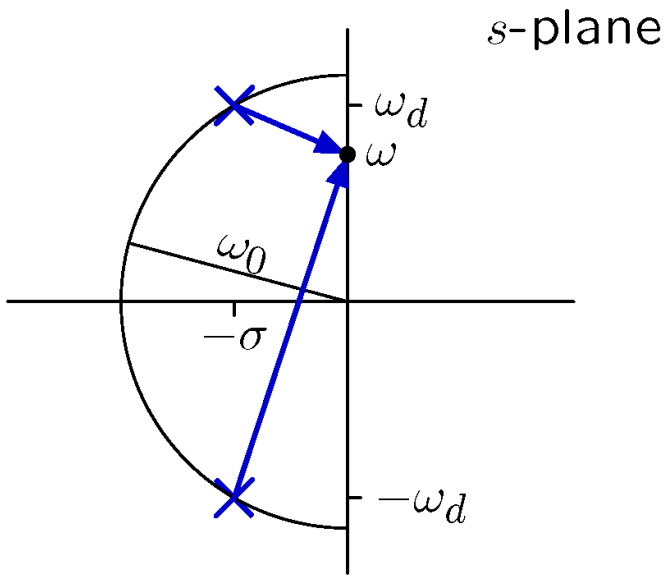


The product of the lengths is $\left(\sqrt{(\omega+\omega_d)^2+\sigma^2}\right)\left(\sqrt{(\omega-\omega_d)^2+\sigma^2}\right)$.

Decreasing ω from ω_d to $\omega_d-\epsilon$ decreases the product since length of bottom vector decreases as ϵ while length of top vector increases only ϵ^2.

Check Yourself: Frequency Response

More mathematically ...



The product of the lengths is $\left(\sqrt{(\omega+\omega_d)^2+\sigma^2}\right)\left(\sqrt{(\omega-\omega_d)^2+\sigma^2}\right)$.

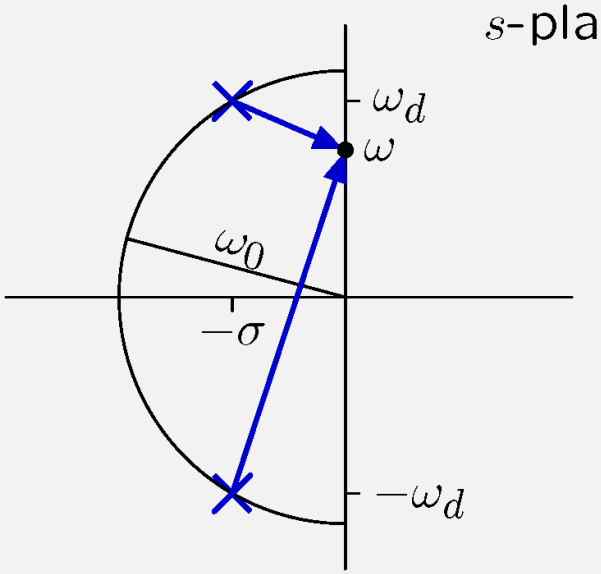
Maximum occurs where derivative of squared lengths is zero.

$$\frac{d}{d\omega}\left((\omega+\omega_d)^2+\sigma^2\right)\left((\omega-\omega_d)^2+\sigma^2\right)=0$$

$$\rightarrow \quad \omega^2=\omega_d^2-\sigma^2=\omega_0^2-2\sigma^2\,.$$

Check Yourself

Consider the system represented by the following poles.

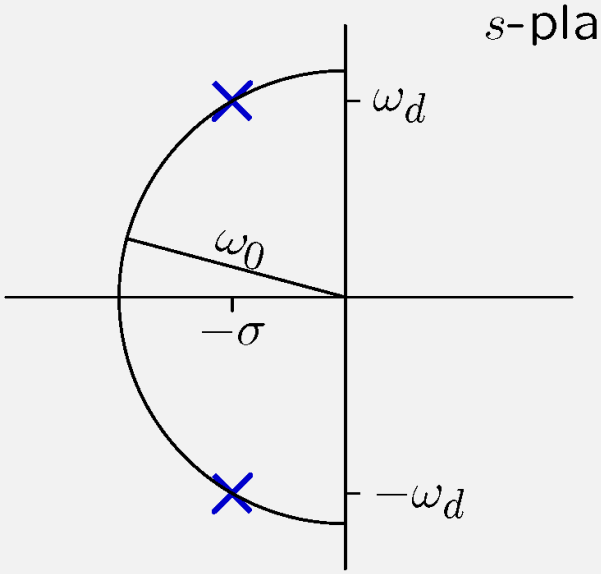


Find the frequency ω at which the magnitude of the response $y(t)$ is greatest if $x(t) = \cos \omega t$. 3

1. $\omega = \omega_d$
2. $\omega_d < \omega < \omega_0$
3. $0 < \omega < \omega_d$
4. none of the above

Check Yourself

Consider the system represented by the following poles.

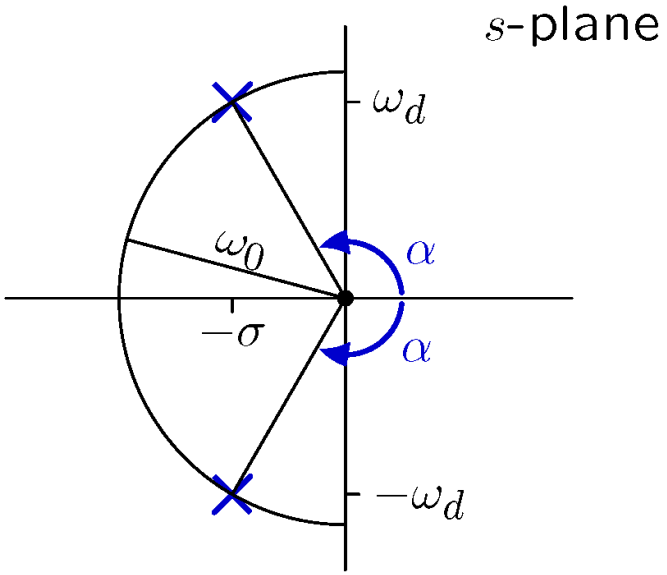


Find the frequency ω at which the phase of the response $y(t)$ is $-\pi/2$ if $x(t) = \cos \omega t$.

0. $0 < \omega < \omega_d$
1. $\omega = \omega_d$
2. $\omega_d < \omega < \omega_0$
3. $\omega = \omega_0$
4. $\omega > \omega_0$
5. none

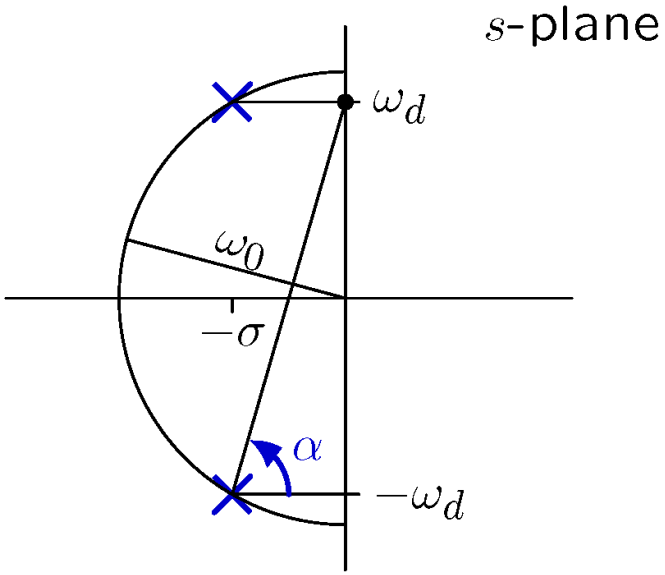
Check Yourself

The phase is 0 when $\omega = 0$.



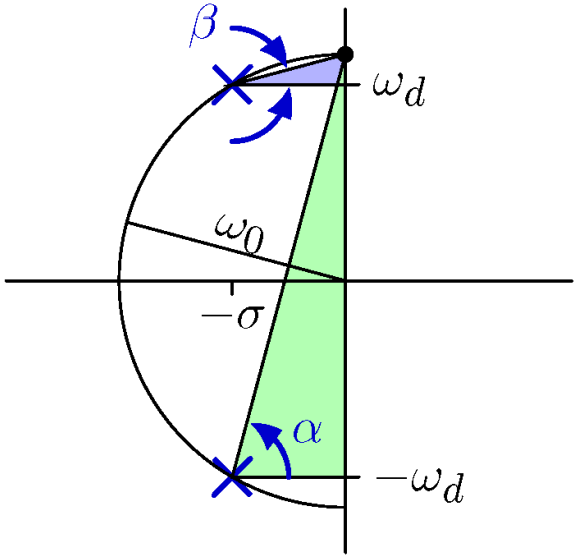
Check Yourself

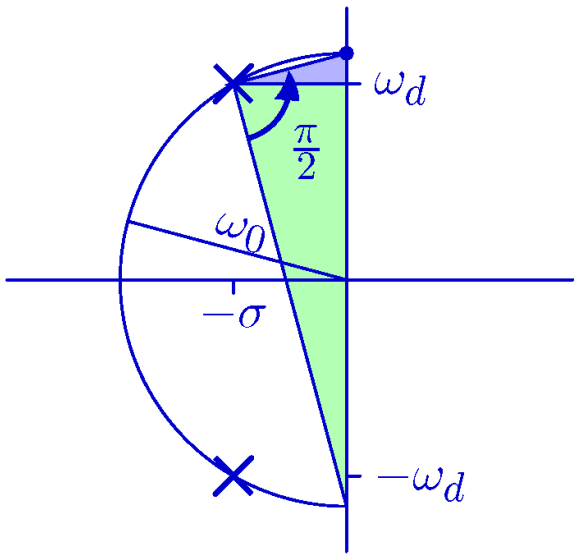
The phase is less than $\pi/2$ when $\omega = \omega_d$.



Check Yourself

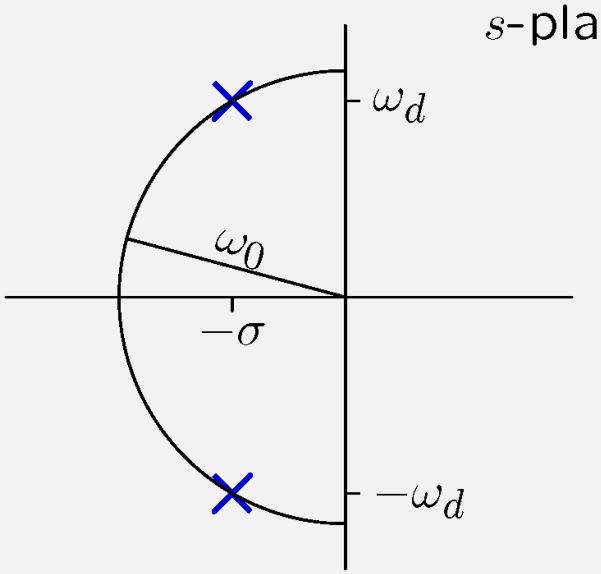
The phase at $\omega = \omega_0$ is $-\pi/2$.





Check Yourself

Consider the system represented by the following poles.



Find the frequency ω at which the phase of the response $y(t)$ is $-\pi/2$ if $x(t) = \cos \omega t$. 3

0. $0 < \omega < \omega_d$
1. $\omega = \omega_d$
2. $\omega_d < \omega < \omega_0$
3. $\omega = \omega_0$
4. $\omega > \omega_0$
5. none

Review: Frequency Response Summary

LTI systems can be characterized by responses to eternal sinusoids.

Many systems are naturally described by their frequency response.
– audio systems
– mass, spring, dashpot system

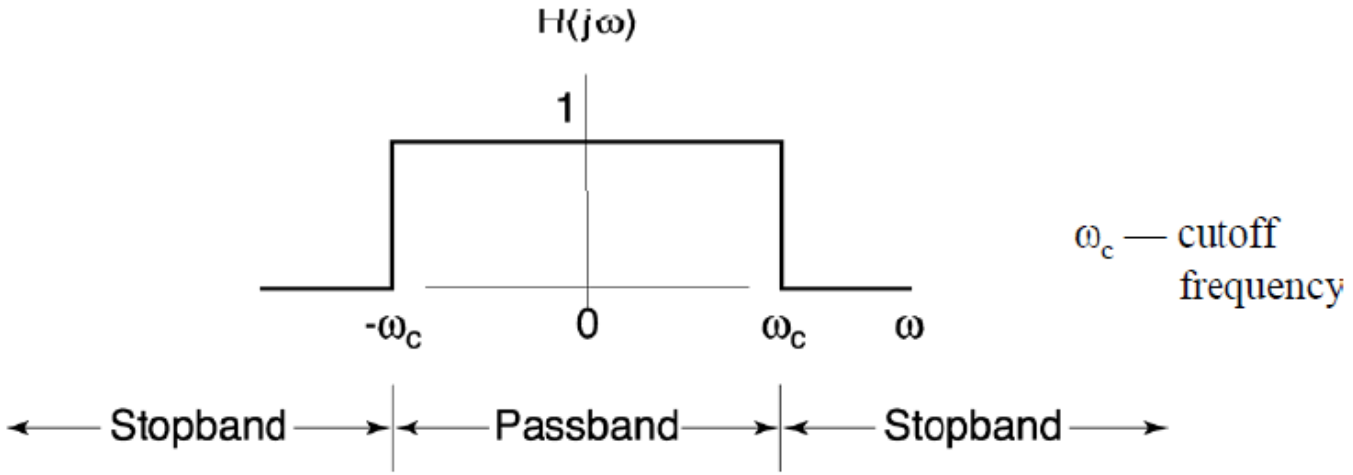
Frequency response is easy to calculate from the system function.

Frequency response lives on the $j\omega$ axis of the Laplace transform.

Idealized Filters

Lowpass filter

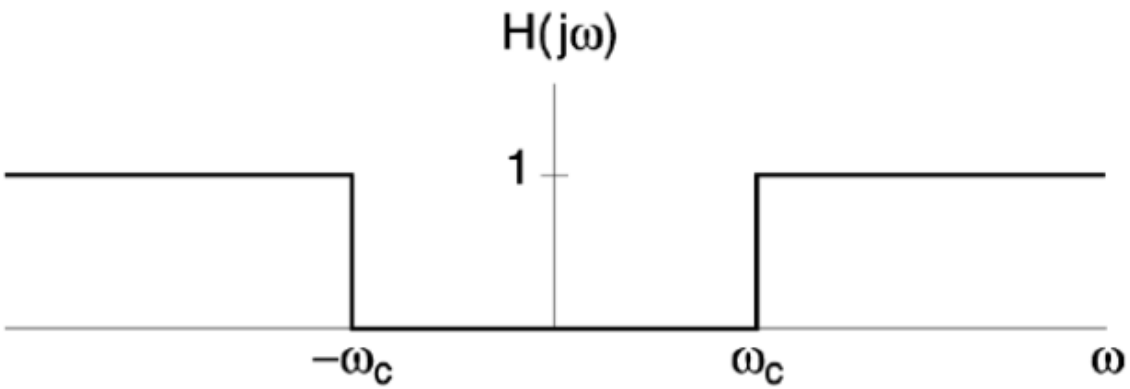
CT



Filtering

Highpass

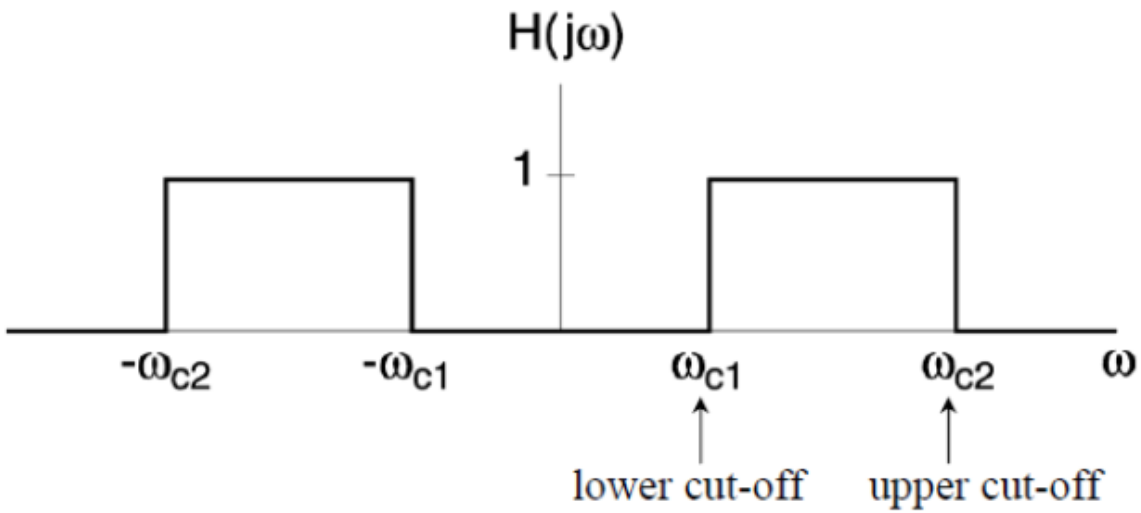
CT



Filtering

Bandpass

CT



Poles and Zeros

Thinking about systems as collections of poles and zeros is an important design concept.

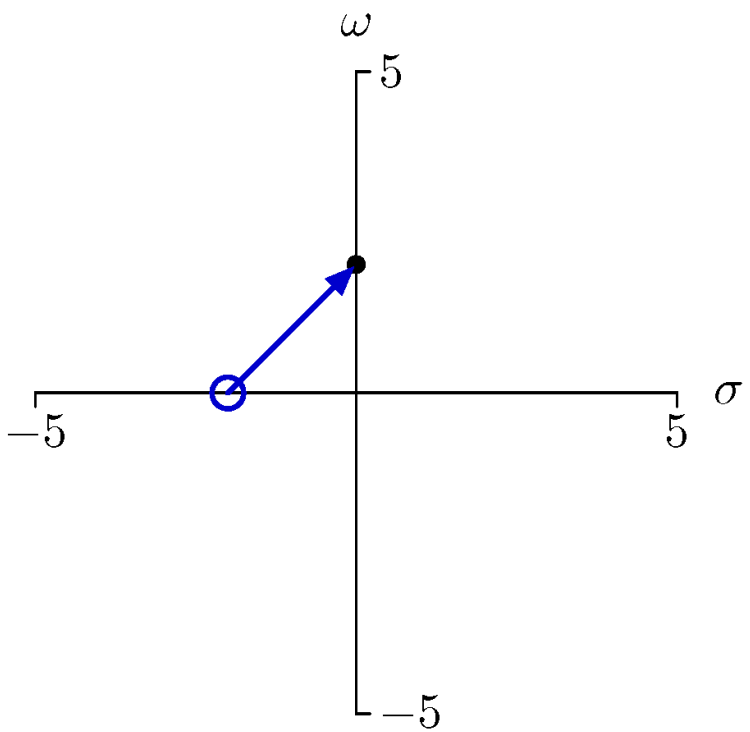
- simple: just a few numbers characterize entire system
- powerful: complete information about frequency response

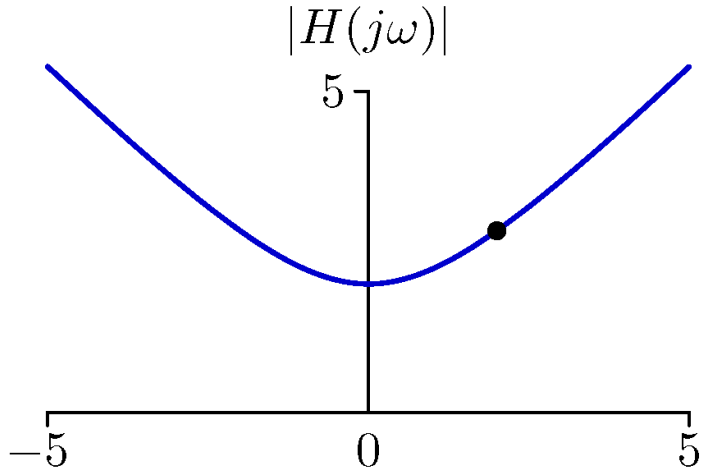
Today: poles, zeros, frequency responses, and Bode plots.

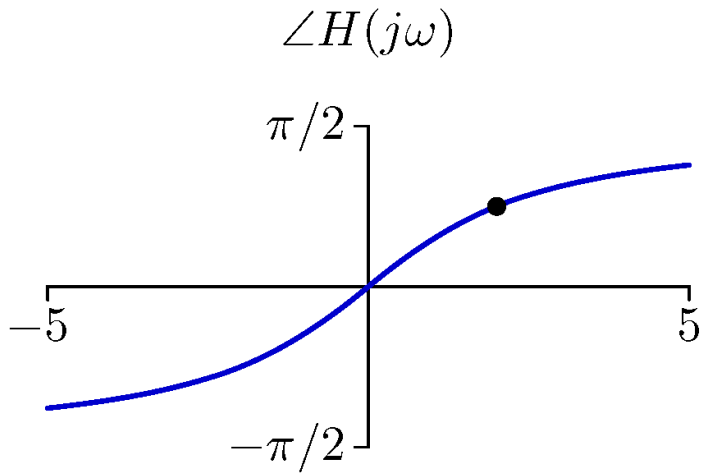
Asymptotic Behavior: Isolated Zero

The magnitude response is simple at low and high frequencies.

$$H(j\omega) = j\omega - z_1$$



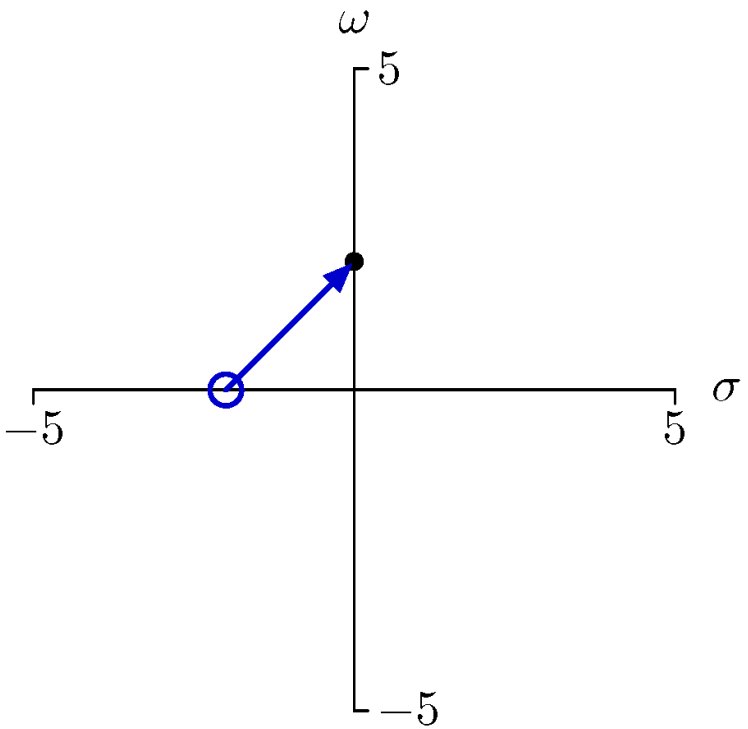


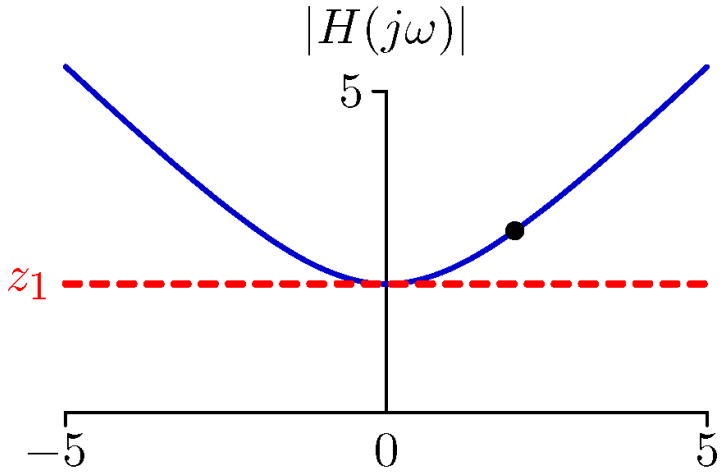


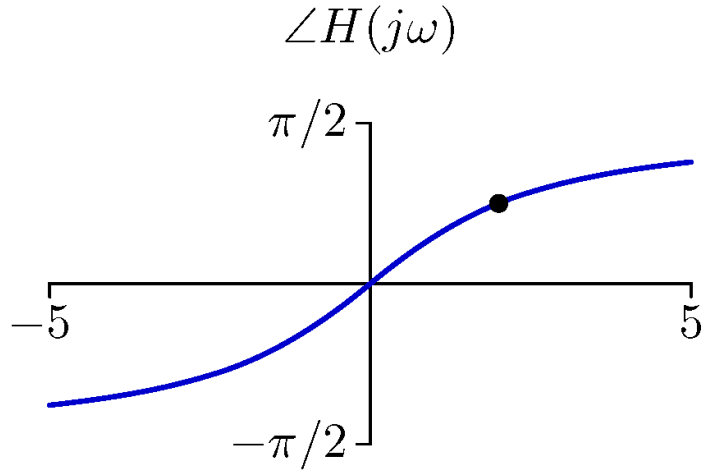
Asymptotic Behavior: Isolated Zero

The magnitude response is simple at low and high frequencies.

$$H(j\omega) = j\omega - z_1$$



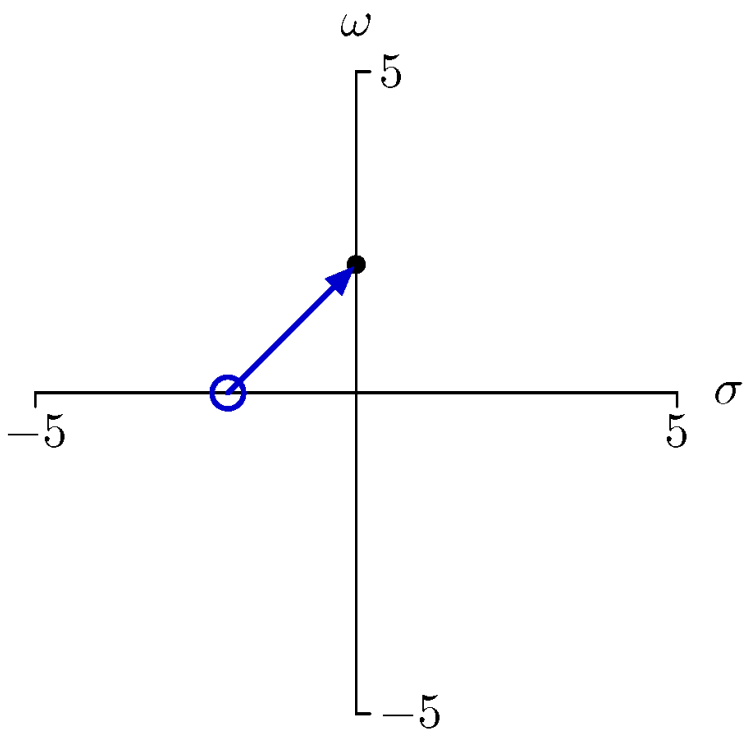


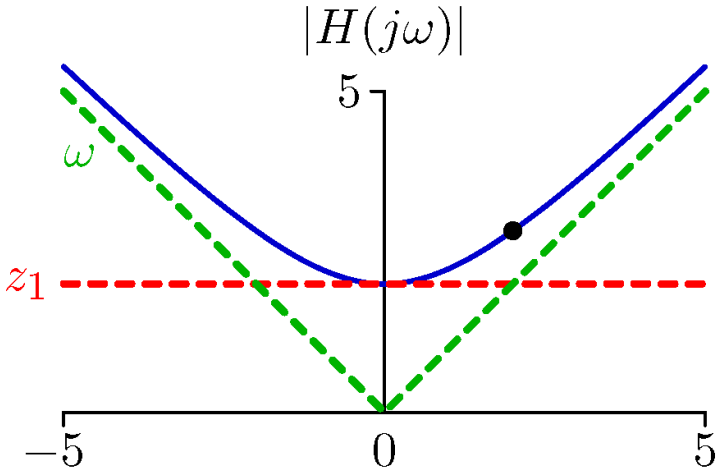


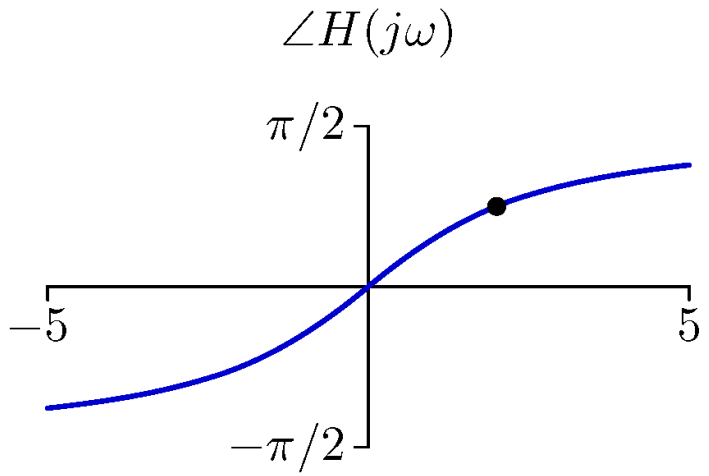
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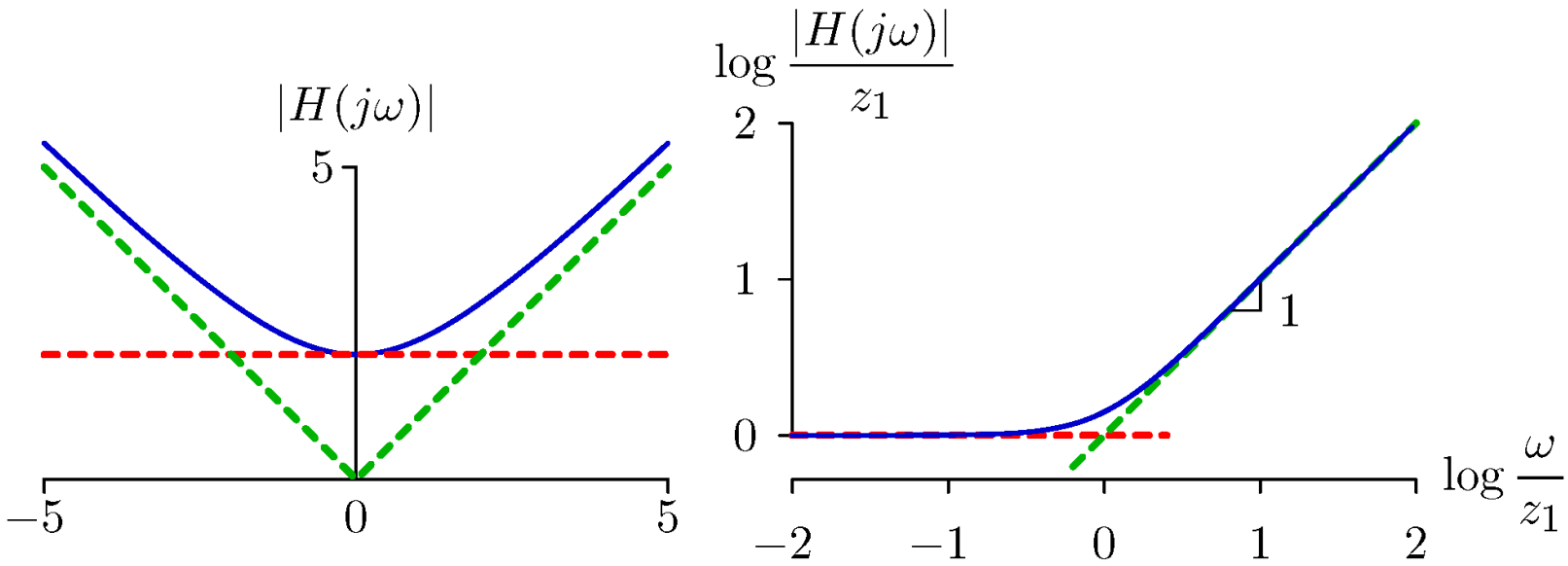




Asymptotic Behavior: Isolated Zero

Two asymptotes provide a good approxmation on log-log axes.

$$H(s) = s - z_1$$



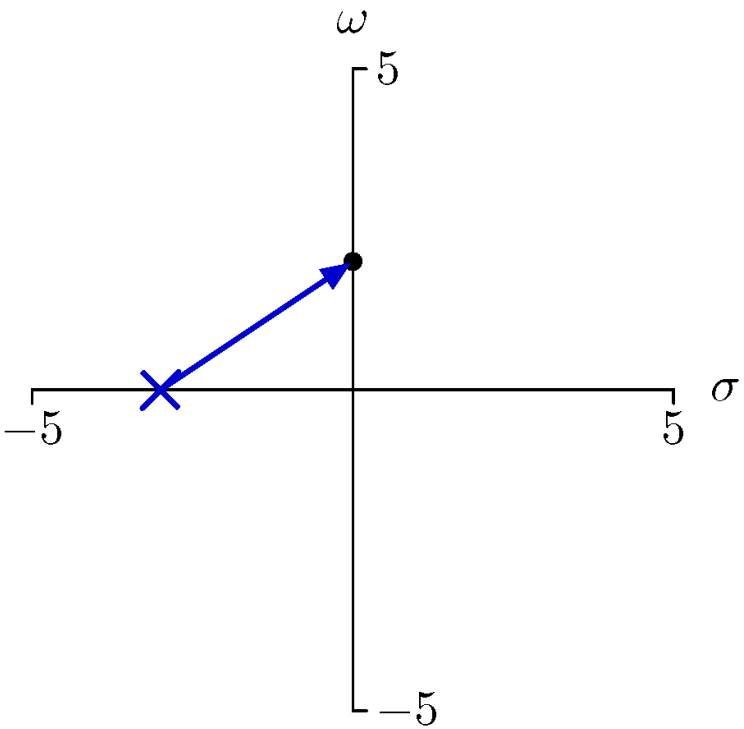
$$\lim_{\omega \to 0} |H(j\omega)| = z_1$$

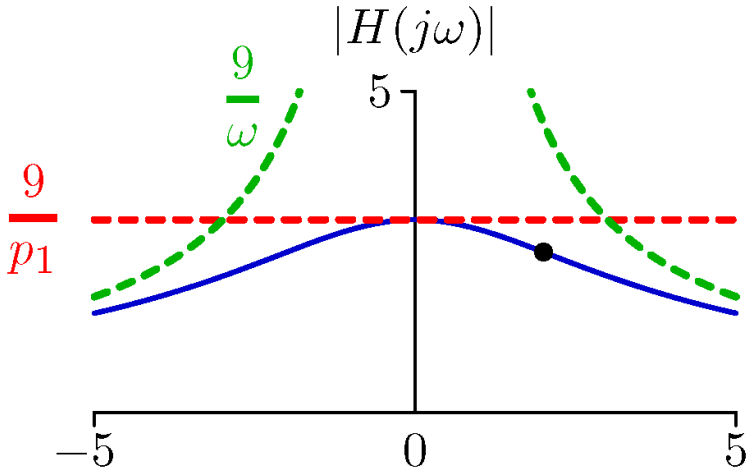
$$\lim_{\omega \to \infty} |H(j\omega)| = \omega$$

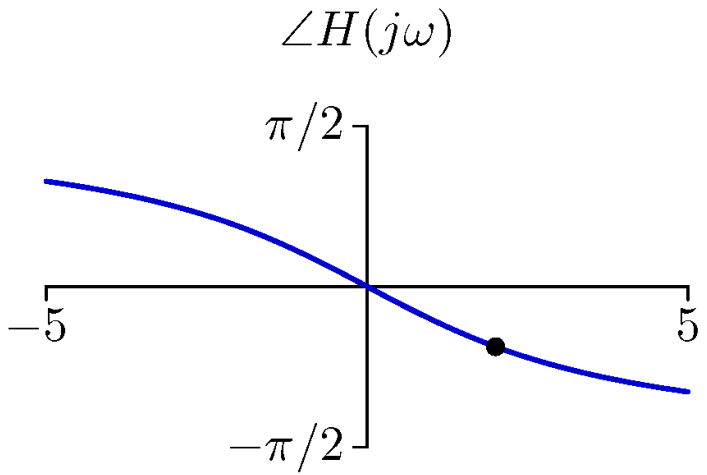
Asymptotic Behavior: Isolated Pole

The magnitude response is simple at low and high frequencies.

$$H(s) = \frac{9}{s - p_1}$$



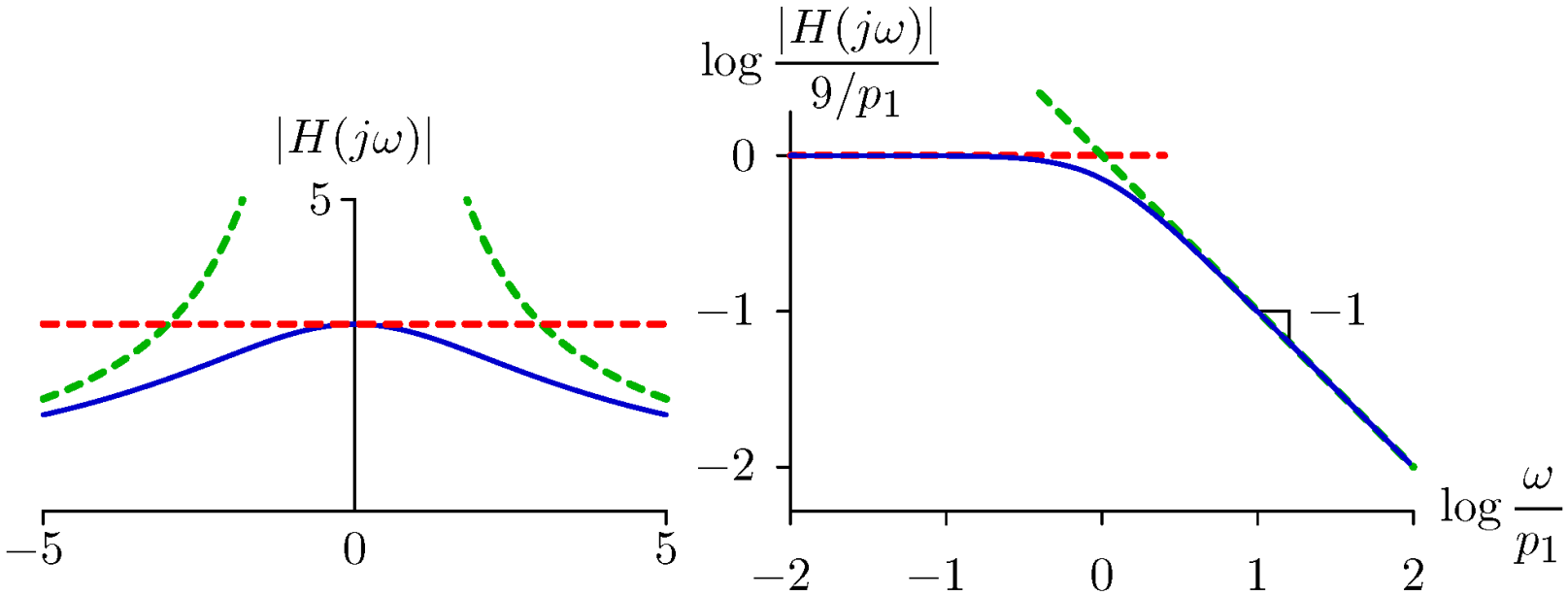




Asymptotic Behavior: Isolated Pole

Two asymptotes provide a good approxmation on log-log axes.

$$H(s) = \frac{9}{s - p_1}$$



$$\lim_{\omega \to 0} |H(j\omega)| = \frac{9}{p_1}$$

$$\lim |H(j\omega)| = \frac{9}{}$$

Check Yourself

Compare log-log plots of the frequency-response magnitudes of the following system functions:

$$H_1(s) = \frac{1}{s+1} \qquad \text{and} \qquad H_2(s) = \frac{1}{s+10}$$

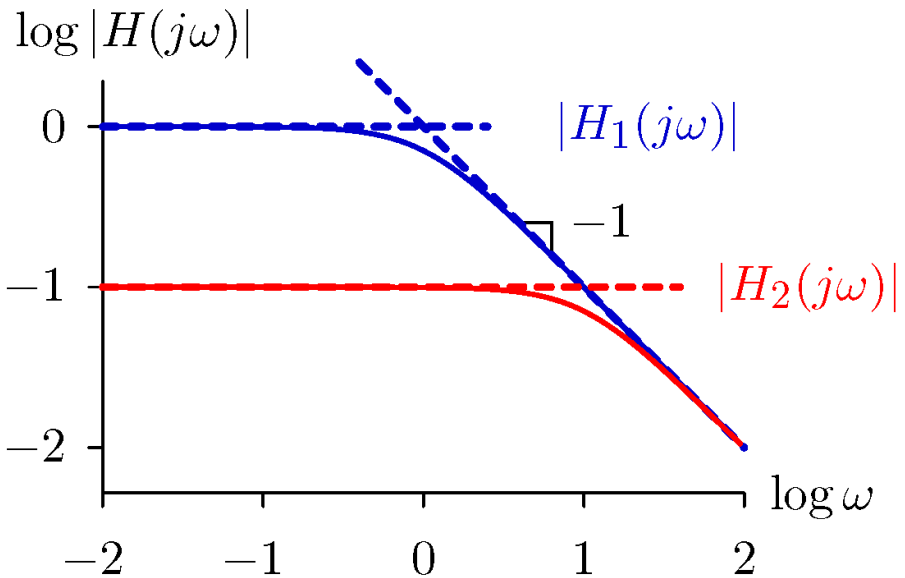
The former can be transformed into the latter by

1. shifting horizontally
2. shifting and scaling horizontally
3. shifting both horizontally and vertically
4. shifting and scaling both horizontally and vertically
5. none of the above

Check Yourself

Compare log-log plots of the frequency-response magnitudes of the following system functions:

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Check Yourself

Compare log-log plots of the frequency-response magnitudes of the following system functions:

$$H_1(s) = \frac{1}{s+1} \qquad \text{and} \qquad H_2(s) = \frac{1}{s+10}$$

The former can be transformed into the latter by 3

1. shifting horizontally
2. shifting and scaling horizontally
3. shifting both horizontally and vertically
4. shifting and scaling both horizontally and vertically
5. none of the above

no scaling in either vertical or horizontal directions!

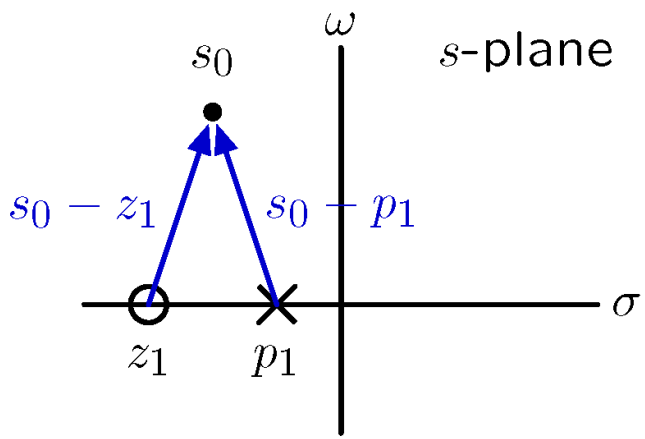
Asymptotic Behavior of More Complicated Systems

Constructing $H(s_0)$.

$$H(s_0) = K \frac{\prod_{q=1}^{Q}(s_0 - z_q)}{\prod_{p=1}^{P}(s_0 - p_p)}$$

$\leftarrow$ product of vectors for zeros

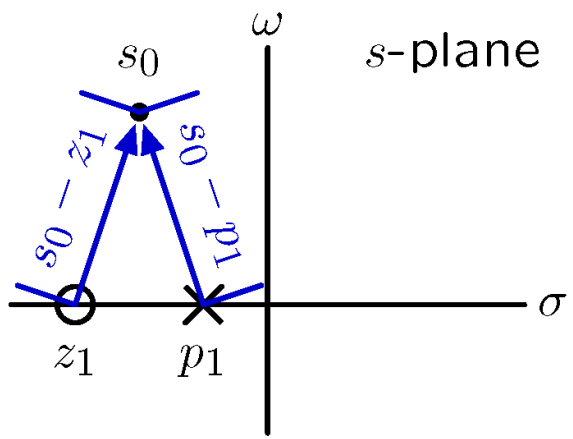
$\leftarrow$ product of vectors for poles



Asymptotic Behavior of More Complicated Systems

The magnitude of a product is the product of the magnitudes.

$$|H(s_0)| = \left| K \frac{\prod_{q=1}^{Q} (s_0 - z_q)}{\prod_{p=1}^{P} (s_0 - p_p)} \right| = |K| \frac{\prod_{q=1}^{Q} |s_0 - z_q|}{\prod_{p=1}^{P} |s_0 - p_p|}$$



Bode Plot

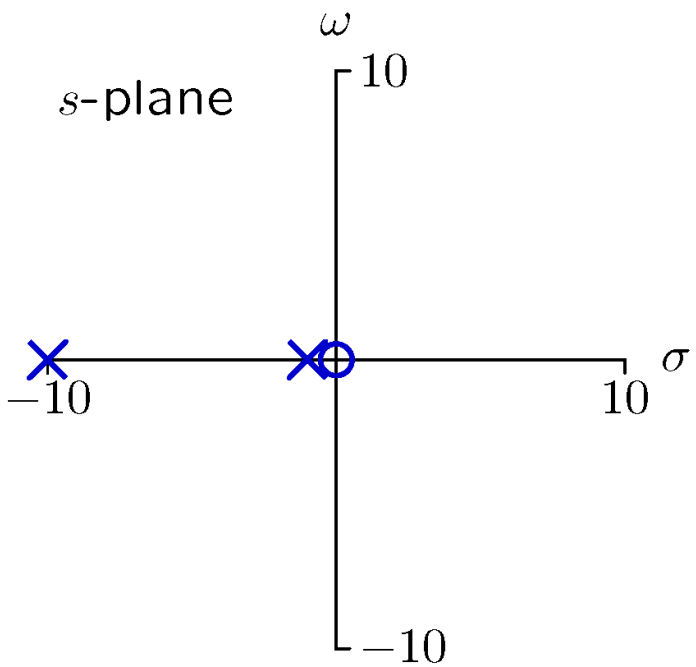
The log of the magnitude is a sum of logs.

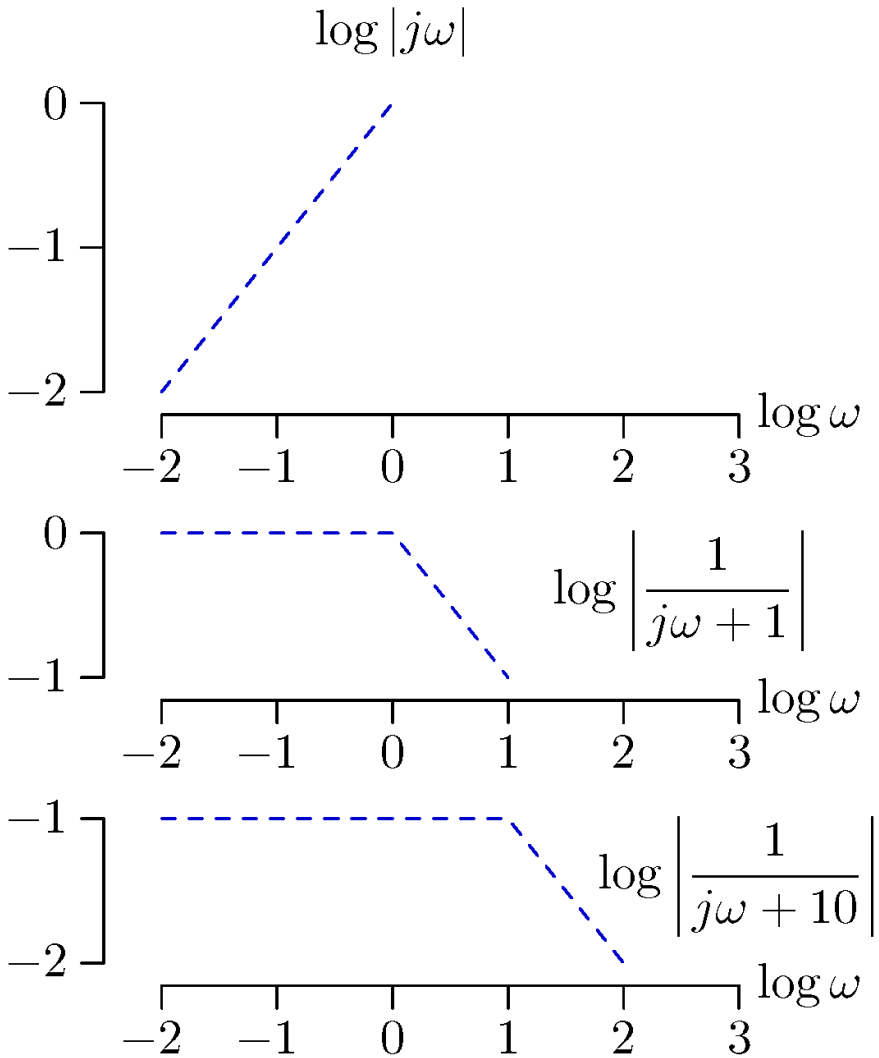
$$|H(s_0)| = \left| K \frac{\prod_{q=1}^{Q} (s_0 - z_q)}{\prod_{p=1}^{P} (s_0 - p_p)} \right| = |K| \frac{\prod_{q=1}^{Q} |s_0 - z_q|}{\prod_{p=1}^{P} |s_0 - p_p|}$$

$$\log |H(j\omega)| = \log |K| + \sum_{q=1}^{Q} \log |j\omega - z_q| - \sum_{p=1}^{P} \log |j\omega - p_p|$$

Bode Plot: Adding Instead of Multiplying

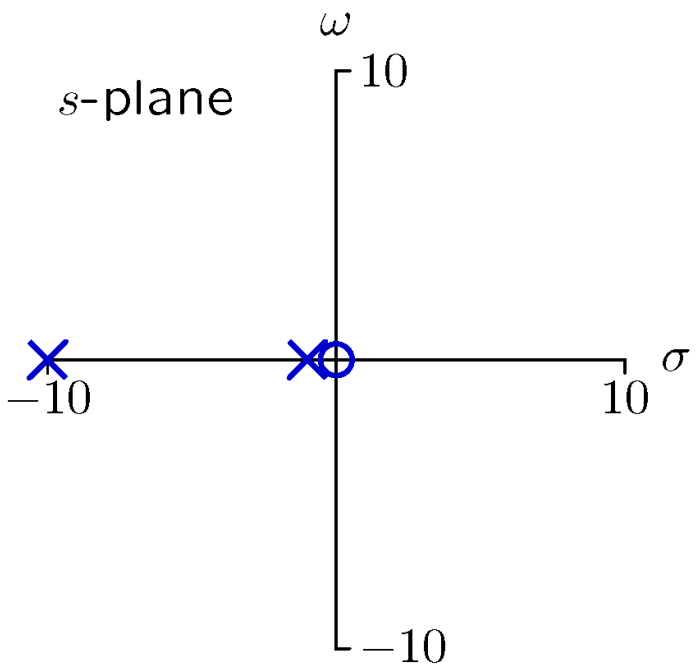
$$H(s) = \frac{s}{(s+1)(s+10)}$$

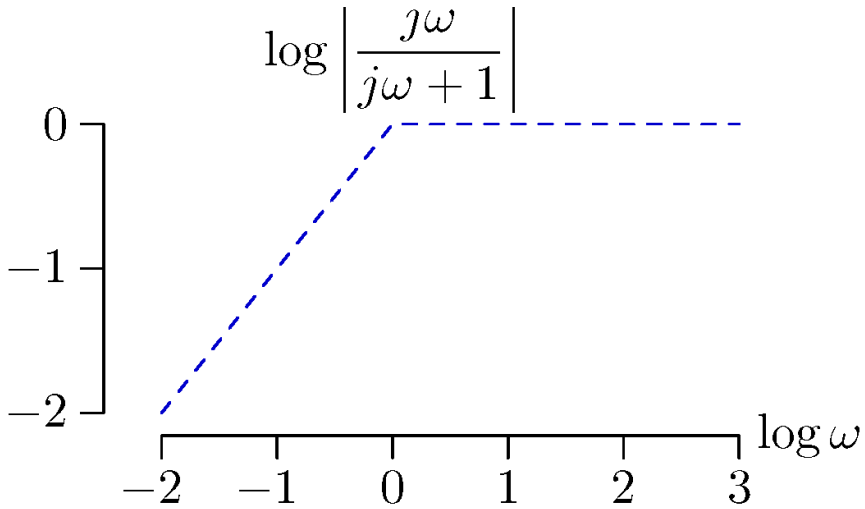


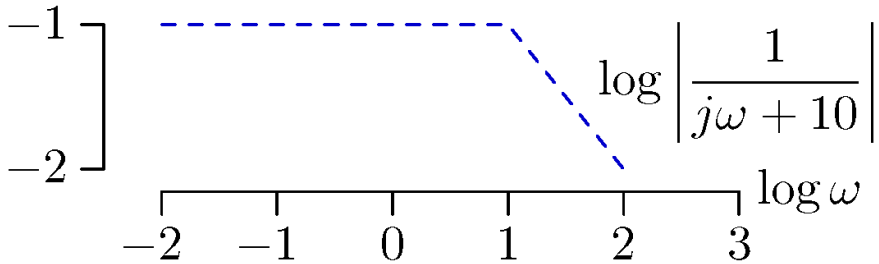


Bode Plot: Adding Instead of Multiplying

$$H(s) = \frac{s}{(s+1)(s+10)}$$

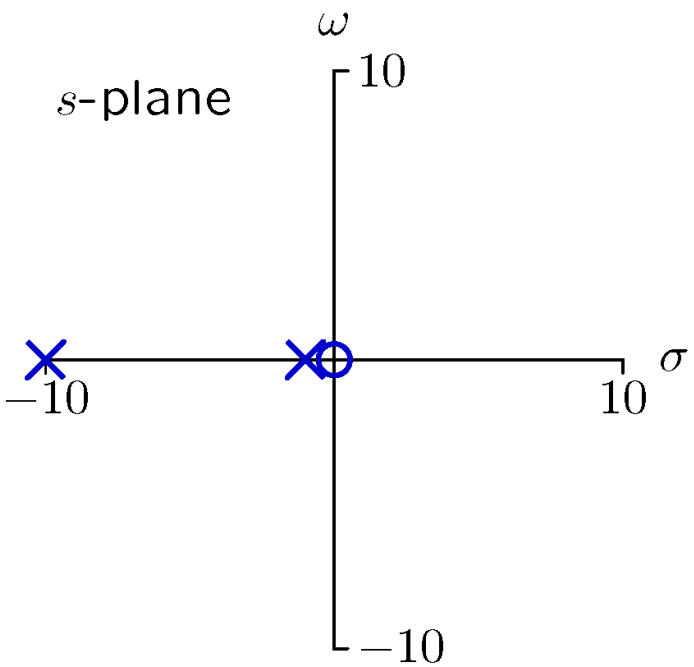


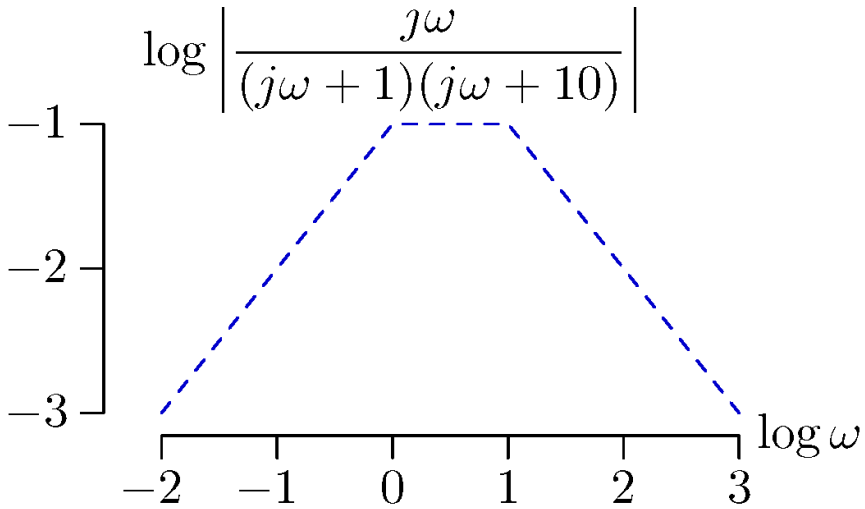




Bode Plot: Adding Instead of Multiplying

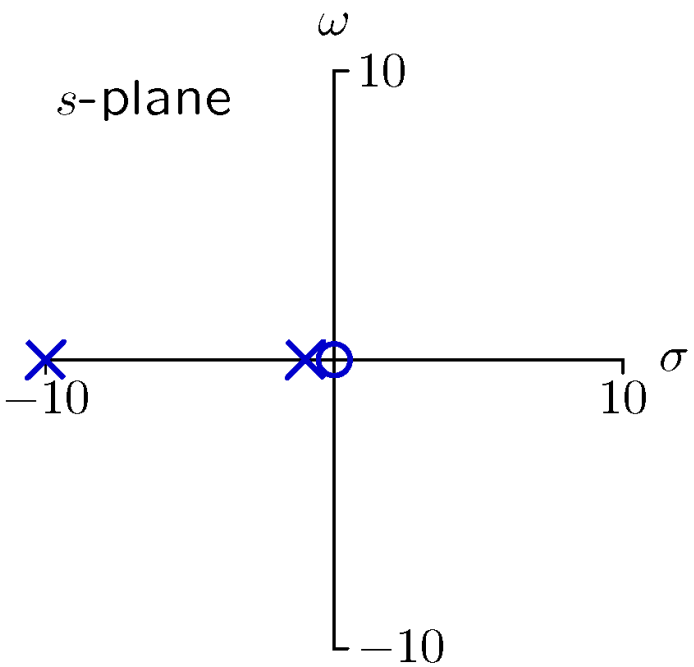
$$H(s) = \frac{s}{(s+1)(s+10)}$$

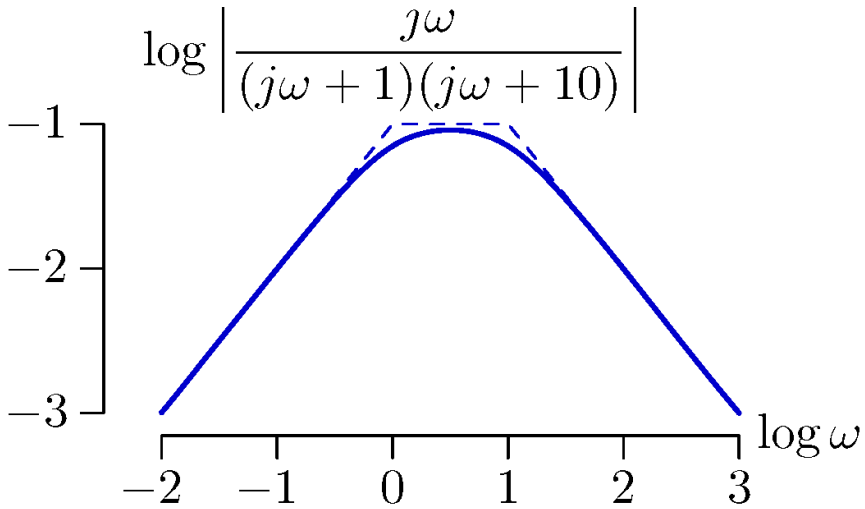




Bode Plot: Adding Instead of Multiplying

$$H(s) = \frac{s}{(s+1)(s+10)}$$

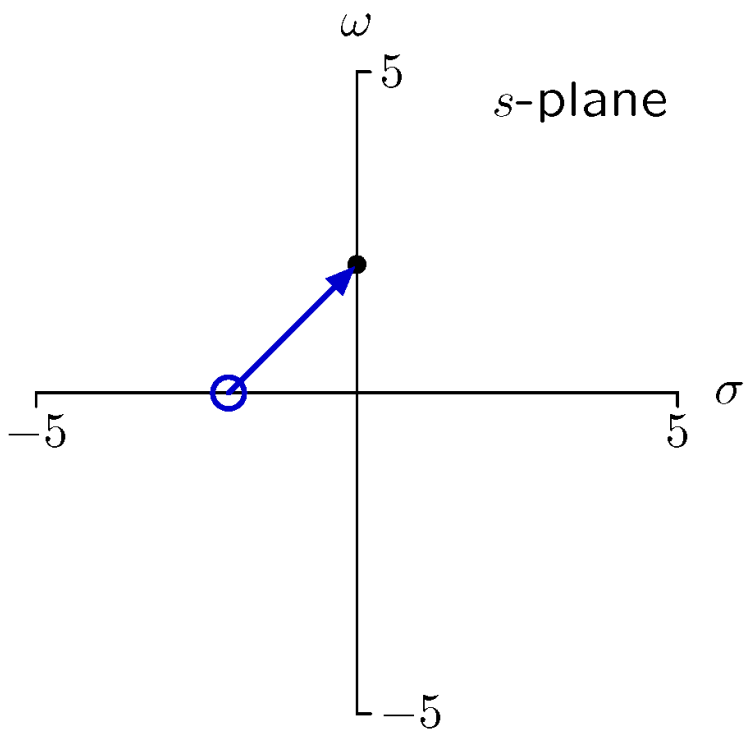


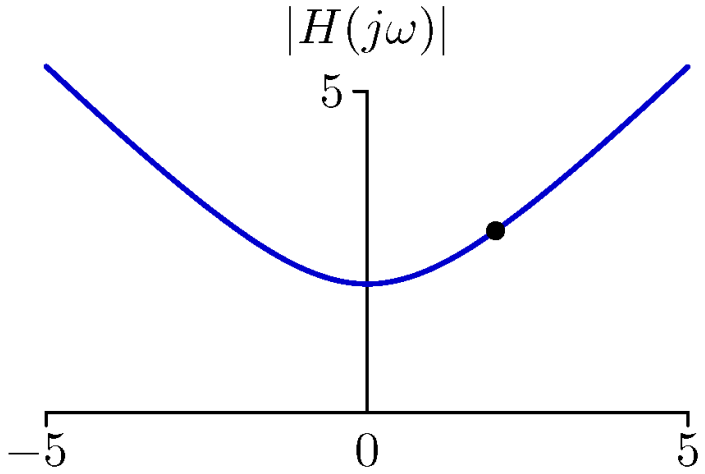


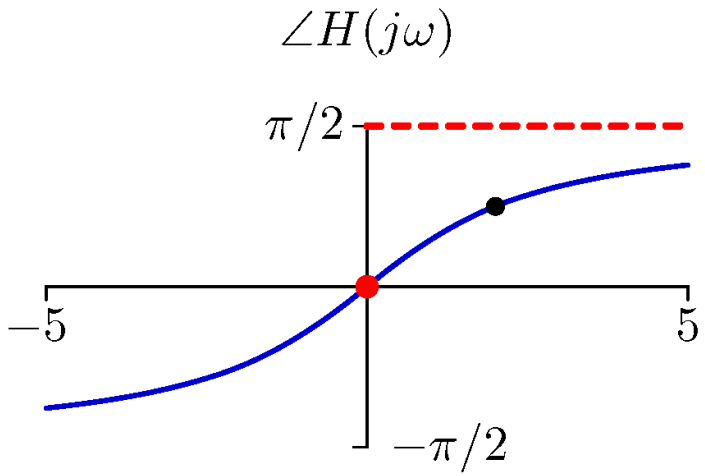
Asymptotic Behavior: Isolated Zero

The angle response is simple at low and high frequencies.

$H(s) = s - z_1$



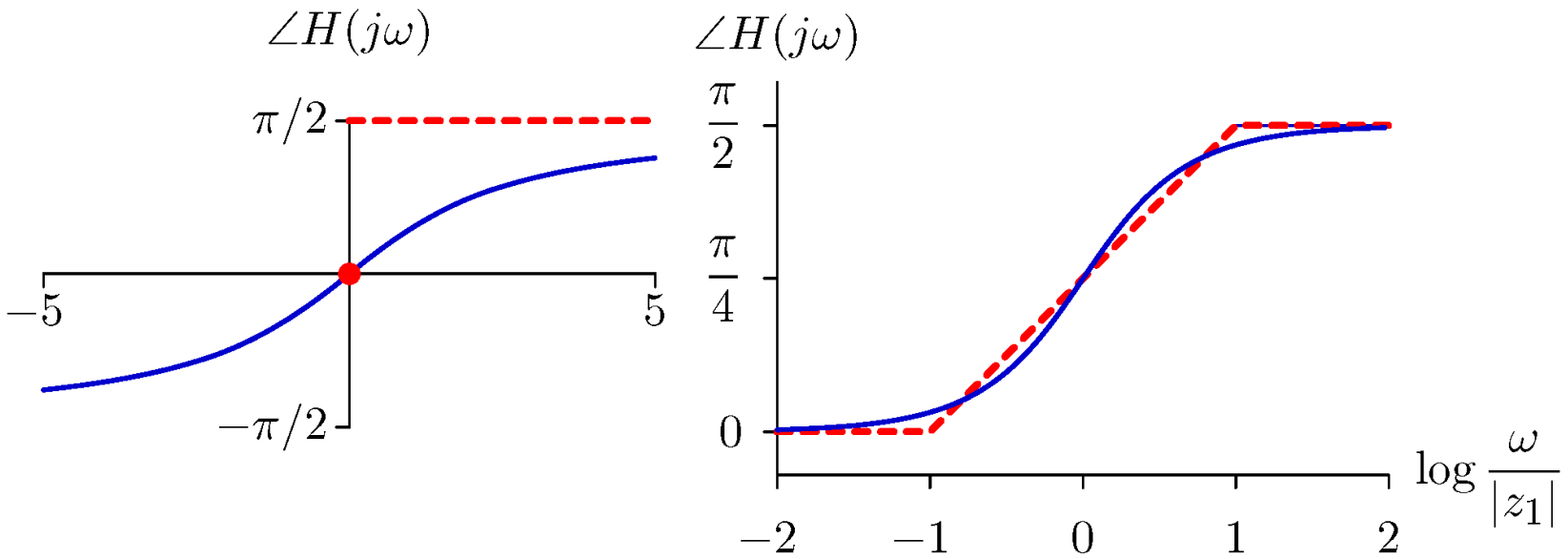




Asymptotic Behavior: Isolated Zero

Three straight lines provide a good approxmation versus log ω.

$$H(s) = s - z_1$$



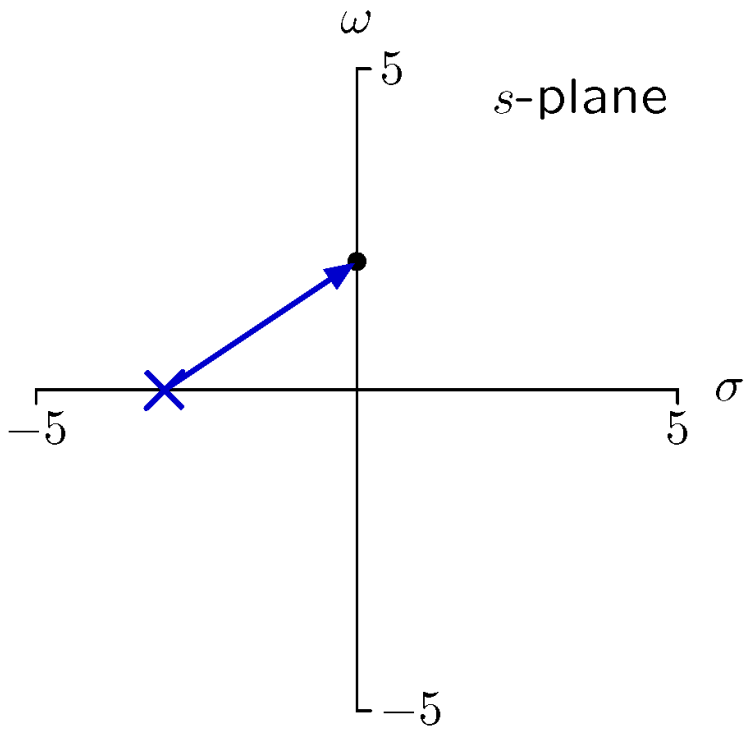
$$\lim_{\omega \to 0} \angle H(j\omega) = 0$$

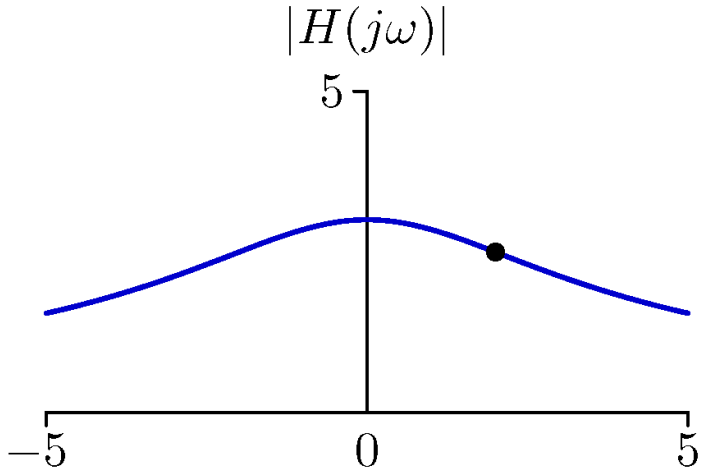
$$\lim_{\omega \to \infty} \angle H(j\omega) = \pi/2$$

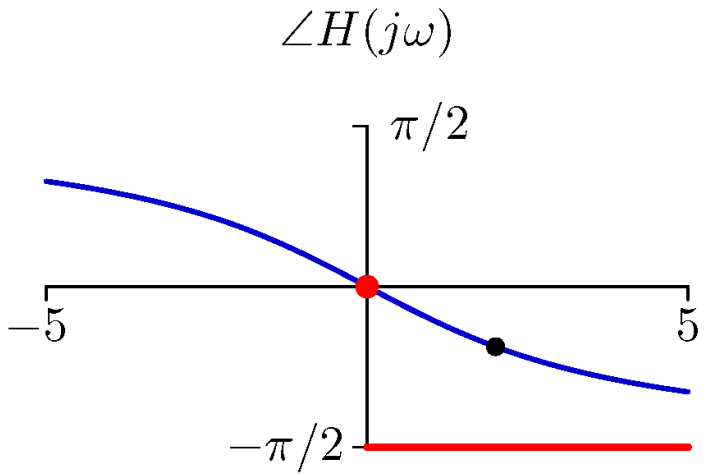
Asymptotic Behavior: Isolated Pole

The angle response is simple at low and high frequencies.

$$H(s) = \frac{9}{s - p_1}$$



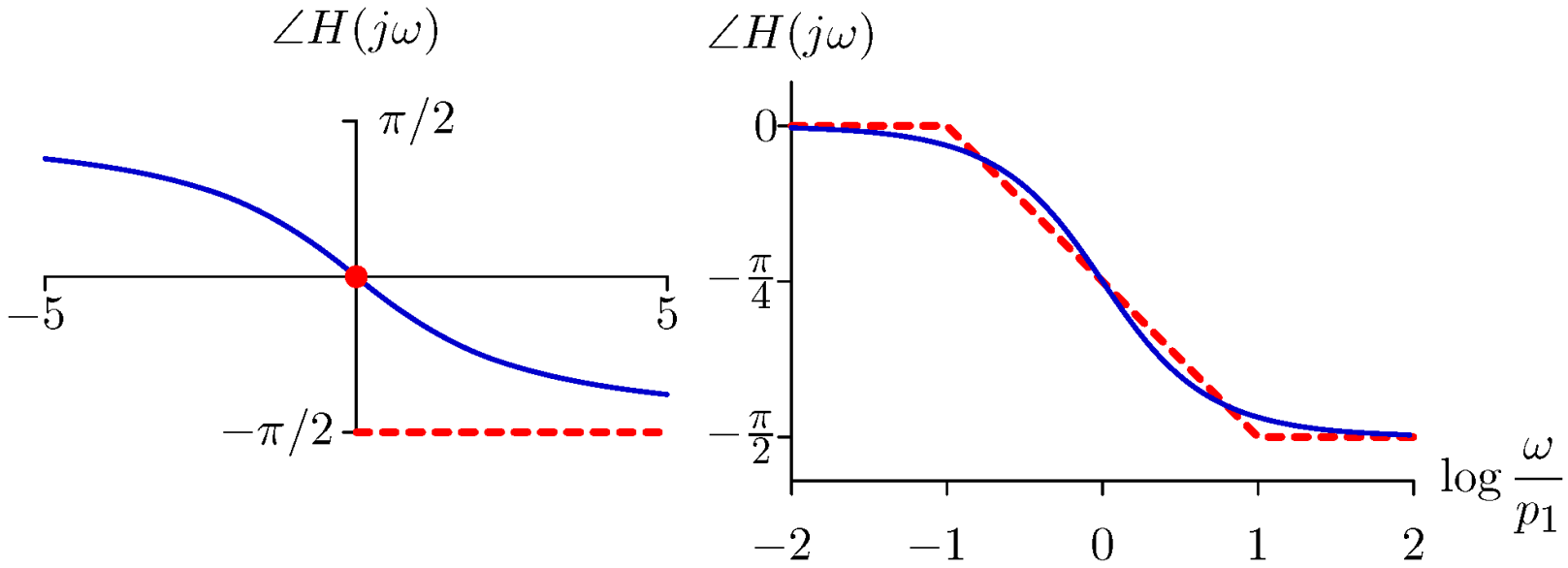




Asymptotic Behavior: Isolated Pole

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$$H(s) = \frac{9}{s - p_1}$$



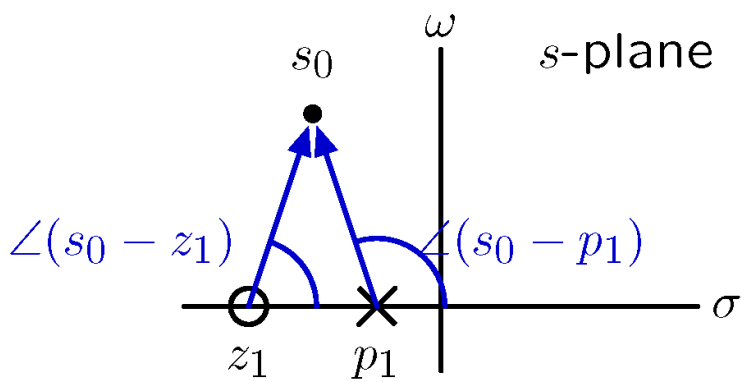
$$\lim_{\omega \to 0} \angle H(j\omega) = 0$$

$$\lim_{\omega \to \infty} \angle H(j\omega) = -\pi/2$$

Bode Plot

The angle of a product is the sum of the angles.

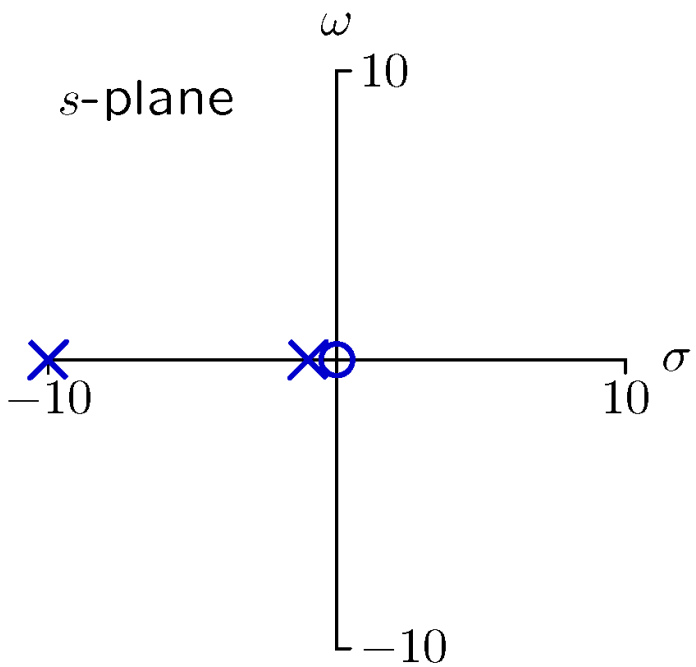
$$\angle H(s_0) = \angle \left(K \frac{\prod_{q=1}^{Q} (s_0 - z_q)}{\prod_{p=1}^{P} (s_0 - p_p)} \right) = \angle K + \sum_{q=1}^{Q} \angle (s_0 - z_q) - \sum_{p=1}^{P} \angle (s_0 - p_p)$$

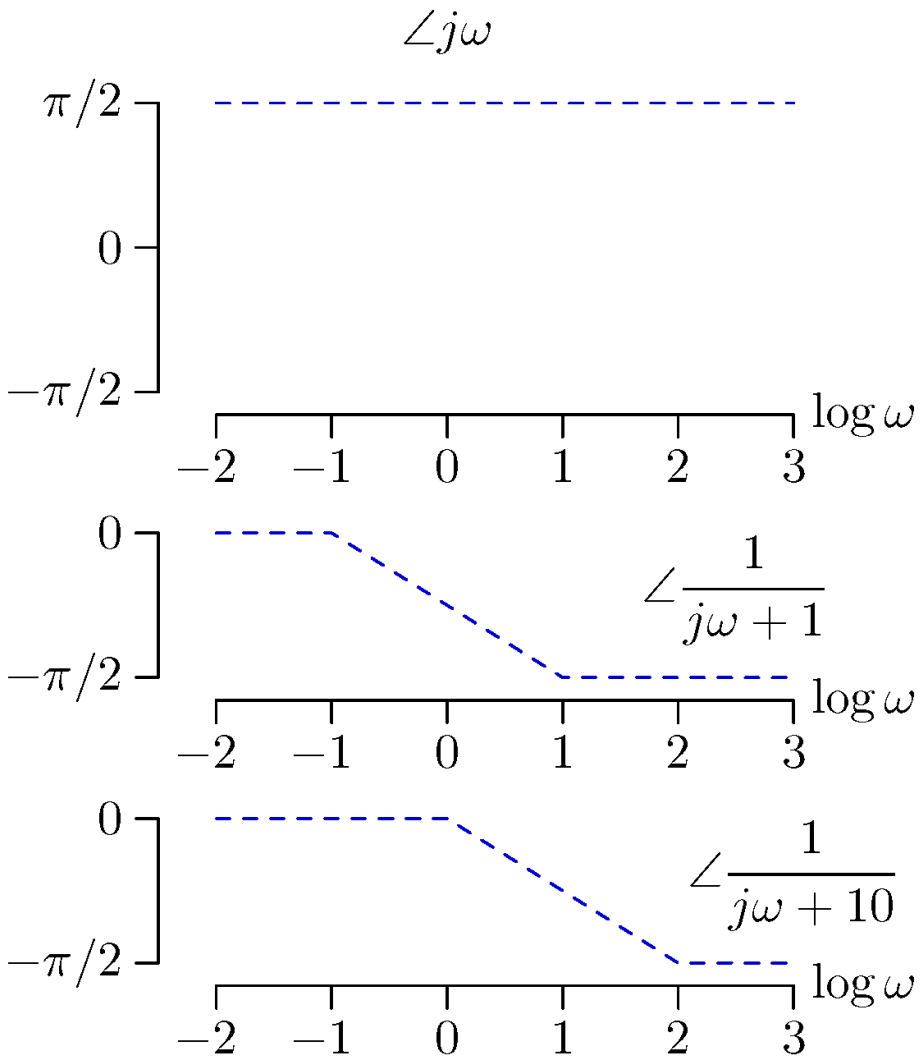


The angle of K can be 0 or π for systems described by linear differential equations with constant, real-valued coefficients.

Bode Plot

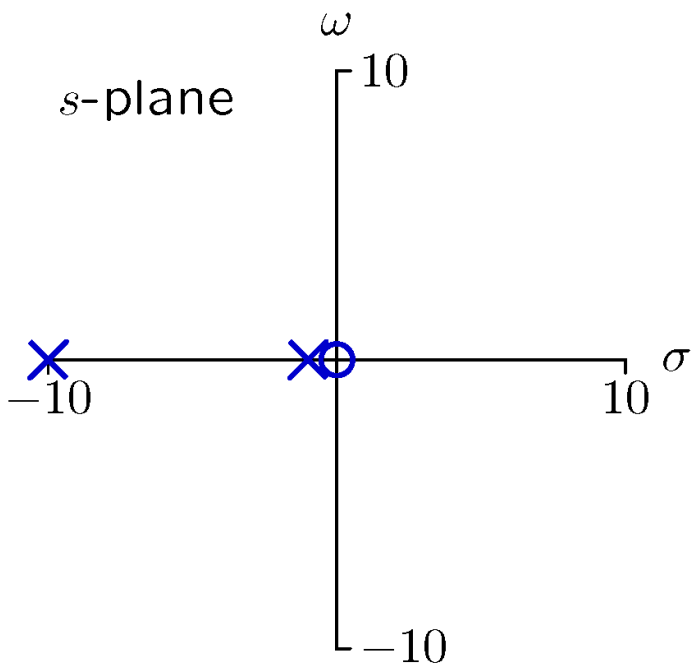
$$H(s) = \frac{s}{(s+1)(s+10)}$$

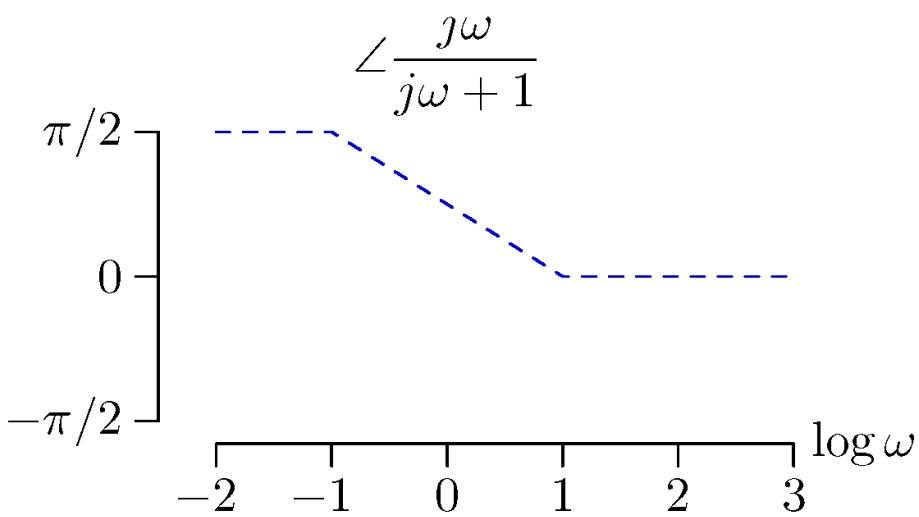


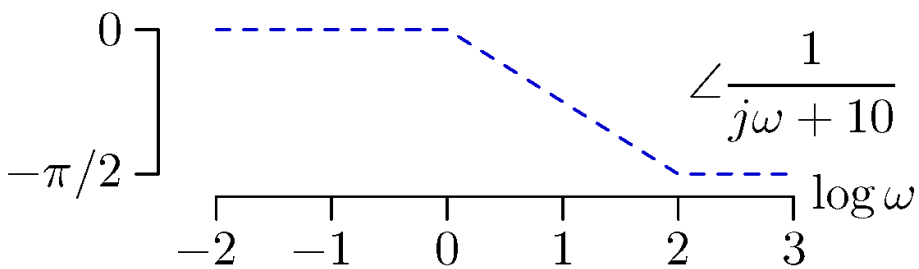


Bode Plot

$$H(s) = \frac{s}{(s+1)(s+10)}$$

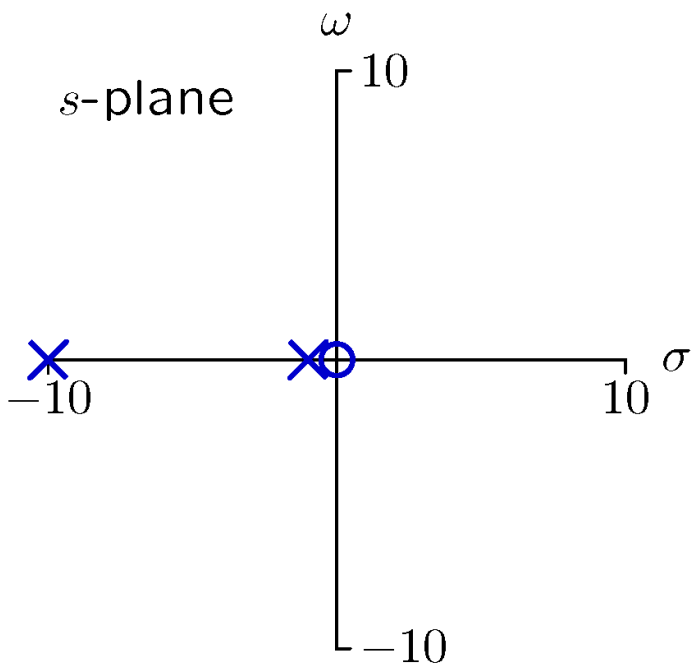


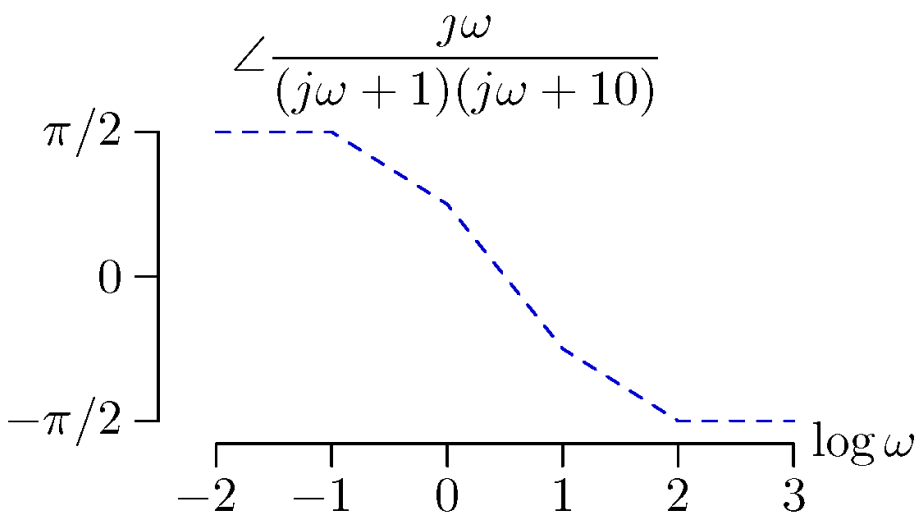




Bode Plot

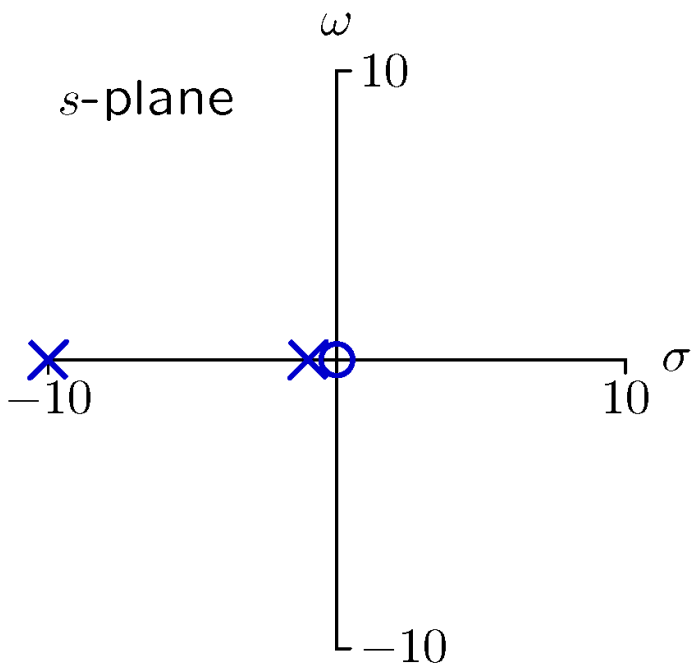
$$H(s) = \frac{s}{(s+1)(s+10)}$$

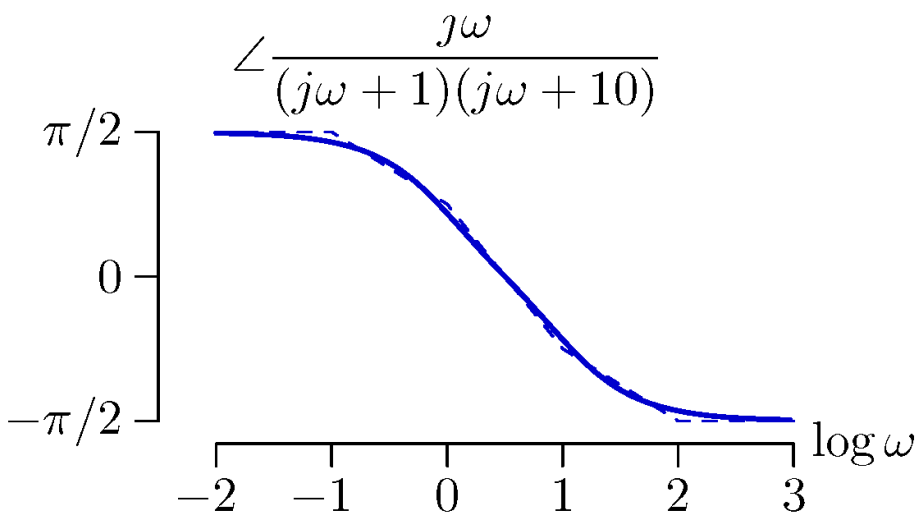




Bode Plot

$$H(s) = \frac{s}{(s+1)(s+10)}$$





From Frequency Response to Bode Plot

The magnitude of $H(j\omega)$ is a product of magnitudes.

$$|H(j\omega)| = |K| \frac{\prod_{q=1}^{Q} |j\omega - z_q|}{\prod_{p=1}^{P} |j\omega - p_p|}$$

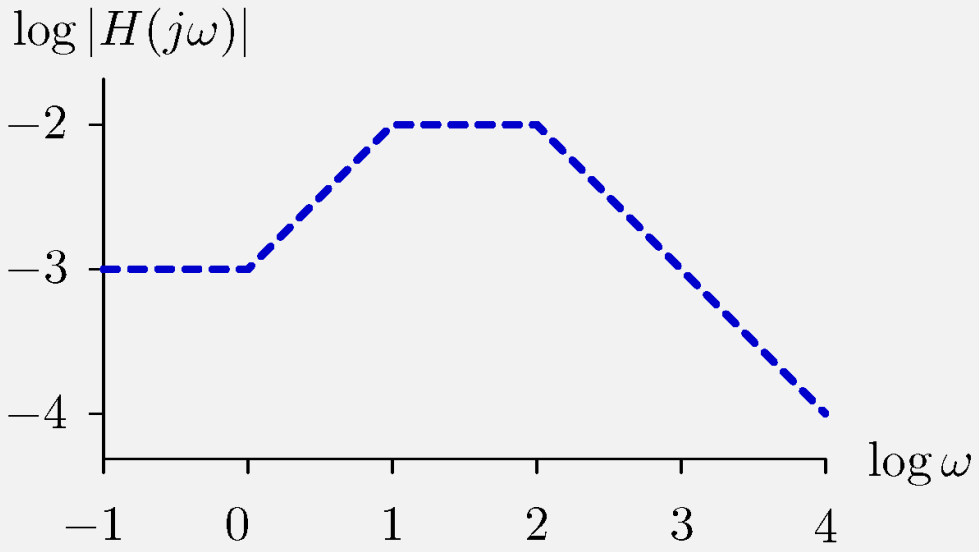
The log of the magnitude is a sum of logs.

$$\log|H(j\omega)| = \log|K| + \sum_{q=1}^{Q} \log\left|j\omega - z_q\right| - \sum_{p=1}^{P} \log\left|j\omega - p_p\right|$$

The angle of $H(j\omega)$ is a sum of angles.

$$\angle H(j\omega) = \angle K + \sum_{q=1}^{Q} \angle\left(j\omega - z_q\right) - \sum_{p=1}^{P} \angle\left(j\omega - p_p\right)$$

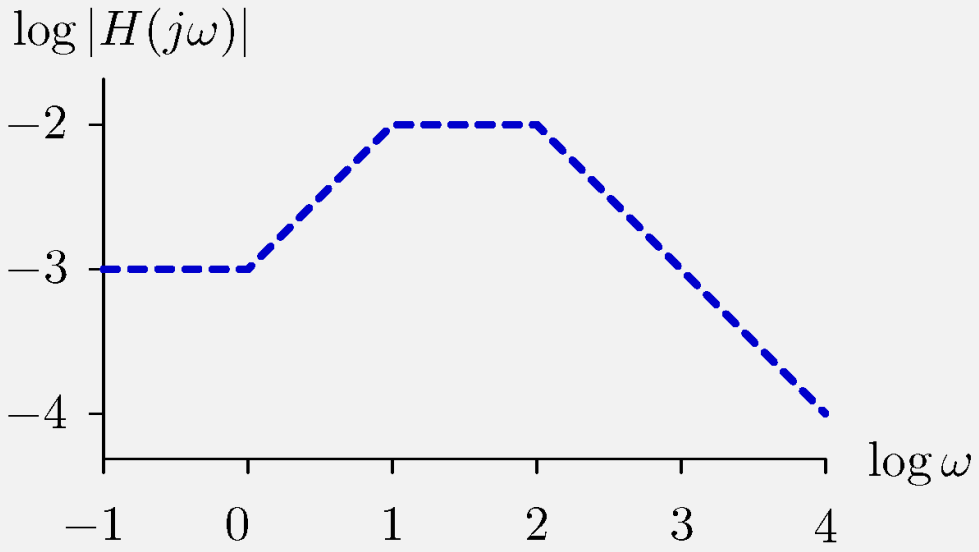
Check Yourself



Which corresponds to the Bode approximation above?

1. $\dfrac{1}{(s+1)(s+10)(s+100)}$
2. $\dfrac{s+1}{(s+10)(s+100)}$
3. $\dfrac{(s+10)(s+100)}{s+1}$
4. $\dfrac{s+100}{(s+1)(s+10)}$
5. none of the above

Check Yourself

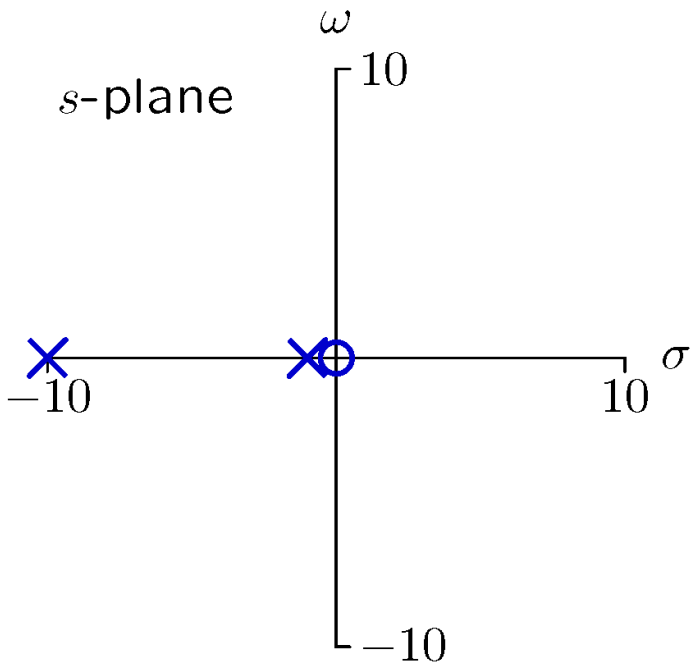


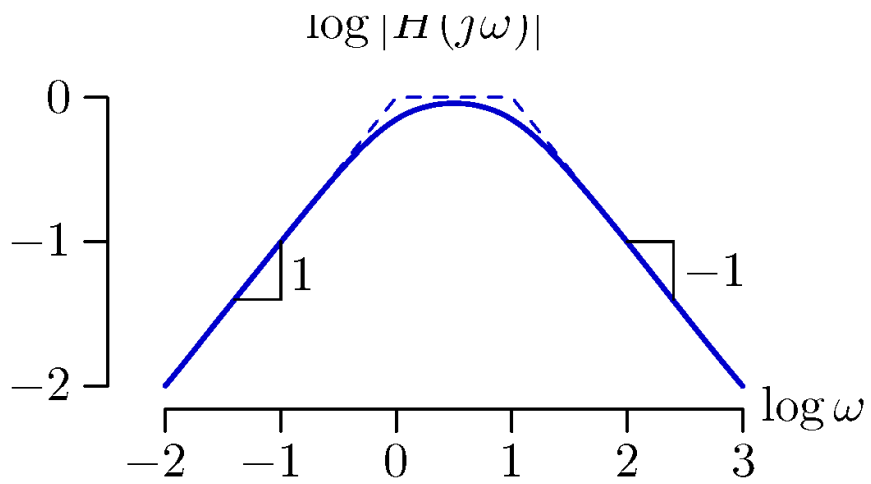
Which corresponds to the Bode approximation above? 2

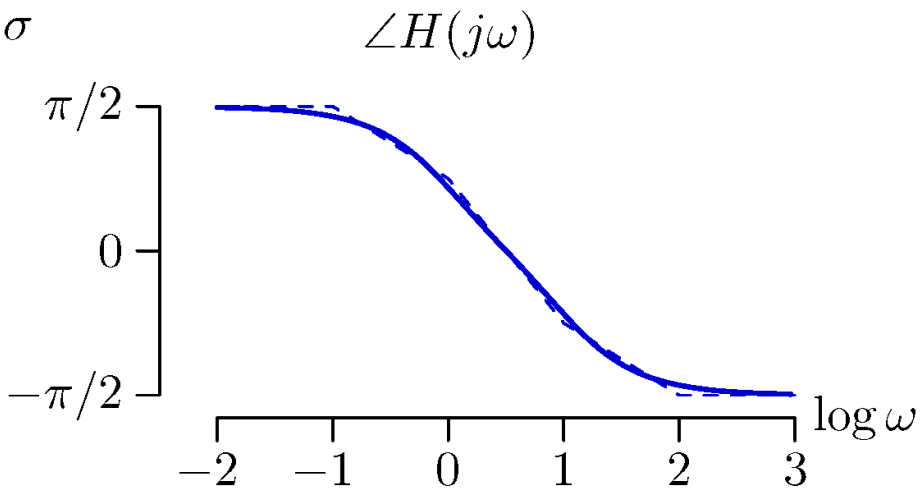
1. $\dfrac{1}{(s+1)(s+10)(s+100)}$
2. $\dfrac{s+1}{(s+10)(s+100)}$
3. $\dfrac{(s+10)(s+100)}{s+1}$
4. $\dfrac{s+100}{(s+1)(s+10)}$
5. none of the above

Bode Plot: dB

$$H(s) = \frac{10s}{(s+1)(s+10)}$$

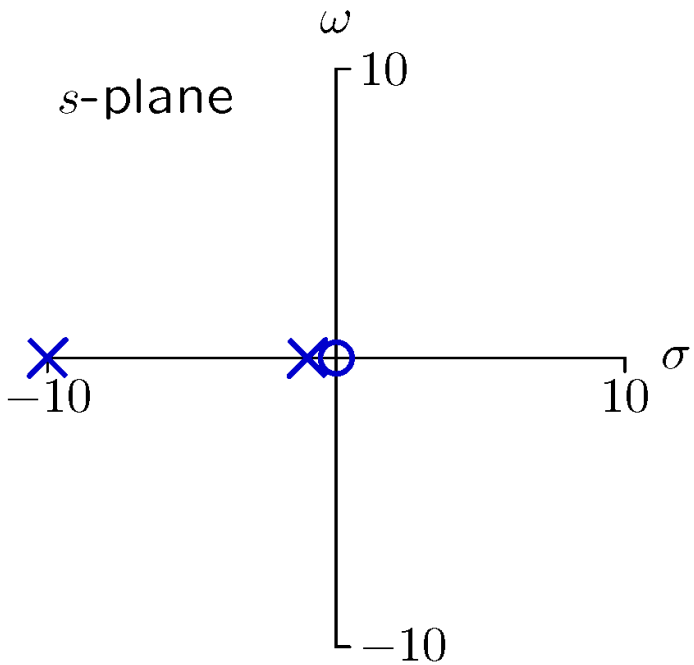


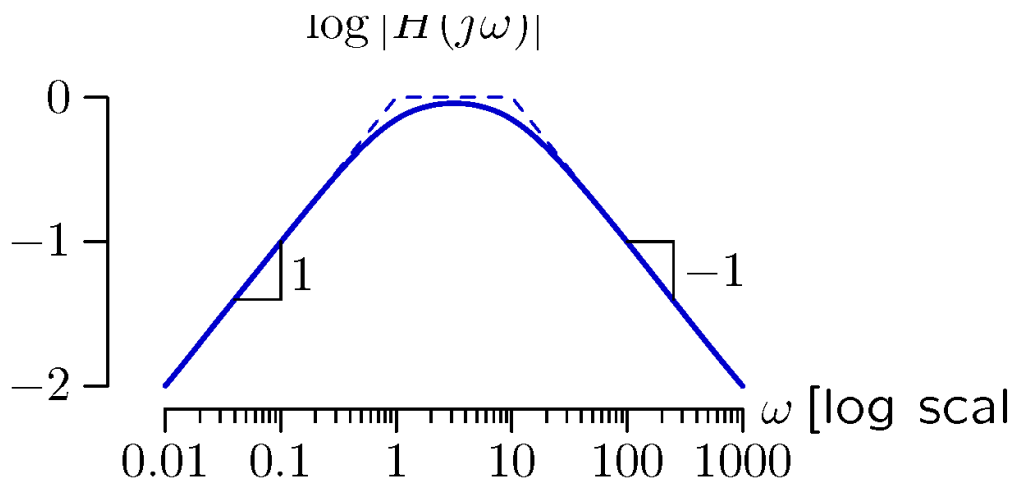


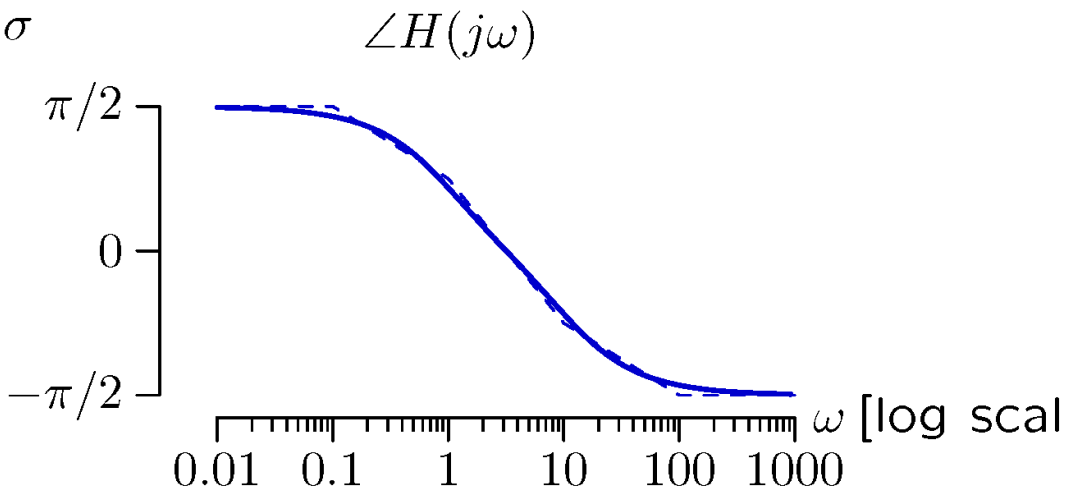


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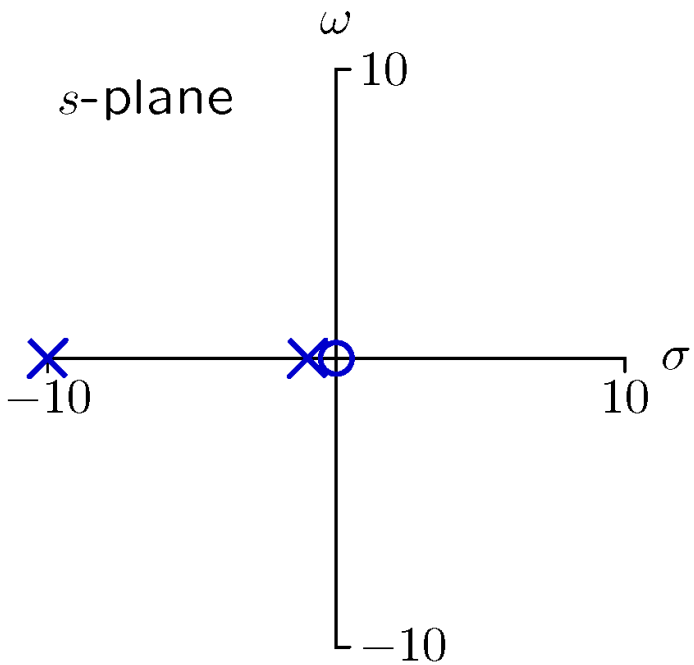


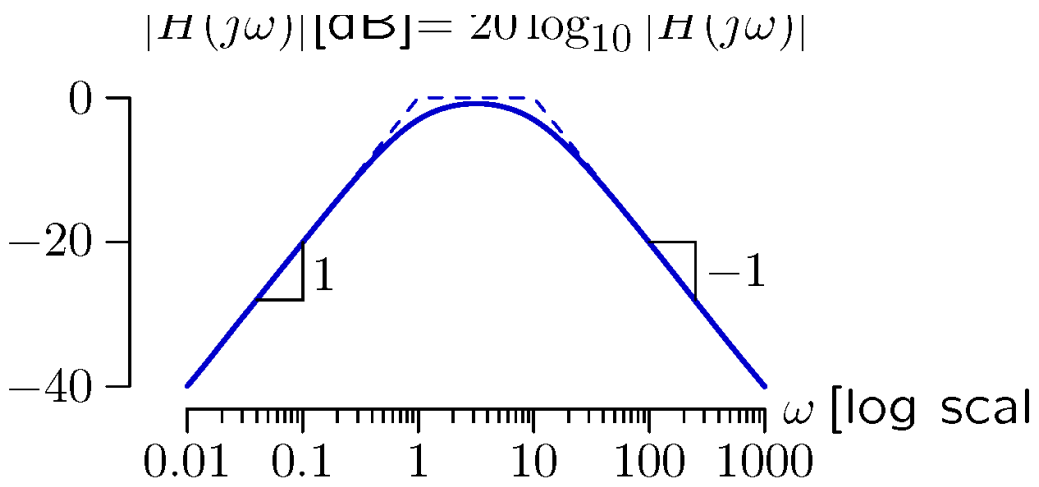


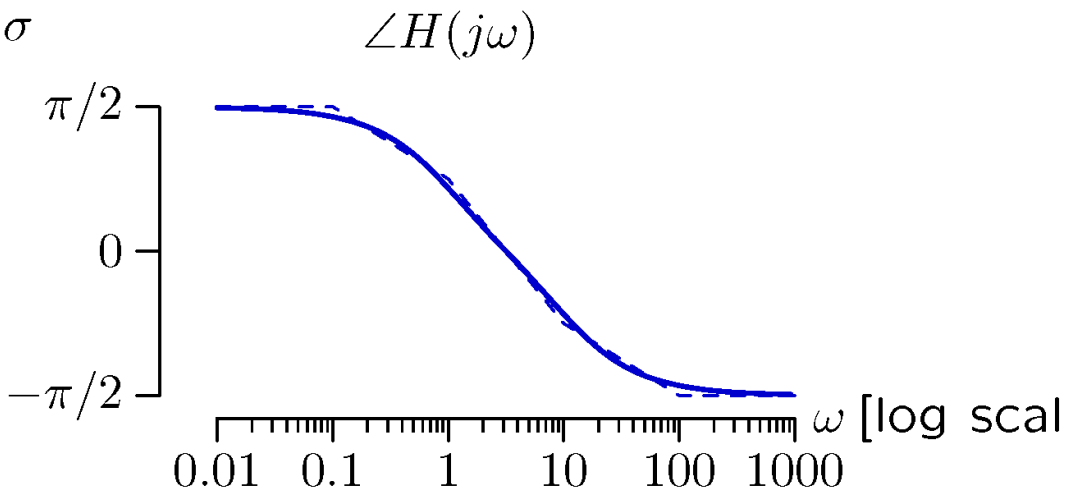


Bode Plot: dB

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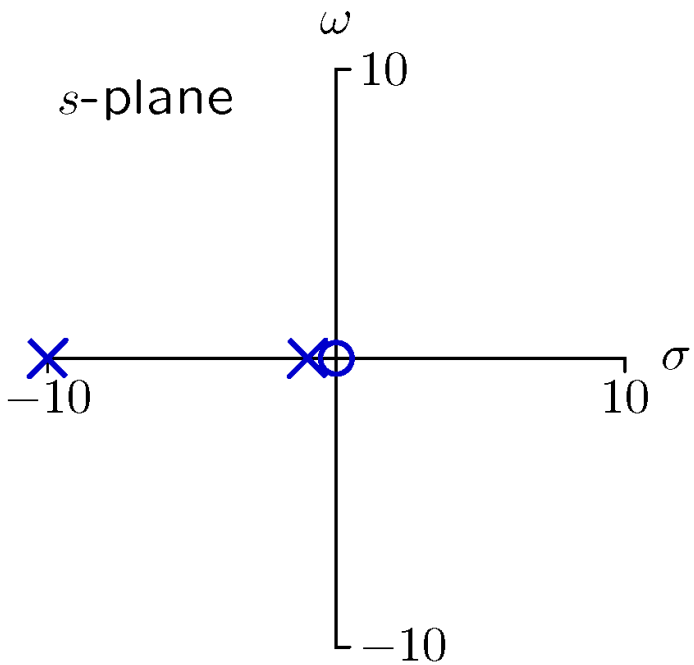


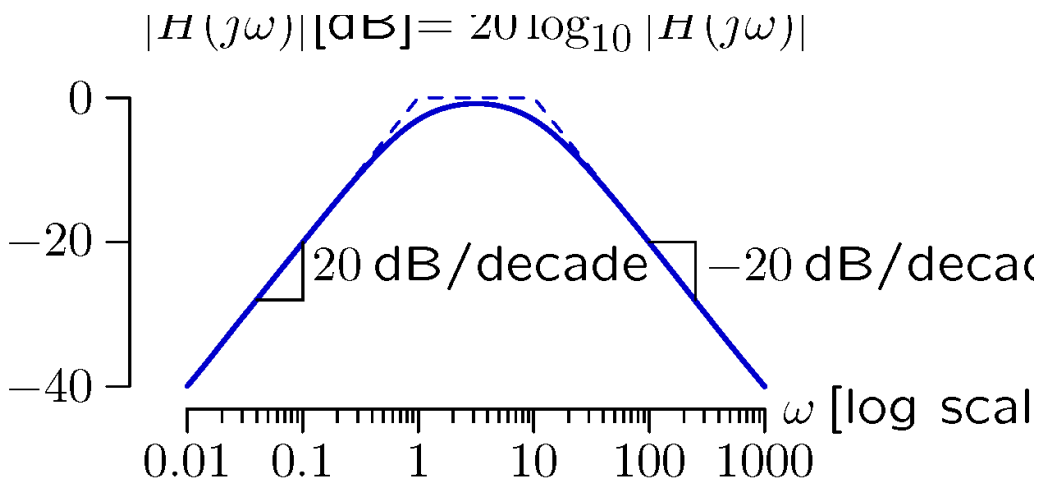


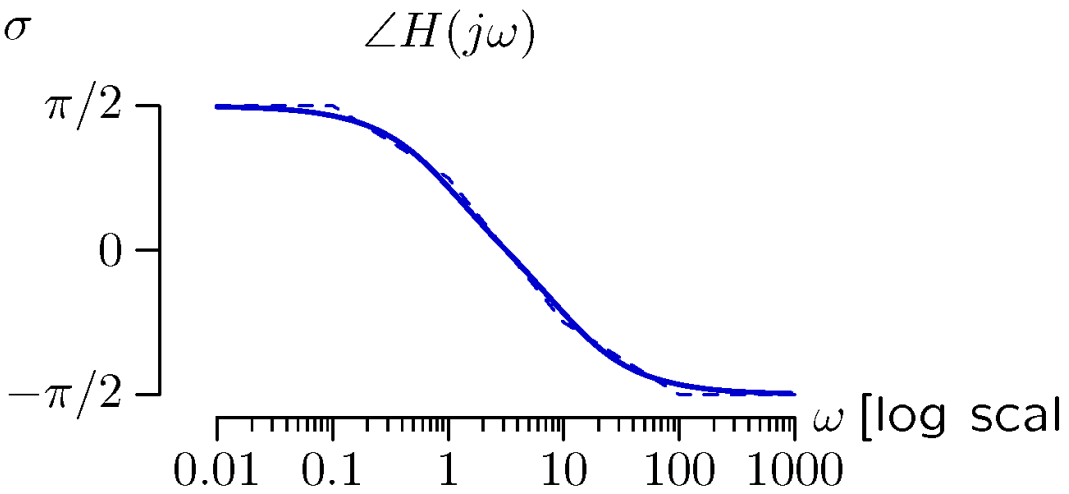


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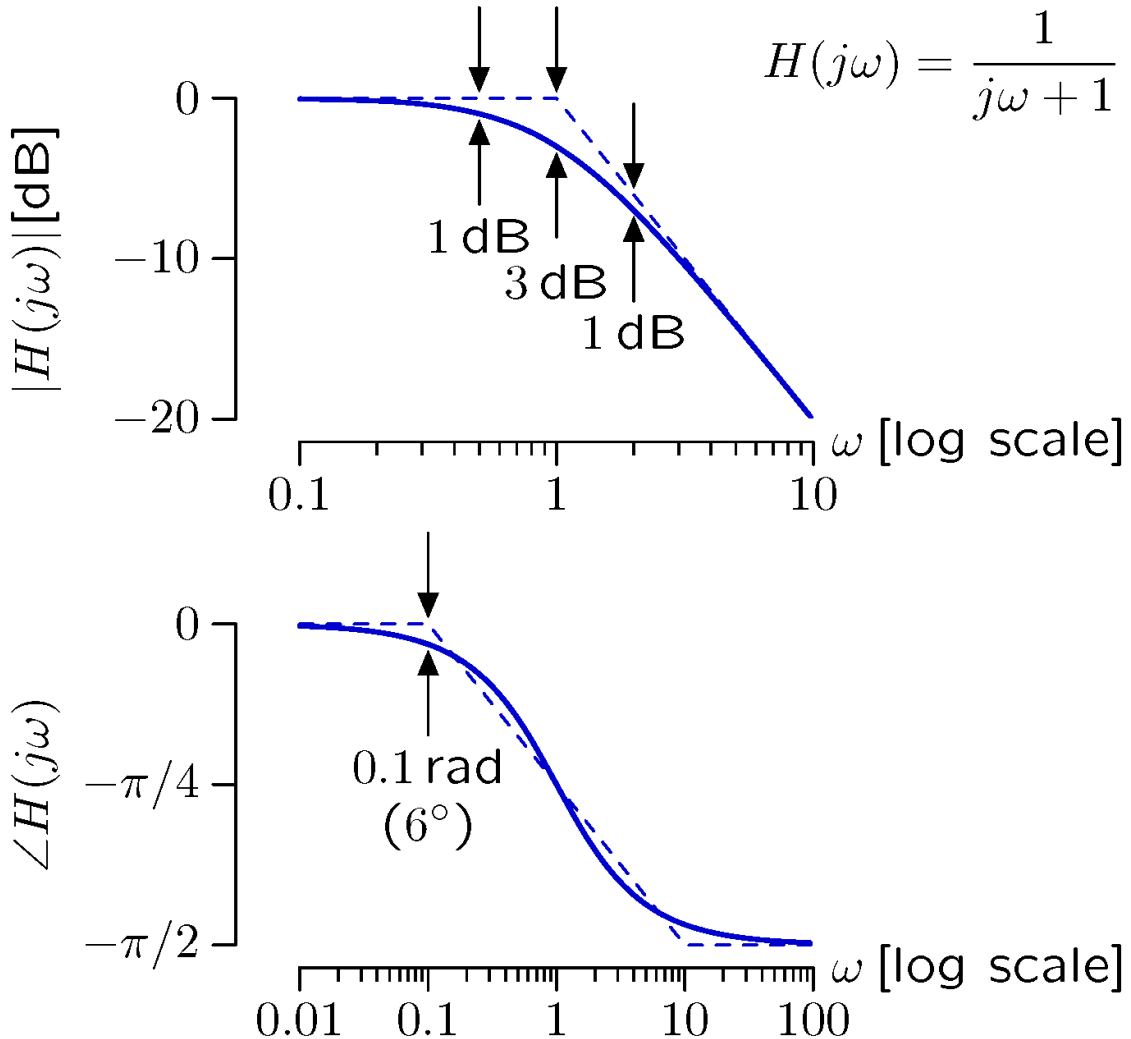






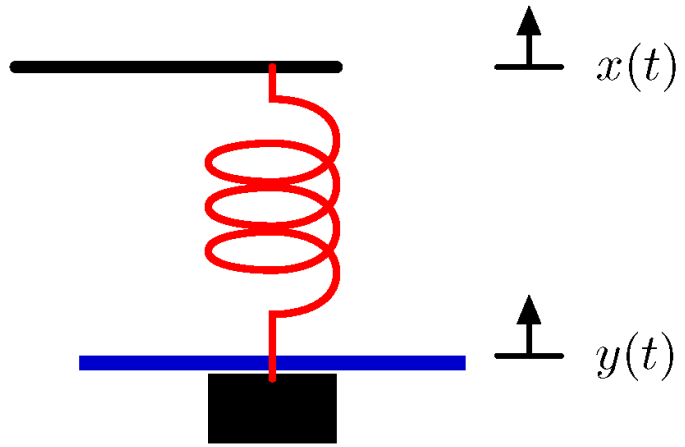
Bode Plot: Accuracy

The straight-line approximations are surprisingly accurate.



X	$20 \log_{10} X$
1	0 dB
$\sqrt{2}$	≈ 3 dB
2	≈ 6 dB
10	20 dB
100	40 dB

Example: Mass, Spring, and Dashpot



$$F = Ma = M\ddot{y}(t) = K(x(t) - y(t)) - B\dot{y}(t)$$

$$M\ddot{y}(t) + B\dot{y}(t) + Ky(t) = Kx(t)$$

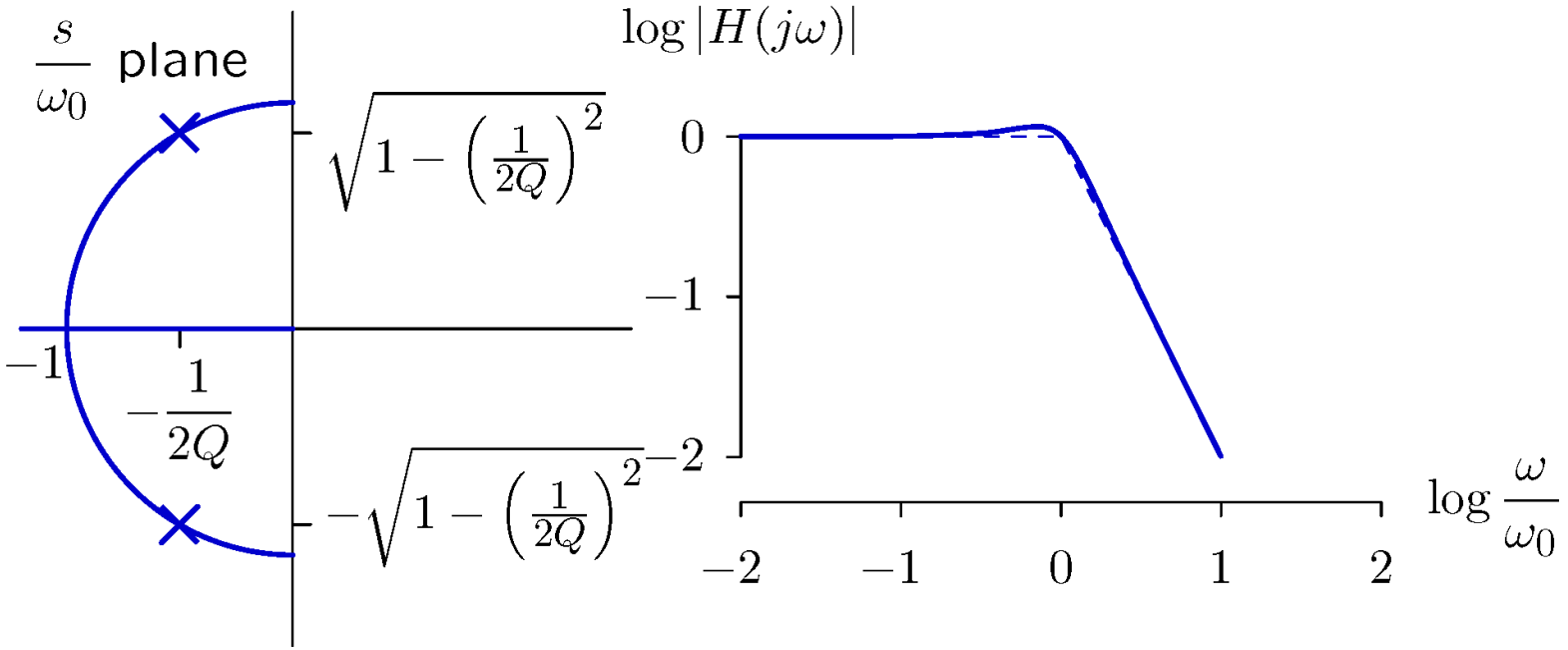
$$(s^2M + sB + K)\ Y(s) = KX(s)$$

$$H(s) = \frac{K}{s^2M + sB + K}$$

Frequency Response of a High-Q System

The frequency-response magnitude of a high-Q system is peaked.

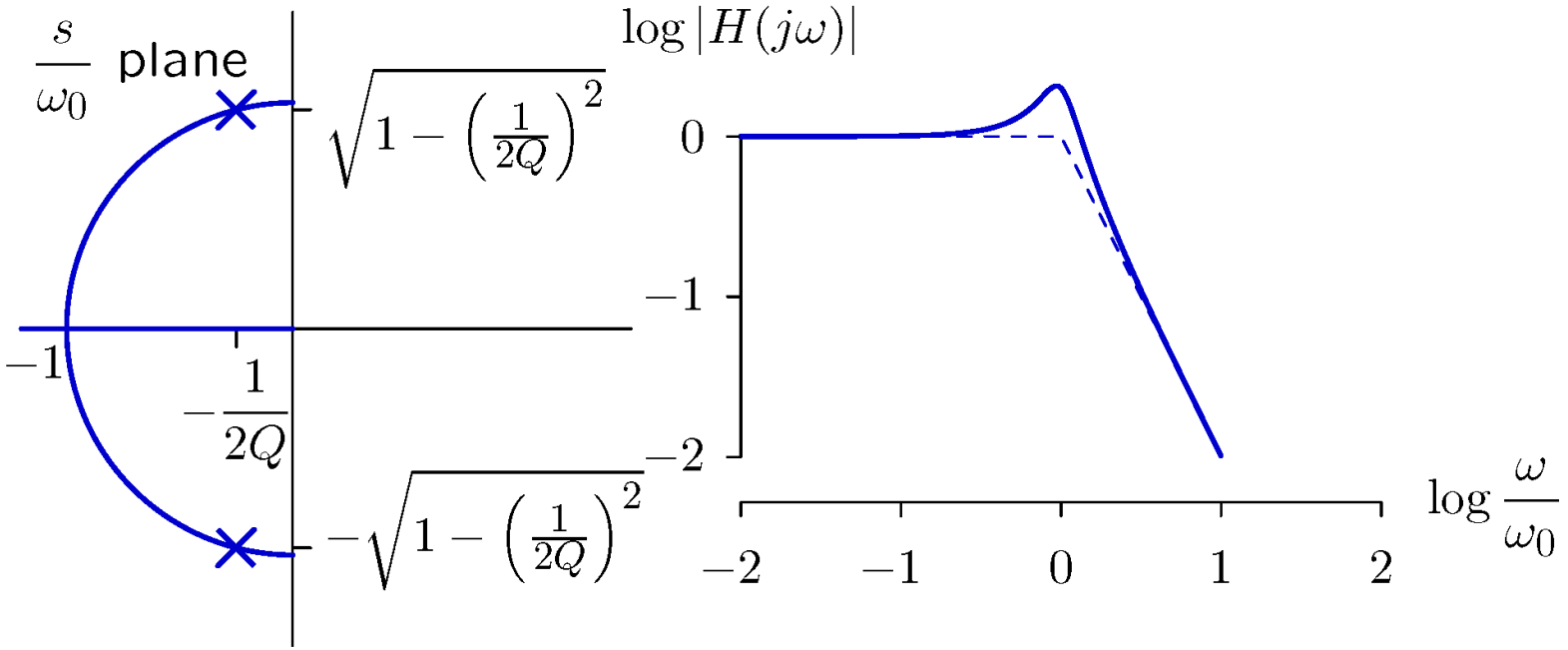
$$H(s) = \frac{1}{1 + \frac{1}{Q}\frac{s}{\omega_0} + \left(\frac{s}{\omega_0}\right)^2}$$



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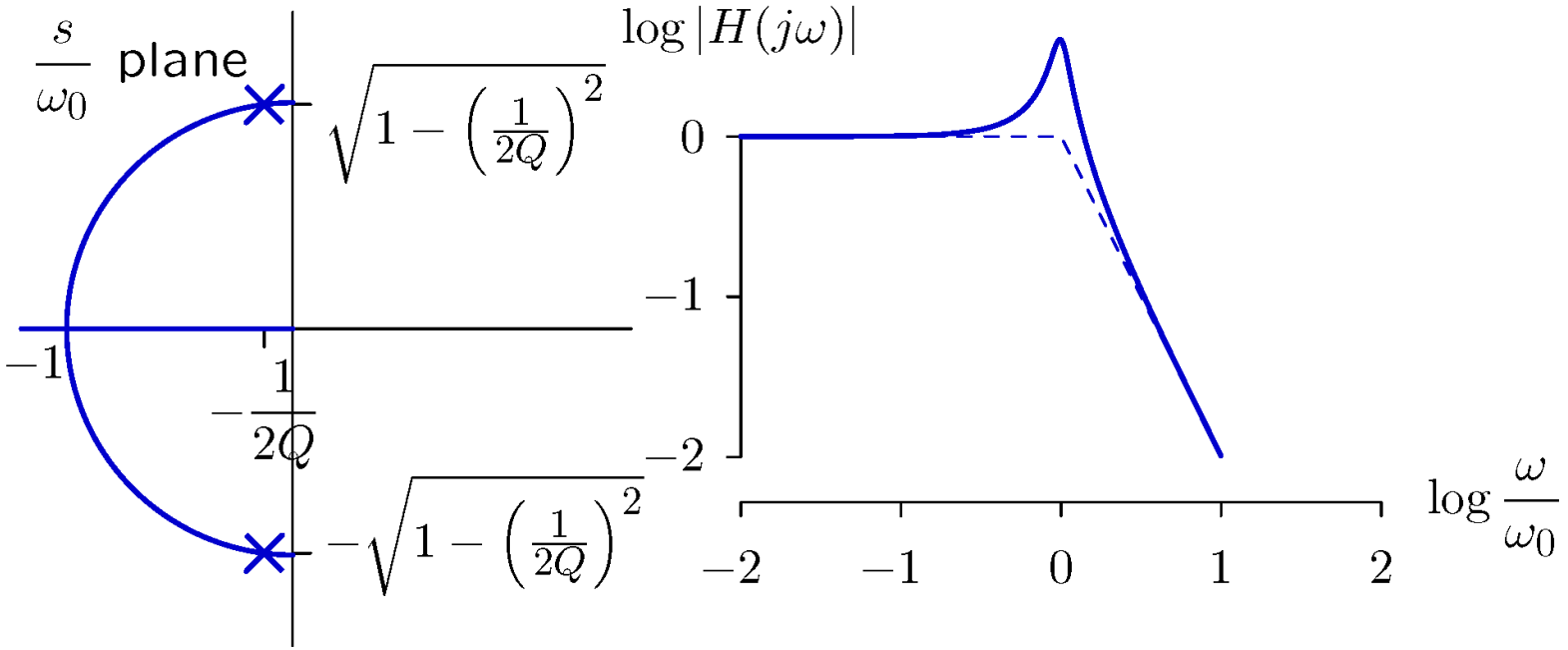
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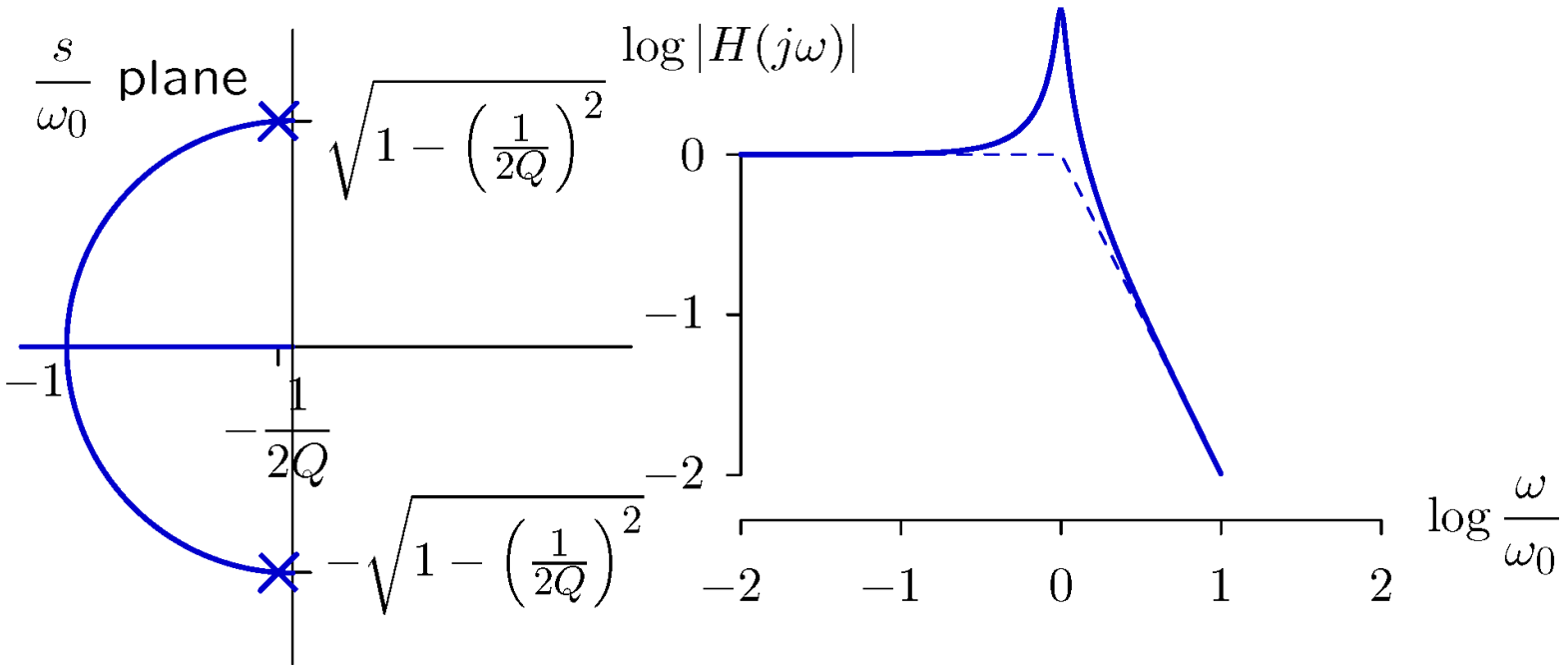
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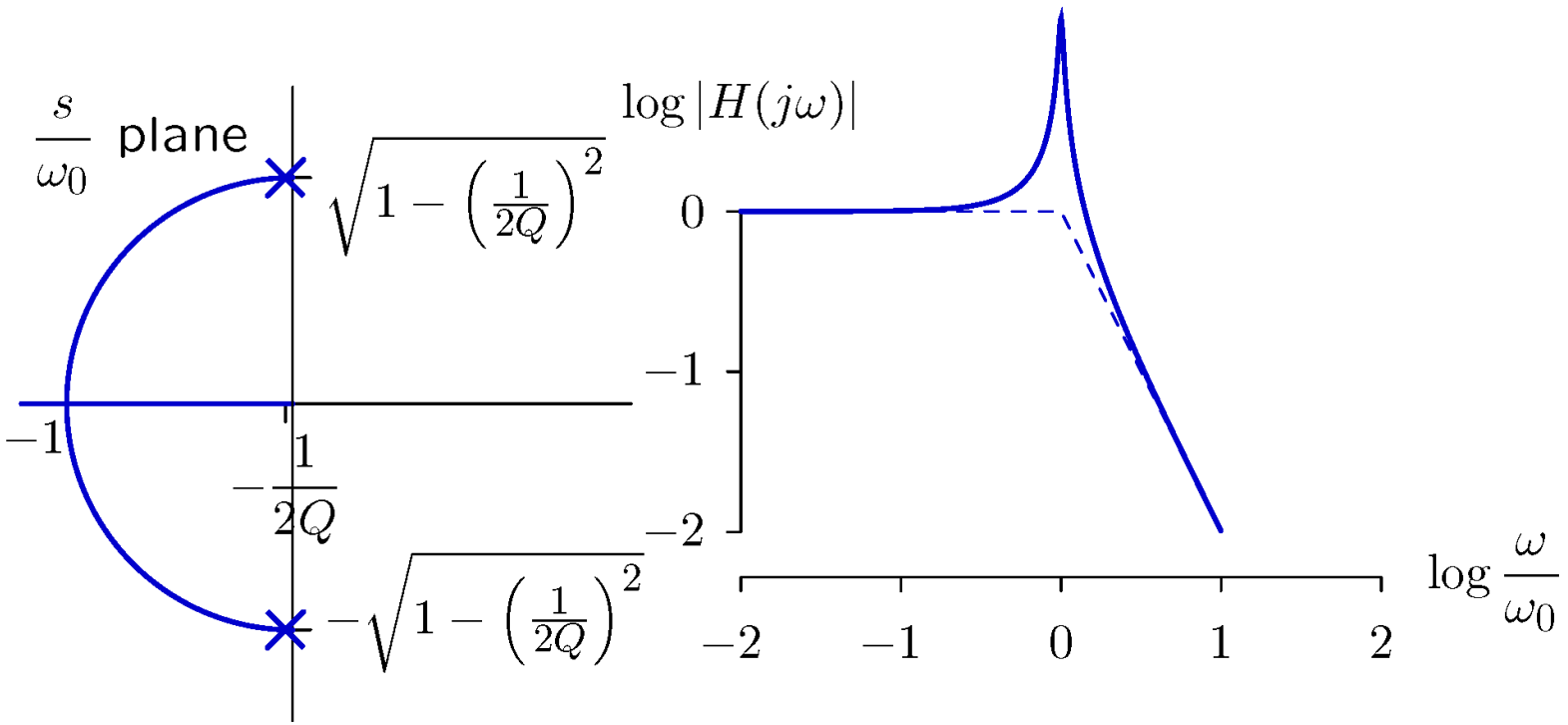
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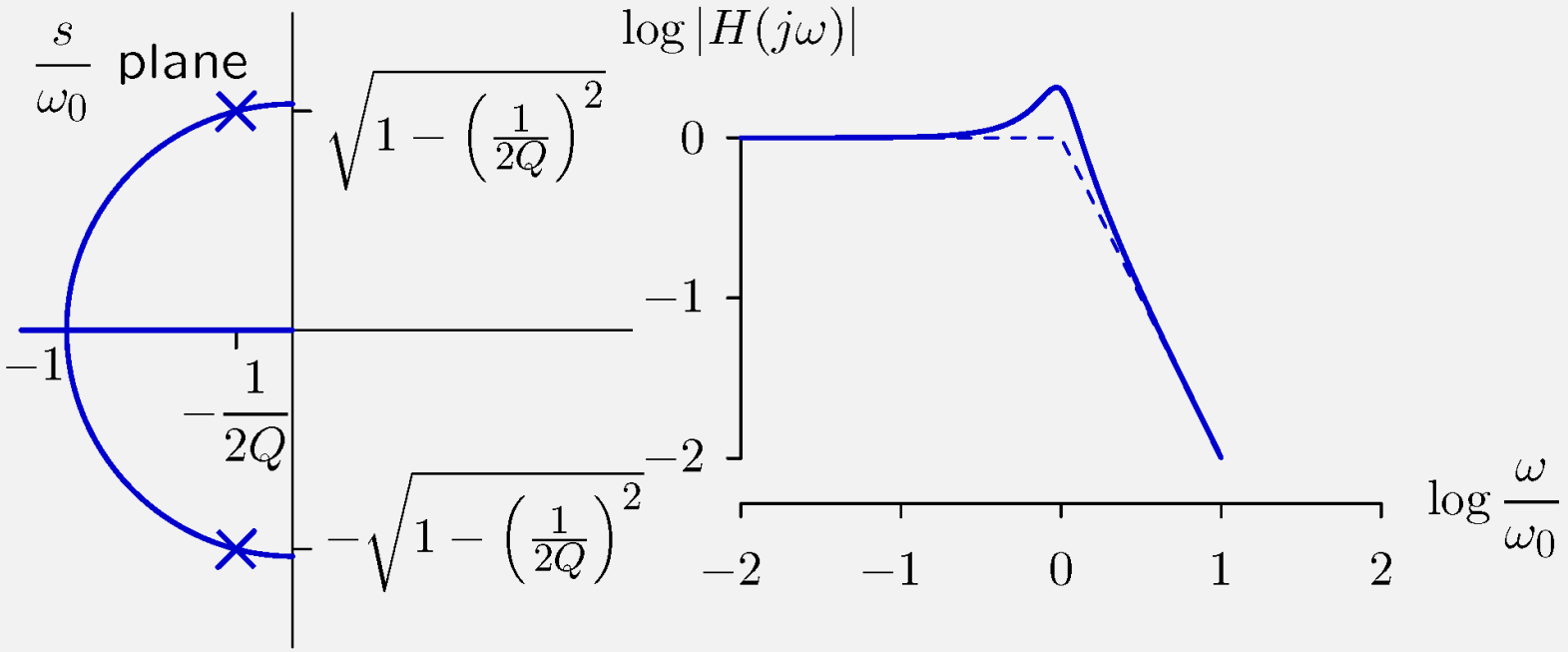
$$H(s) = \frac{1}{1 + \frac{1}{Q}\frac{s}{\omega_0} + \left(\frac{s}{\omega_0}\right)^2}$$



Check Yourself

Find dependence of peak magnitude on Q (assume $Q > 3$).

$$H(s) = \frac{1}{1 + \frac{1}{Q}\frac{s}{\omega_0} + \left(\frac{s}{\omega_0}\right)^2}$$

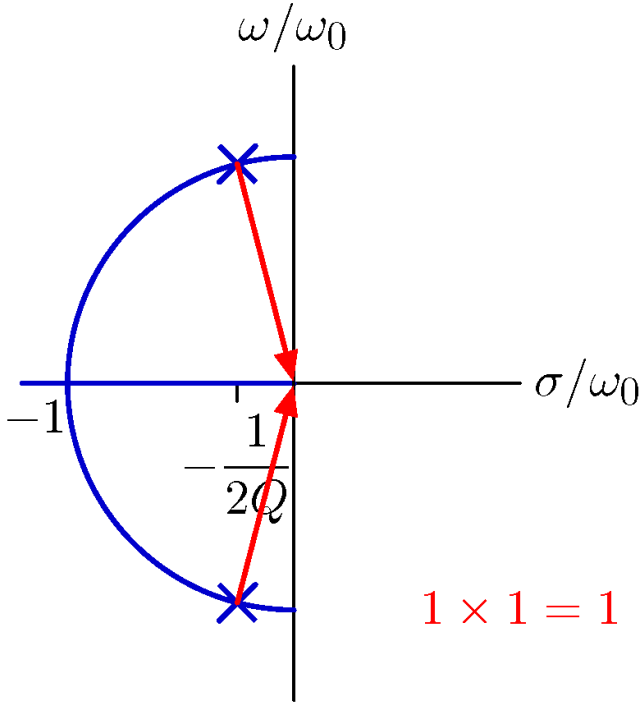


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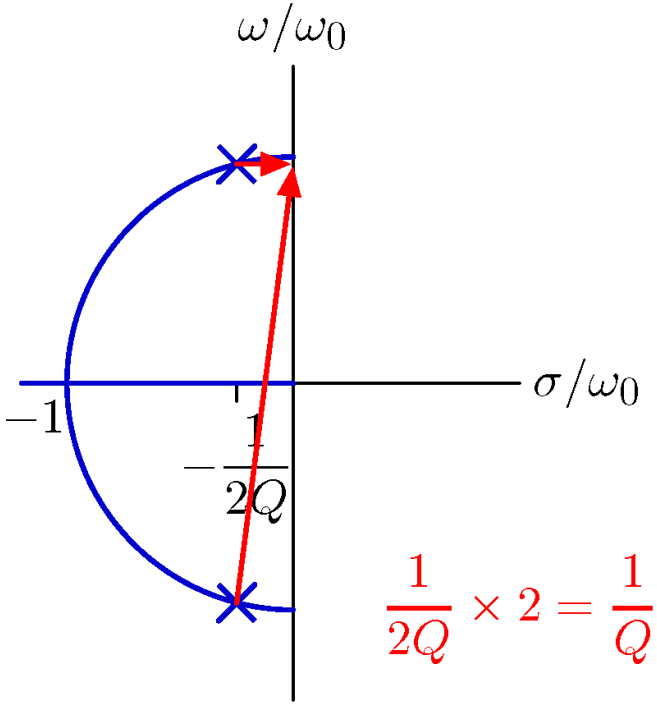
Analyze with vectors.

low frequencies



$1 \times 1 = 1$

high frequencies



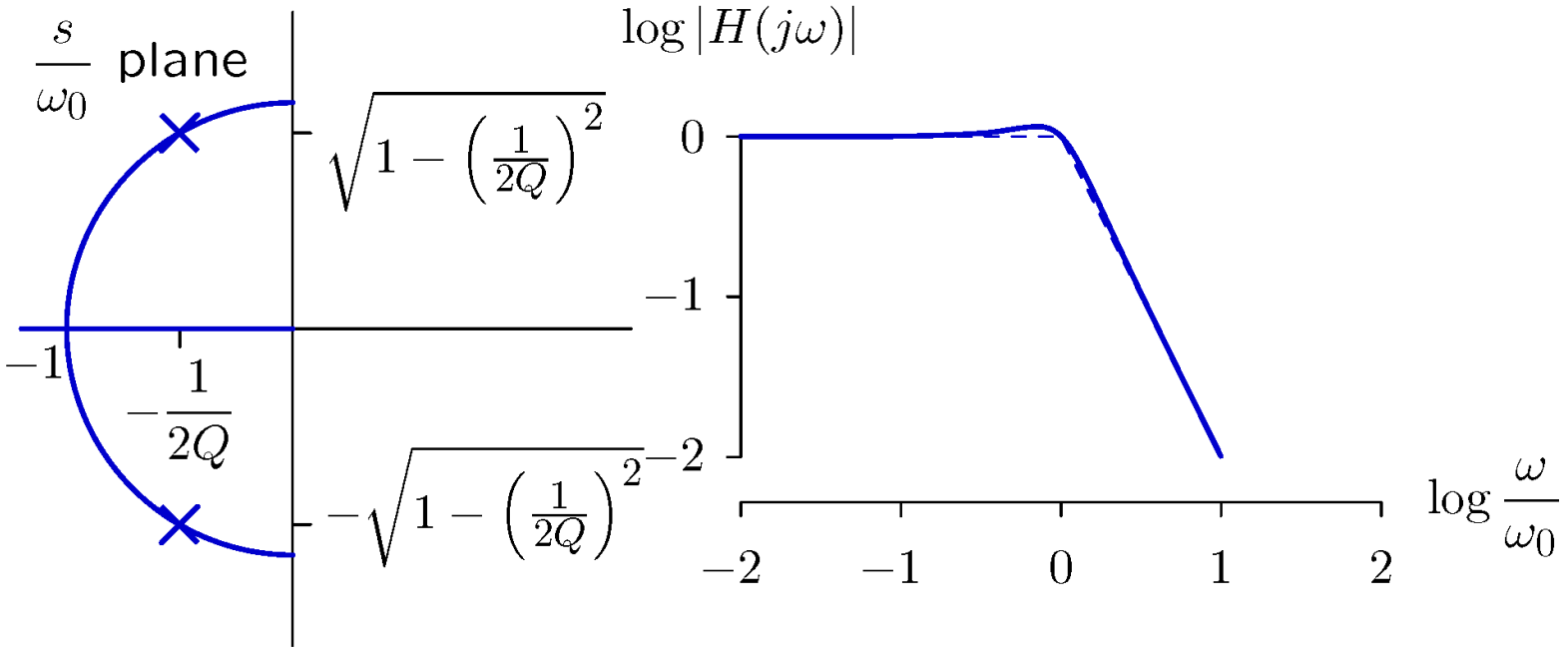
$\frac{1}{2Q} \times 2 = \frac{1}{Q}$

Peak magnitude increases with Q !

Frequency Response of a High-Q System

As Q increases, the width of the peak narrows.

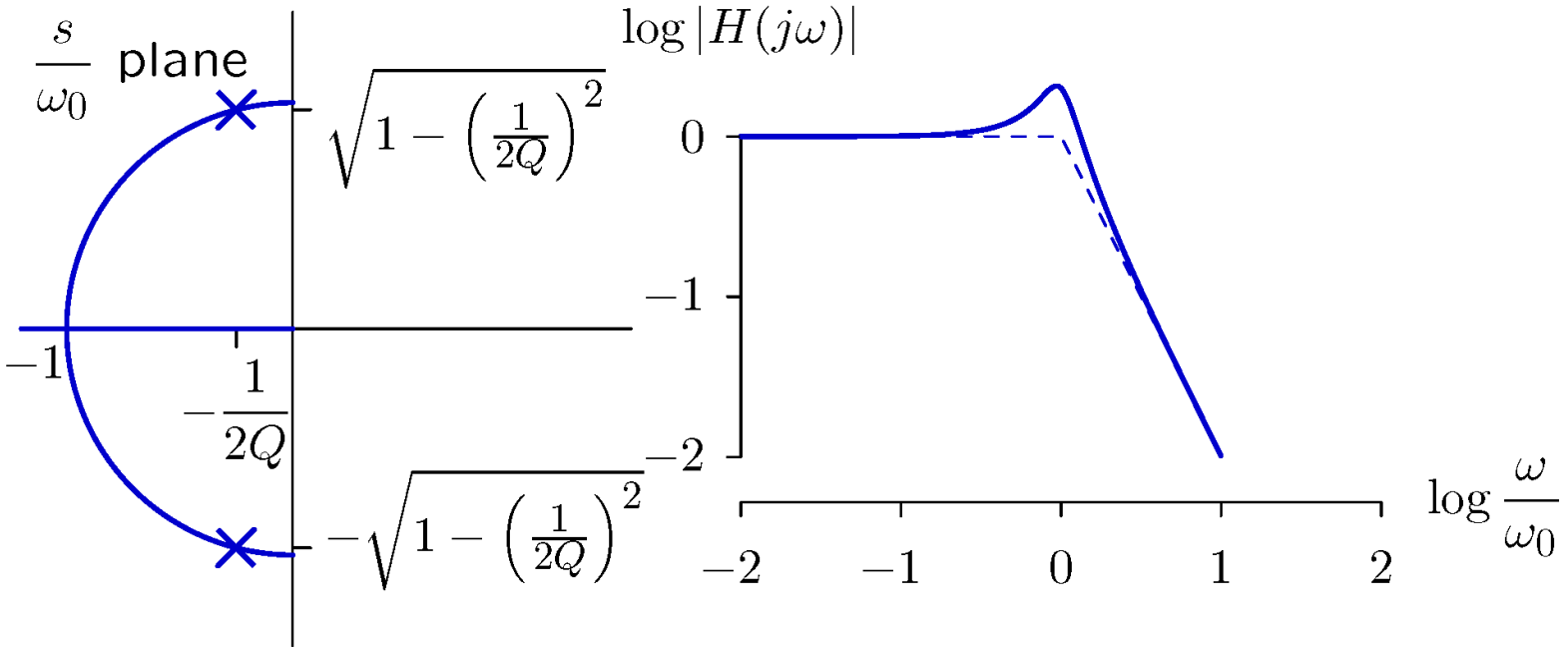
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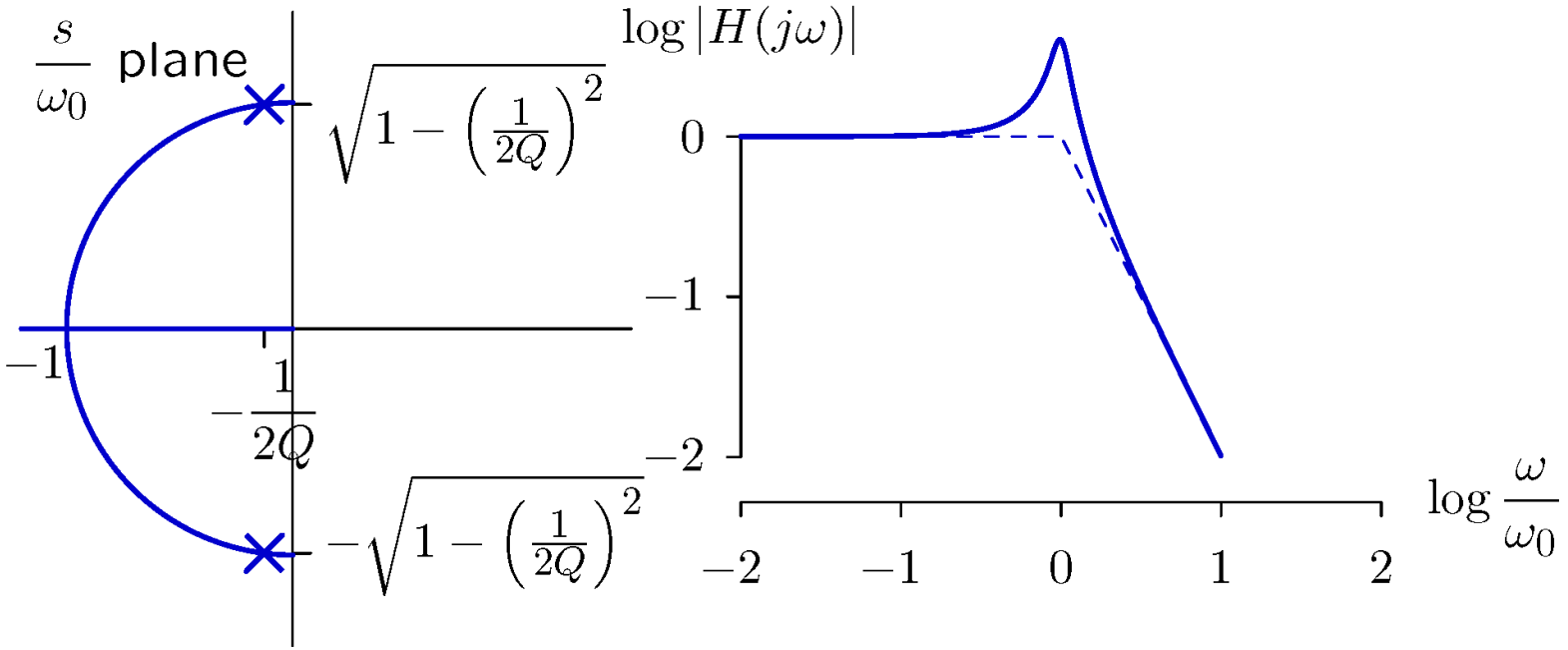
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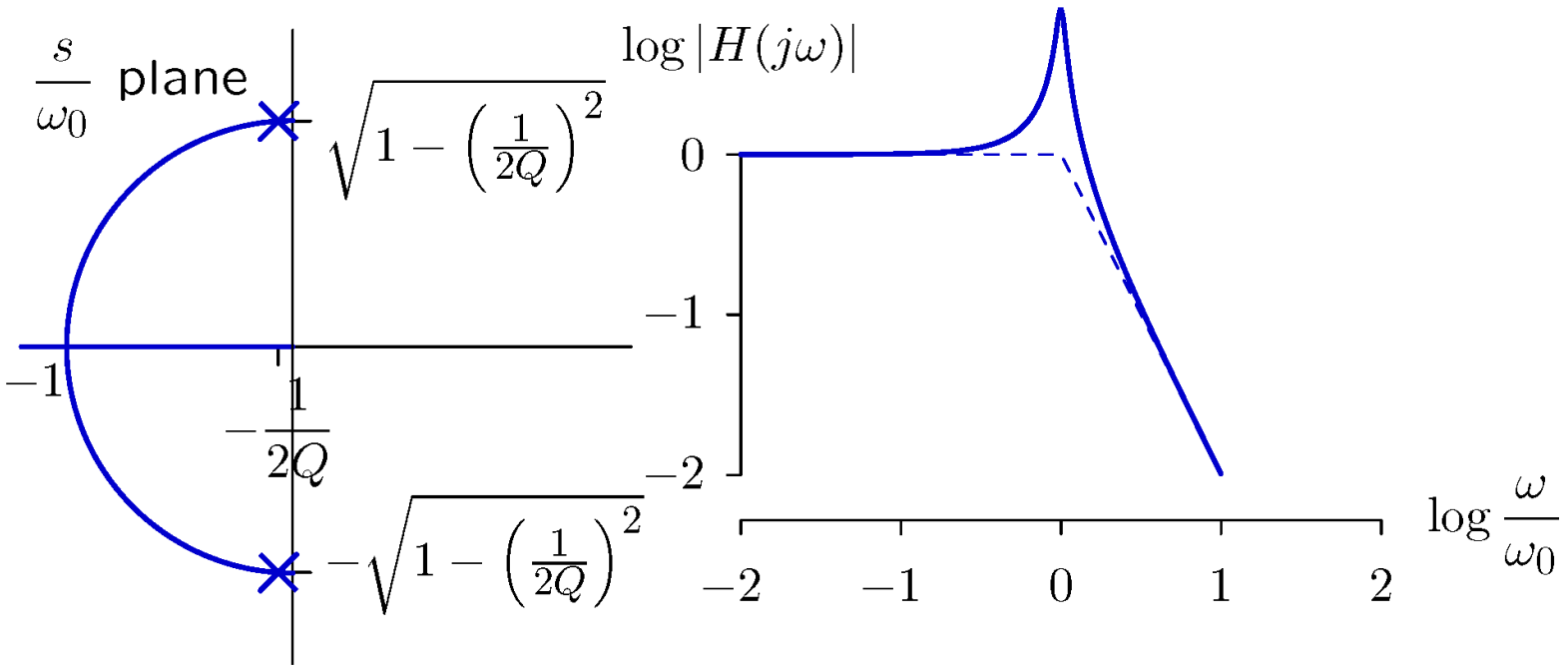
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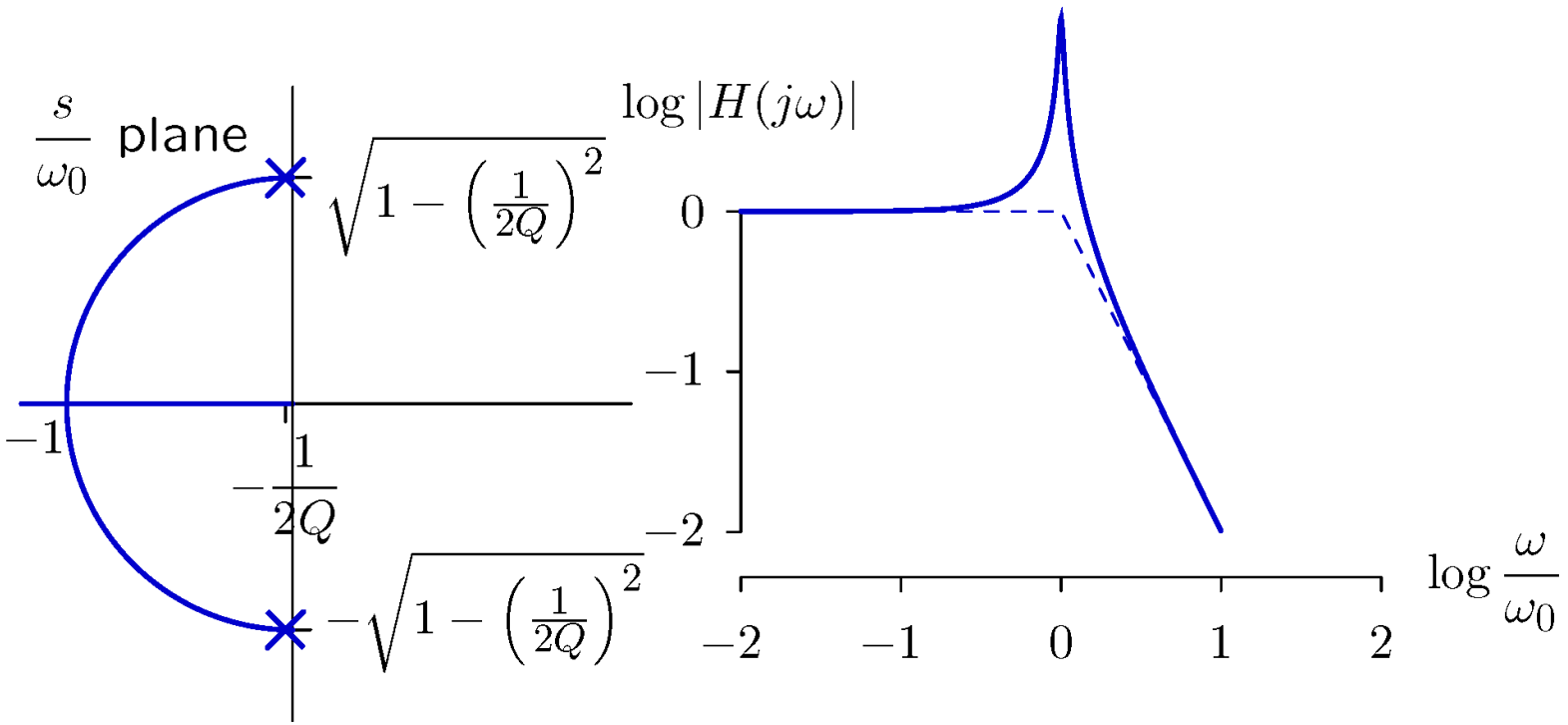
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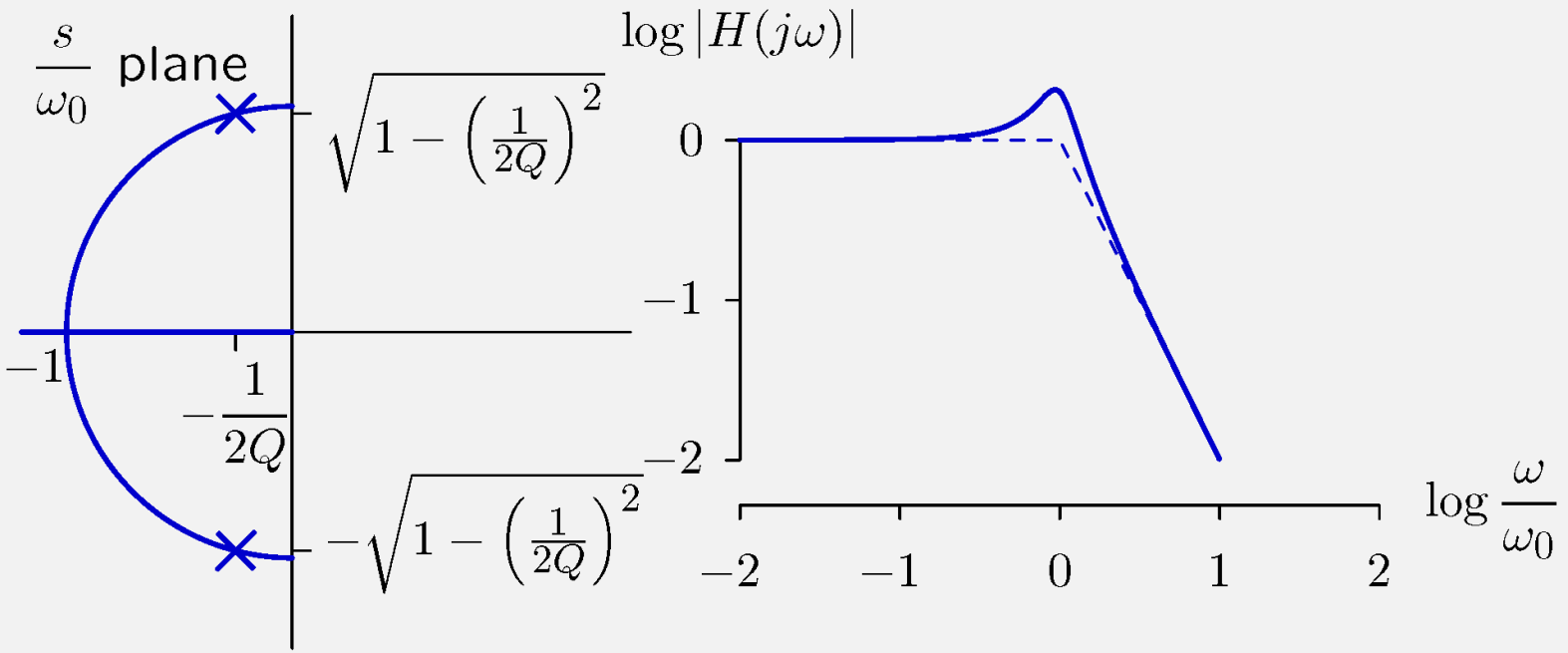
$$H(s) = \frac{1}{1 + \frac{1}{Q}\frac{s}{\omega_0} + \left(\frac{s}{\omega_0}\right)^2}$$



Check Yourself

Estimate the "3dB bandwidth" of the peak (assume $Q > 3$).

Let ω_l (or ω_h) represent the lowest (or highest) frequency for which the magnitude is greater than the peak value divided by $\sqrt{2}$. The 3dB bandwidth is then $\omega_h - \omega_l$.

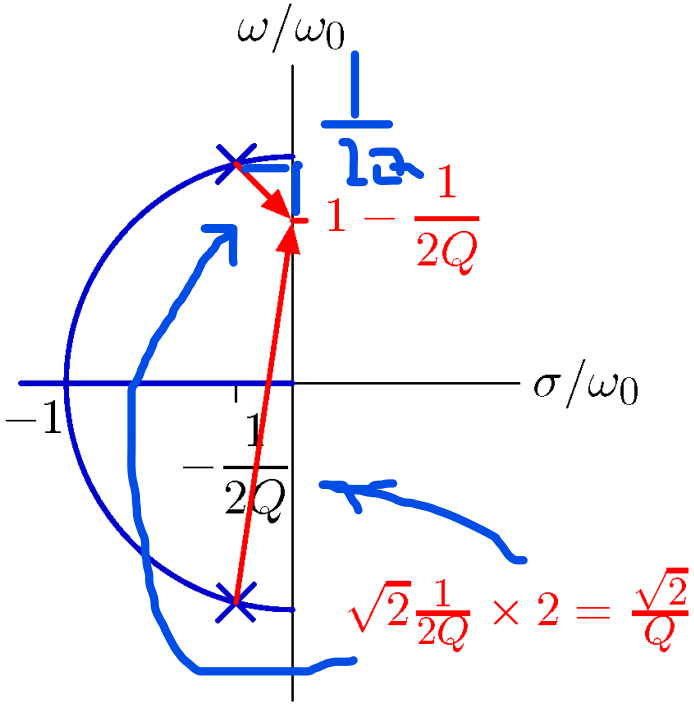


Check Yourself

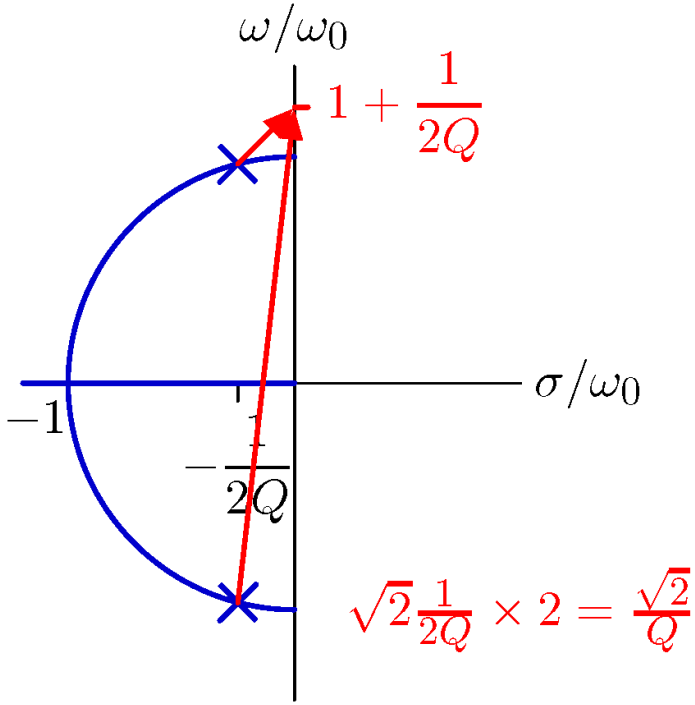
Estimate the "3dB bandwidth" of the peak (assume $Q > 3$).

Analyze with vectors.

low frequencies



high frequencies

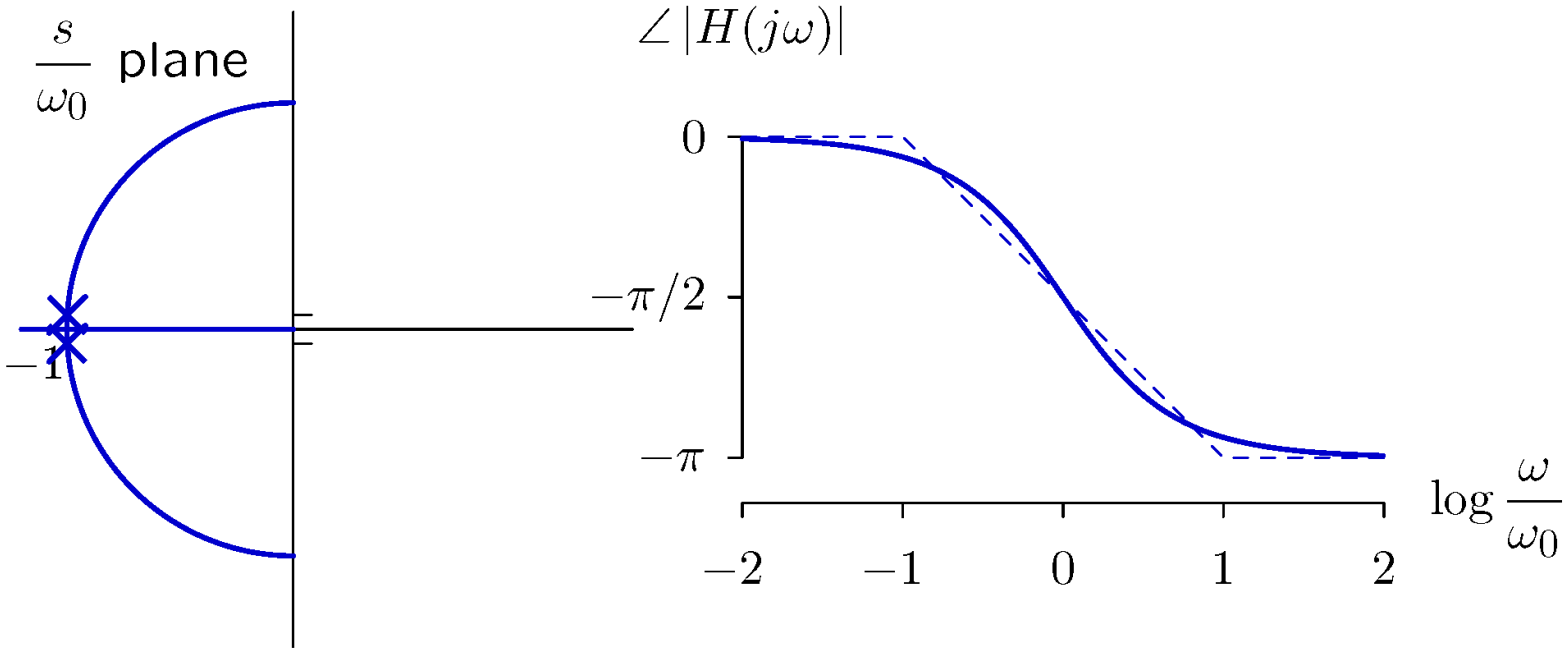


Bandwidth approximately $\frac{1}{Q}$

Frequency Response of a High-Q System

As Q increases, the phase changes more abruptly with ω.

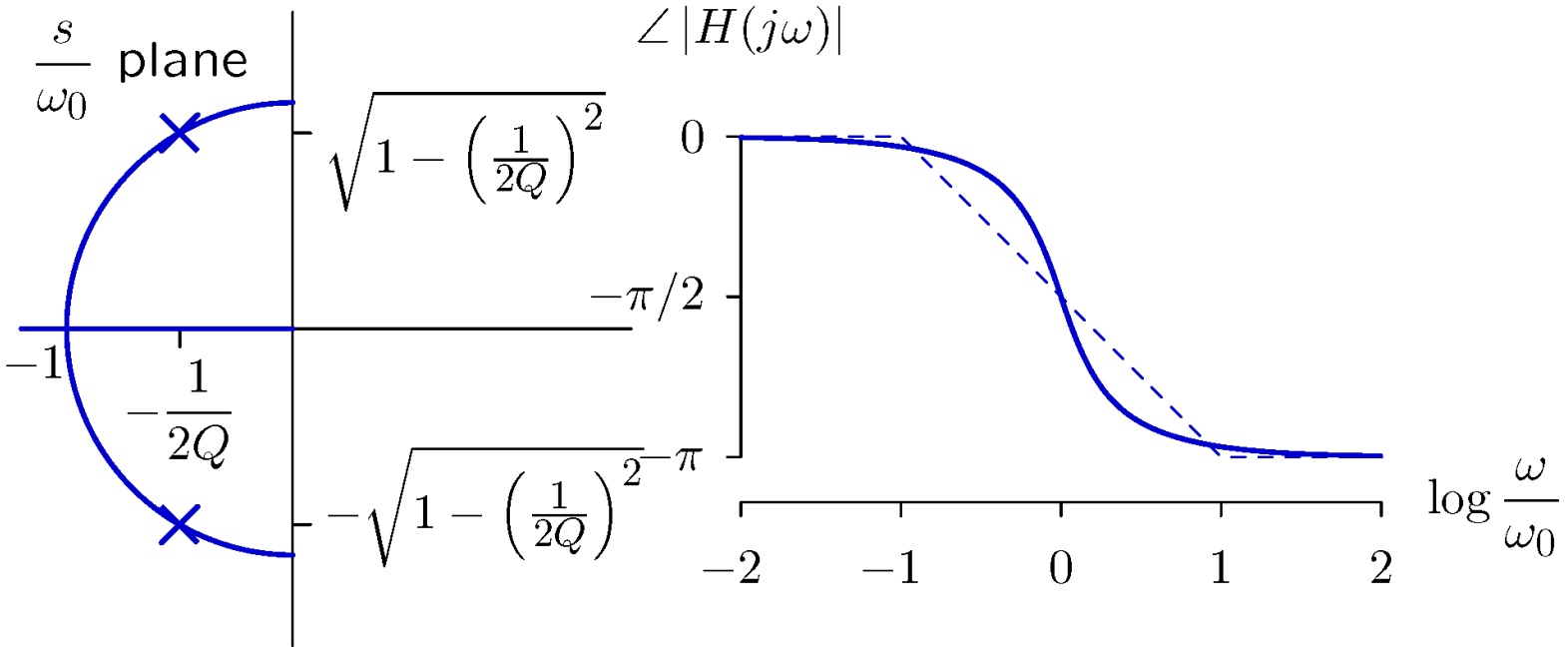
$$H(s) = \frac{1}{1 + \frac{1}{Q}\frac{s}{\omega_0} + \left(\frac{s}{\omega_0}\right)^2}$$



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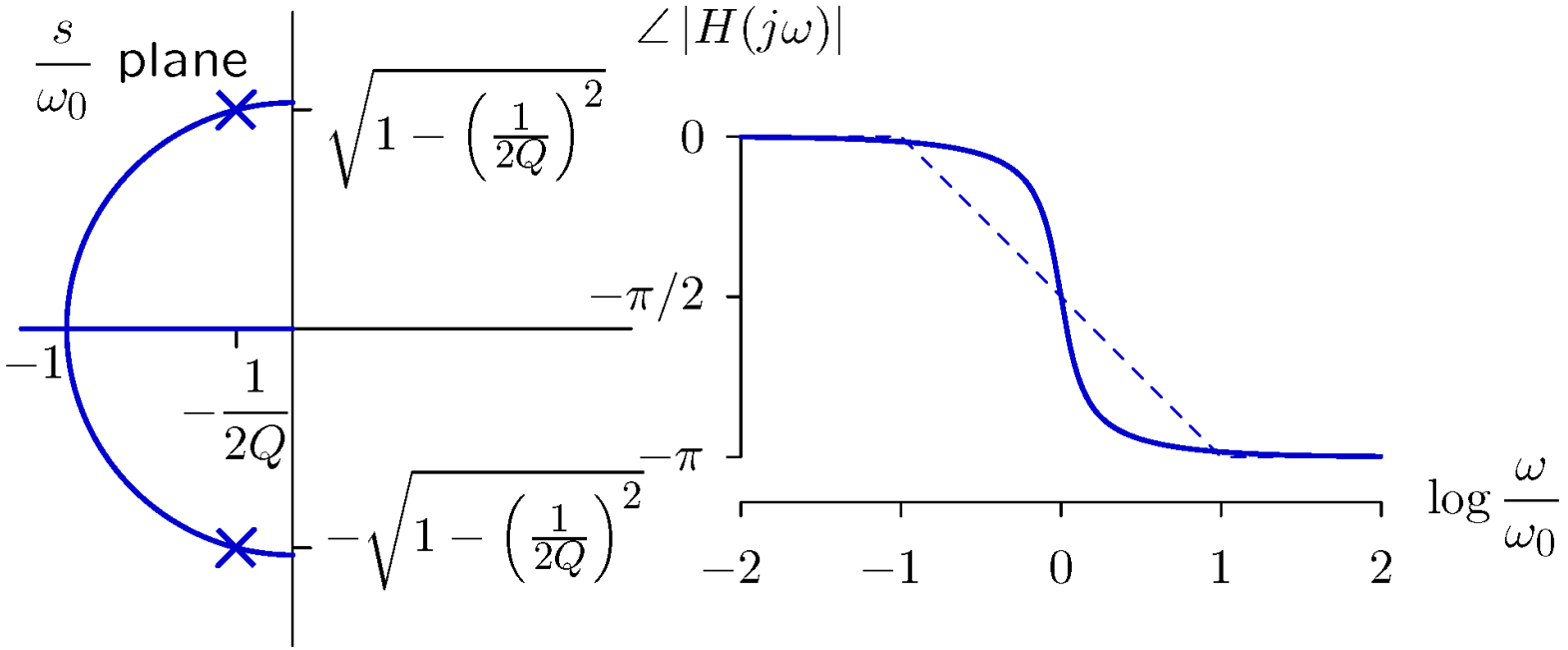
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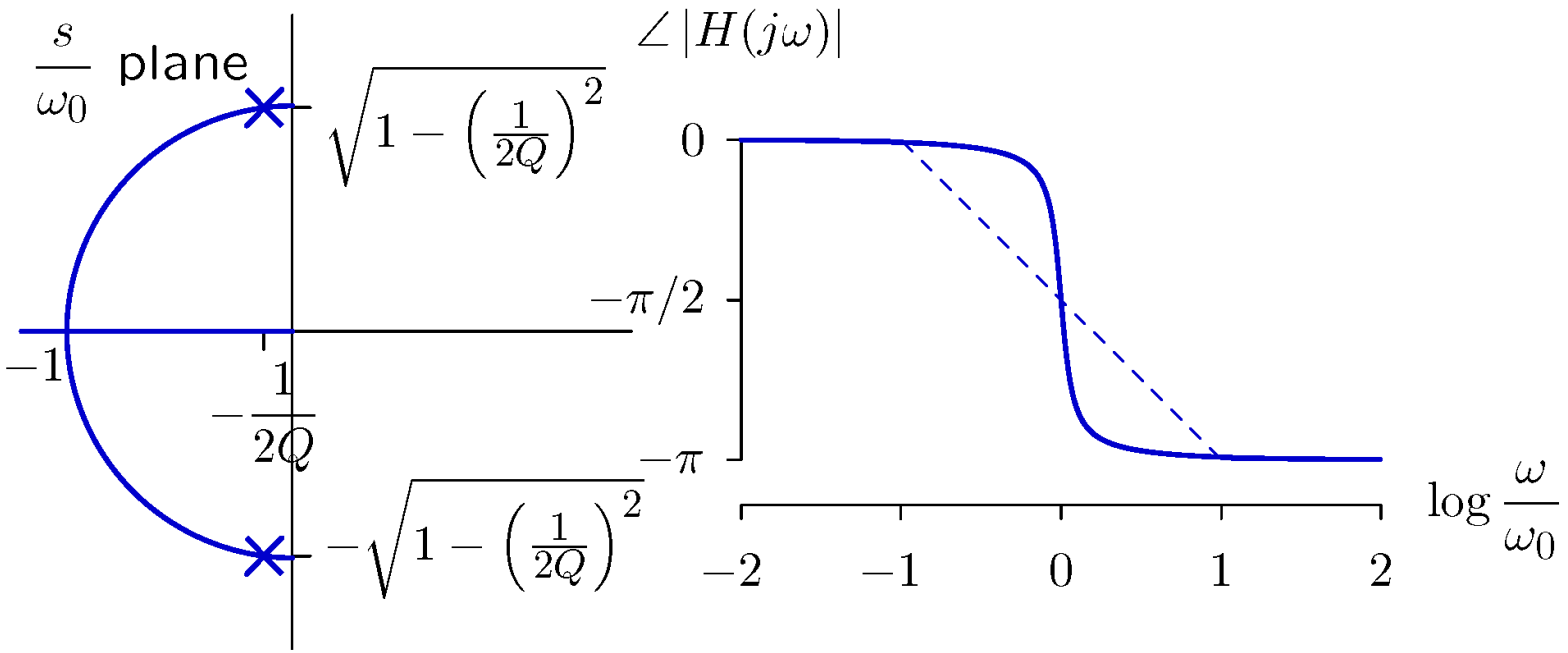
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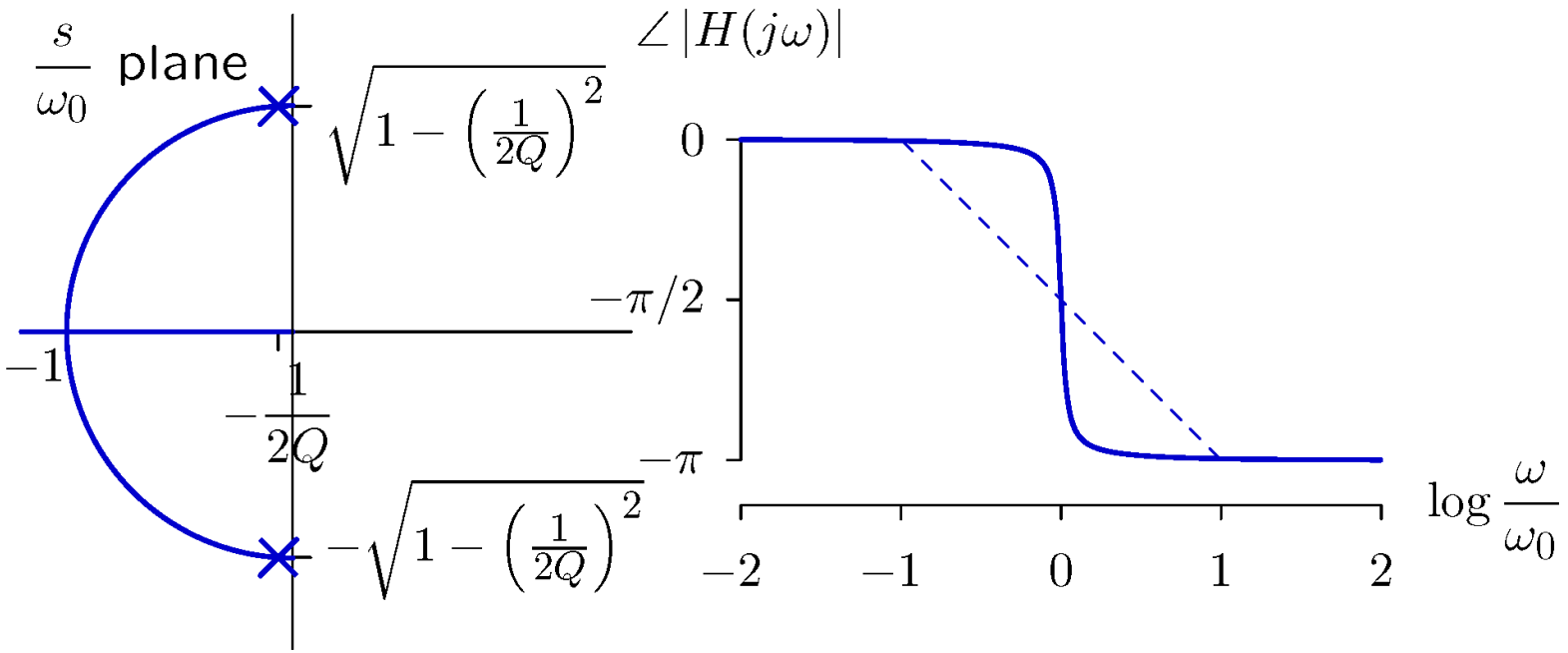
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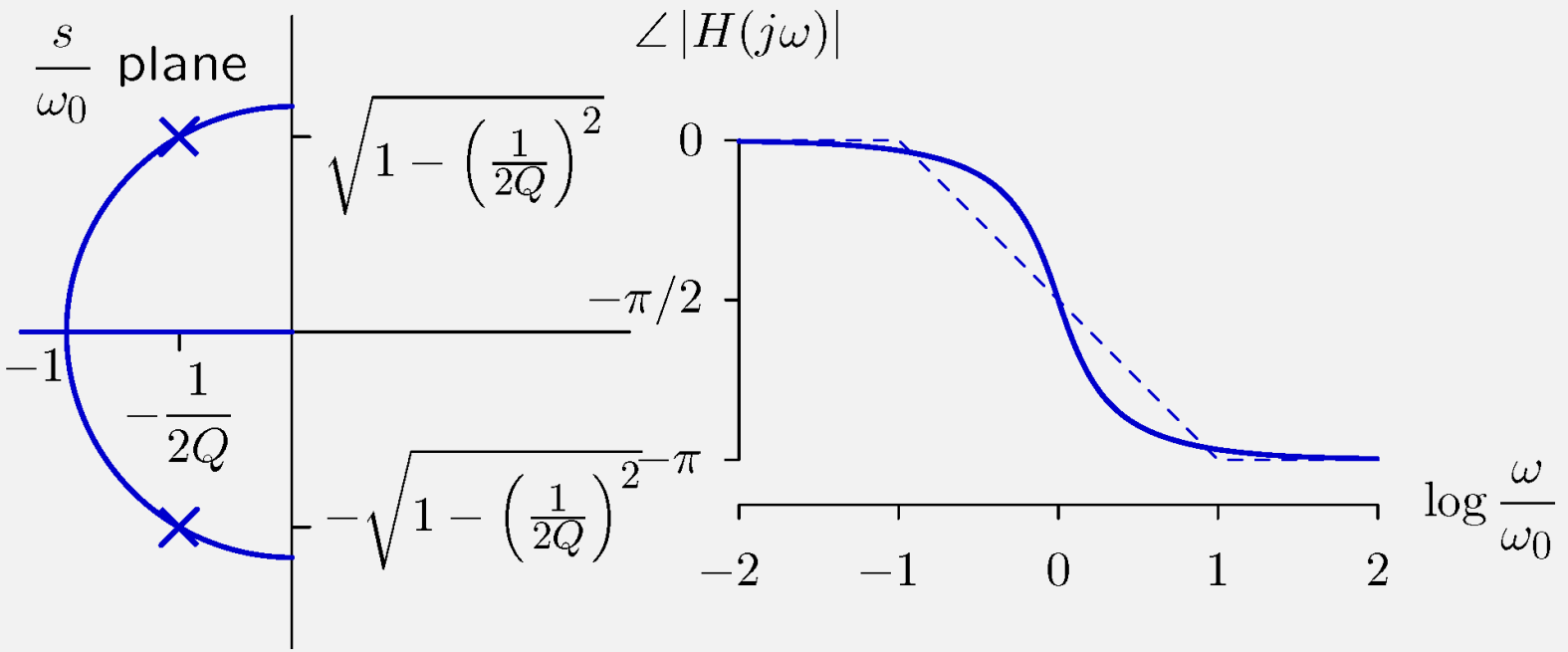
$$H(s) = \frac{1}{1 + \frac{1}{Q}\frac{s}{\omega_0} + \left(\frac{s}{\omega_0}\right)^2}$$



Check Yourself

Estimate change in phase that occurs over the 3dB bandwidth.

$$H(s) = \frac{1}{1 + \frac{1}{Q}\frac{s}{\omega_0} + \left(\frac{s}{\omega_0}\right)^2}$$

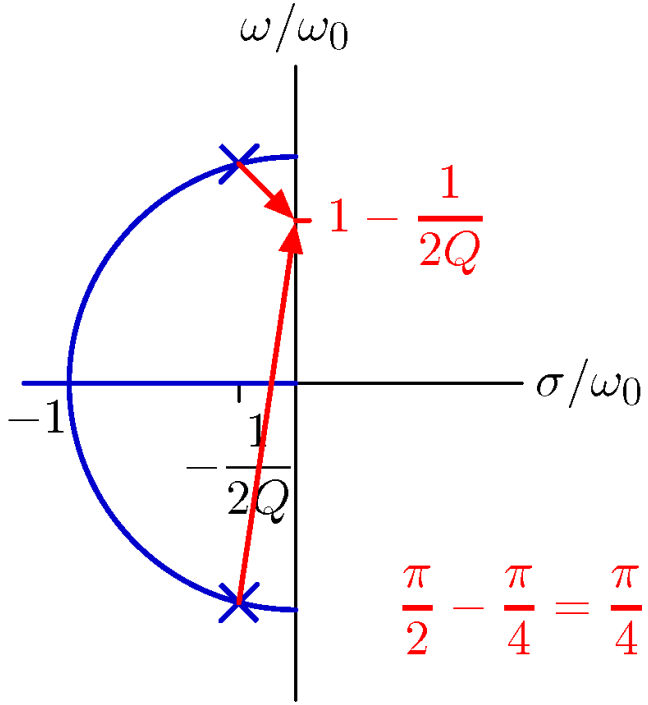


Check Yourself

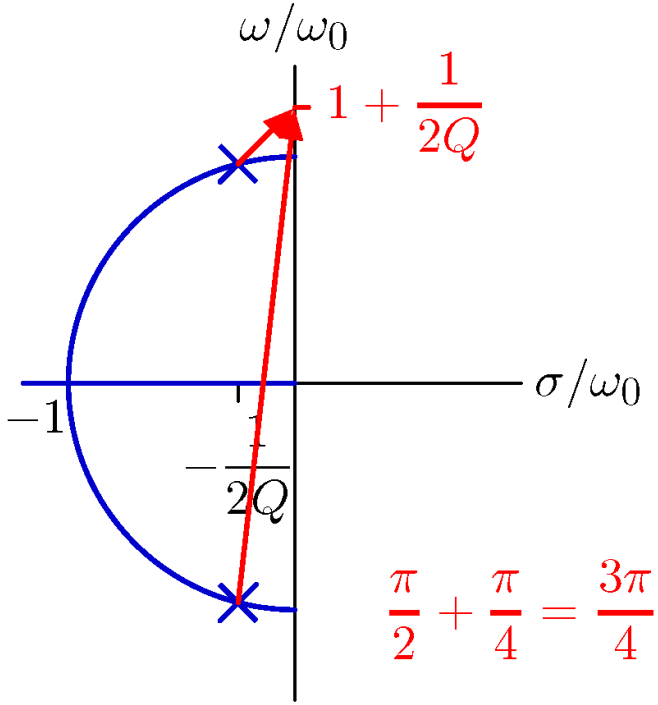
Estimate change in phase that occurs over the 3dB bandwidth.

Analyze with vectors.

low frequencies



high frequencies



Change in phase approximately $\frac{\pi}{2}$.

Summary

The frequency response of a system can be quickly determined using Bode plots.

Bode plots are constructed from sections that correspond to single poles and single zeros.

Responses for each section simply sum when plotted on logarithmic coordinates.

Assignments

- Reading Assignment: Chap. 9.4, 10.4, 3.8-3.11, 6.0-6.3, 6.5